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Volkerink et al.

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(45) **Date of Patent:** **Sep. 9, 2025**

(54) **DISTRIBUTED INTELLIGENT SOFTWARE FOR VIBRATION AND ACOUSTIC MONITORING AND SYSTEMS AND METHODS IMPLEMENTING THE SAME**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 253 days.

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(65) **Prior Publication Data**

US 2022/0095085 A1 Mar. 24, 2022

Related U.S. Application Data

(63) Continuation-in-part of application No. 17/209,192, filed on Mar. 22, 2021, now Pat. No. 11,363,427.
(Continued)

(51) **Int. Cl.**
H04W 4/00 (2018.01)
H04W 4/38 (2018.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04W 4/38** (2018.02); **H04W 4/80** (2018.02); **H04W 56/001** (2013.01); **H04W 4/35** (2018.02)

(58) **Field of Classification Search**
CPC H04W 4/38; H04W 4/80; H04W 56/001; H04W 4/35
See application file for complete search history.

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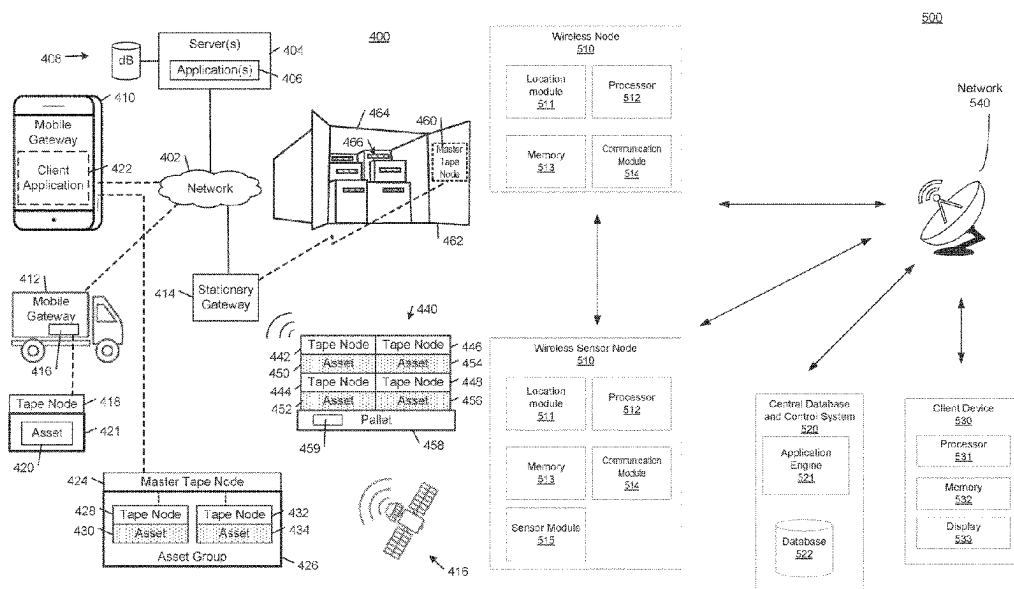
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Primary Examiner — Stephen M D Agosta

(57) **ABSTRACT**

A method includes instructing a first sensing device to enter a first state, the first state comprising a first set of functions and behavior for the sensing device. The first sensing device is instructed to aggregate sensing data, the sensing data comprising measurements of the property of the surroundings of the first sensing device measured by the at least one sensor. Relevant data from the aggregated sensing data is stored on one of a memory of the first sensing device, a database of a central database and control system associated with the first sensing device, and a memory of a client device. It is determined that a first event has occurred based on the aggregated sensing data, and responsive to the determining that the first event has occurred, the first sensing device is instructed to enter a second state.

20 Claims, 30 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 63/087,284, filed on Oct. 4, 2020, provisional application No. 63/081,887, filed on Sep. 22, 2020, provisional application No. 62/992,904, filed on Mar. 21, 2020.

- (51) **Int. Cl.**
H04W 4/80 (2018.01)
H04W 56/00 (2009.01)
H04W 4/35 (2018.01)

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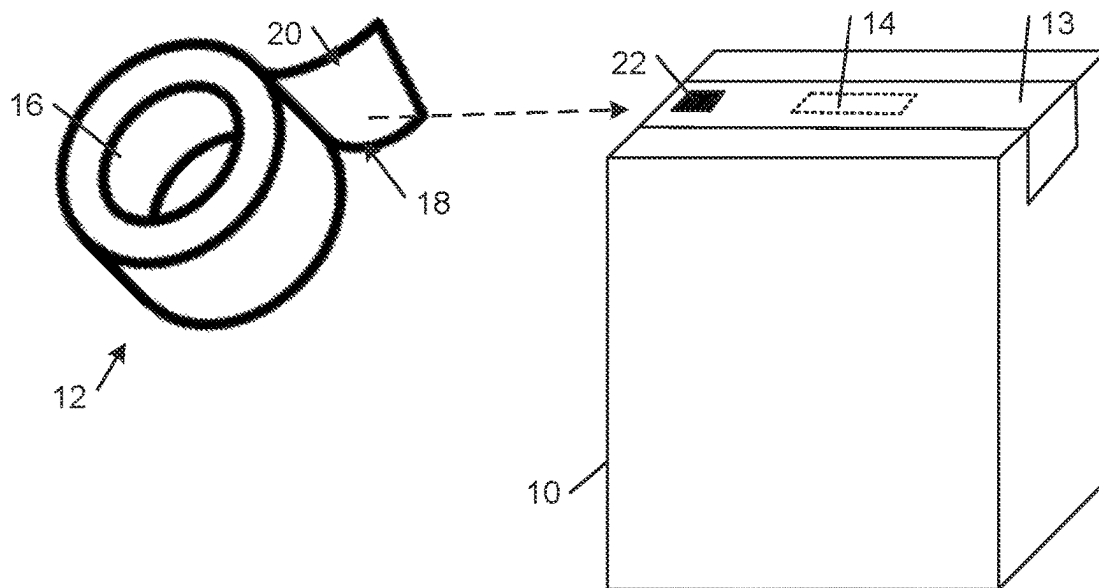


FIG. 1A

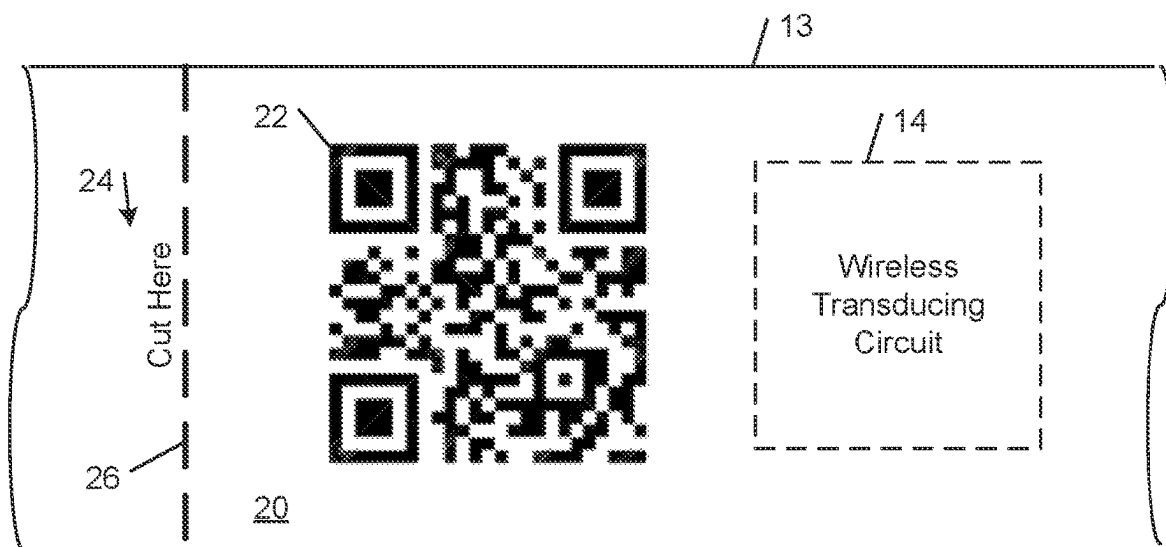
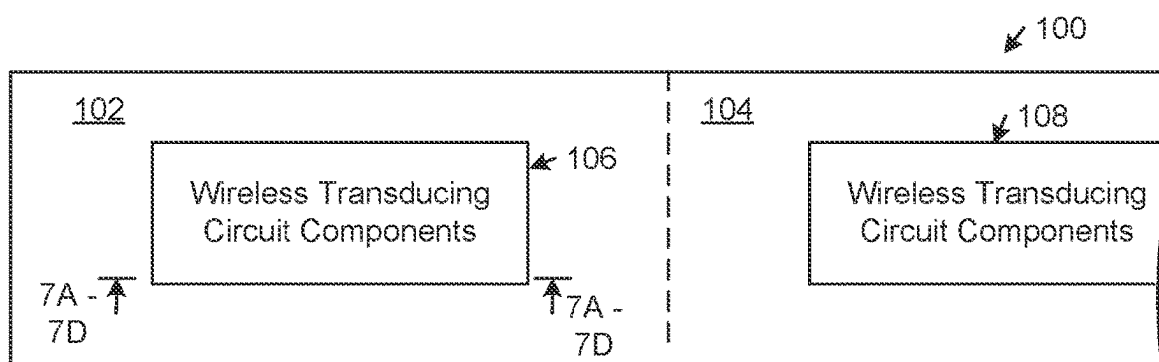
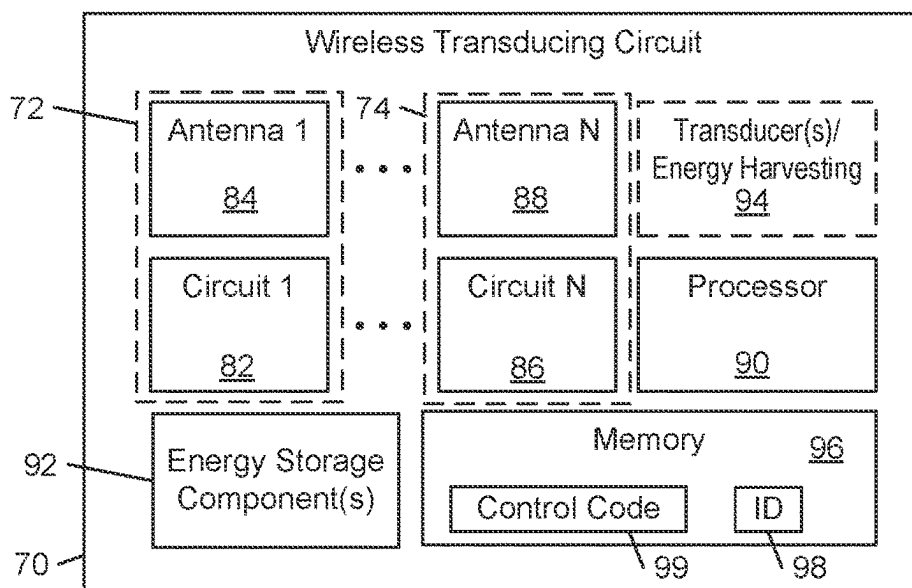
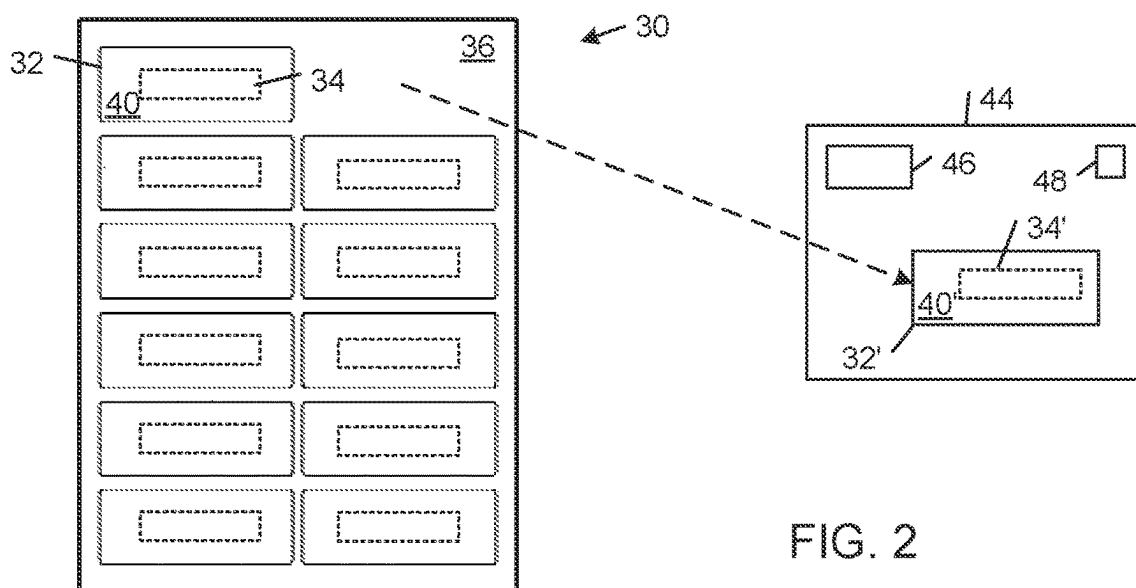


FIG. 1B



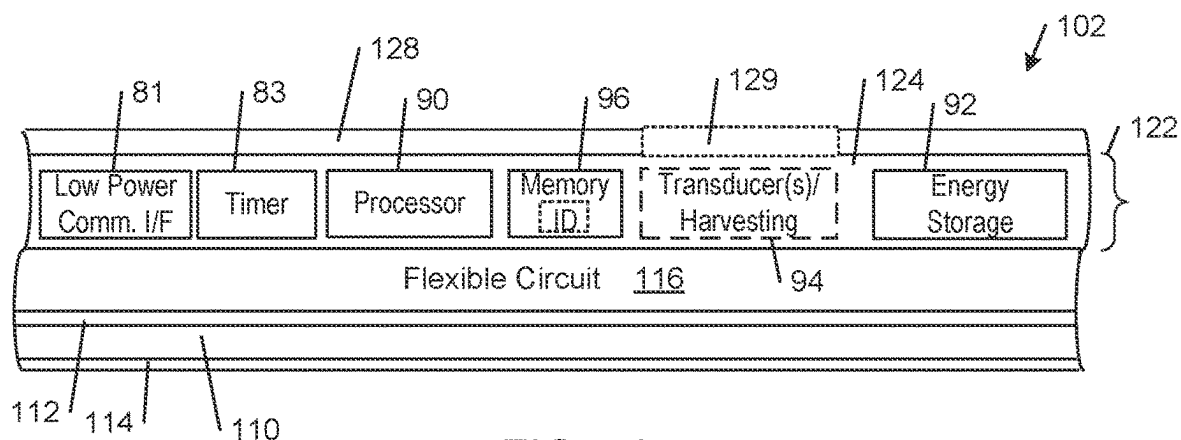


FIG. 5A

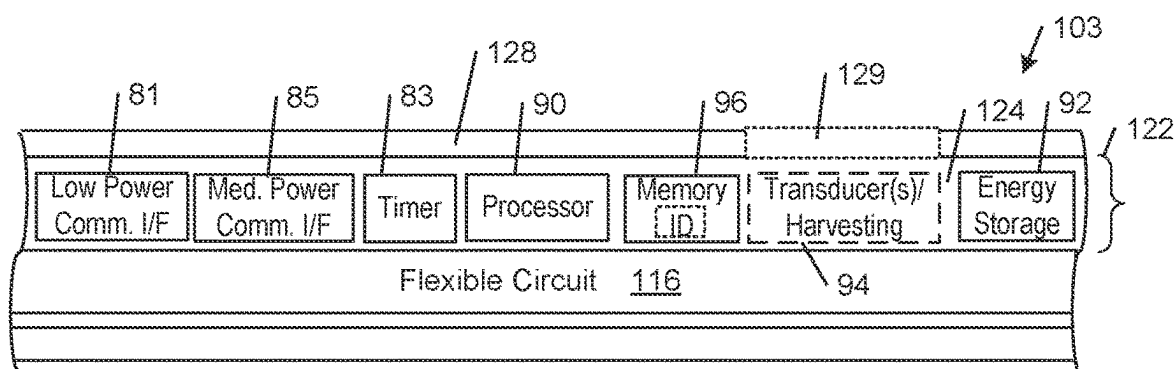


FIG. 5B

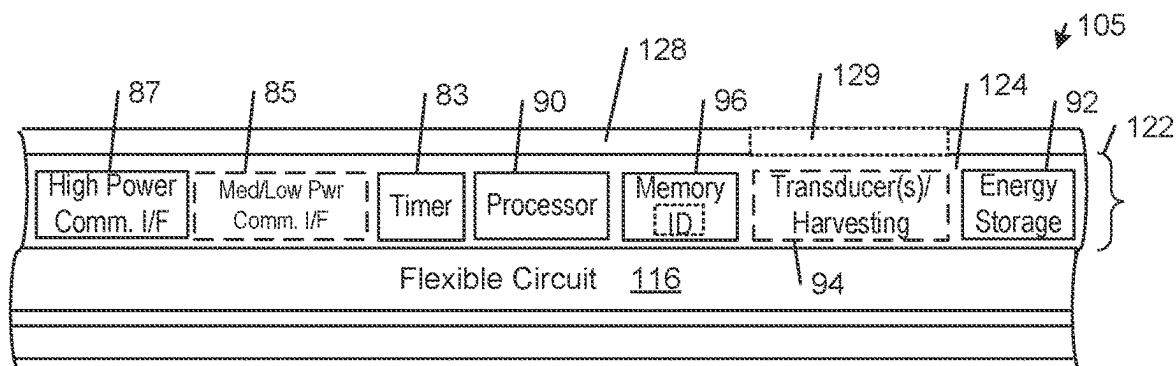


FIG. 5C

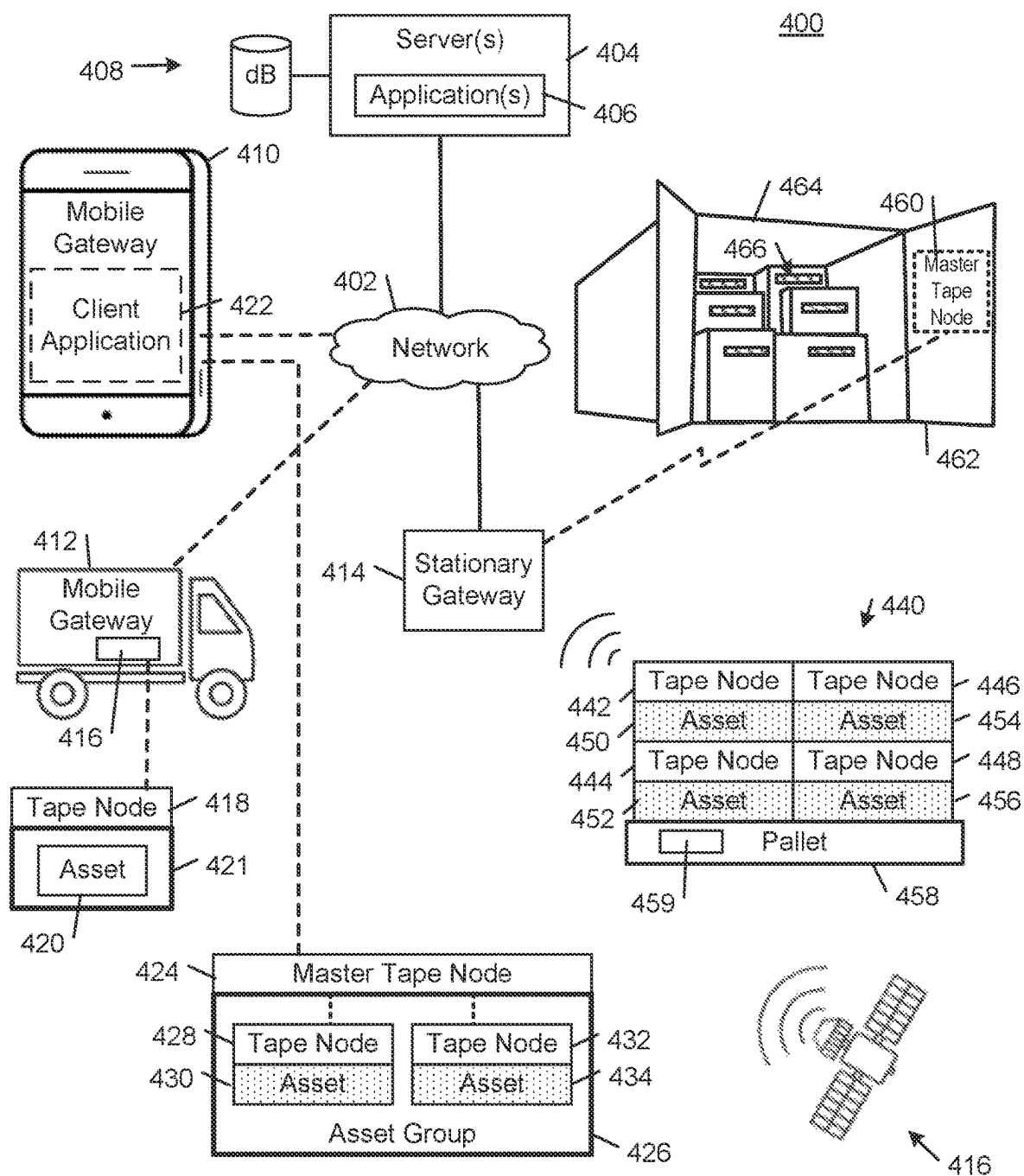


FIG. 6A

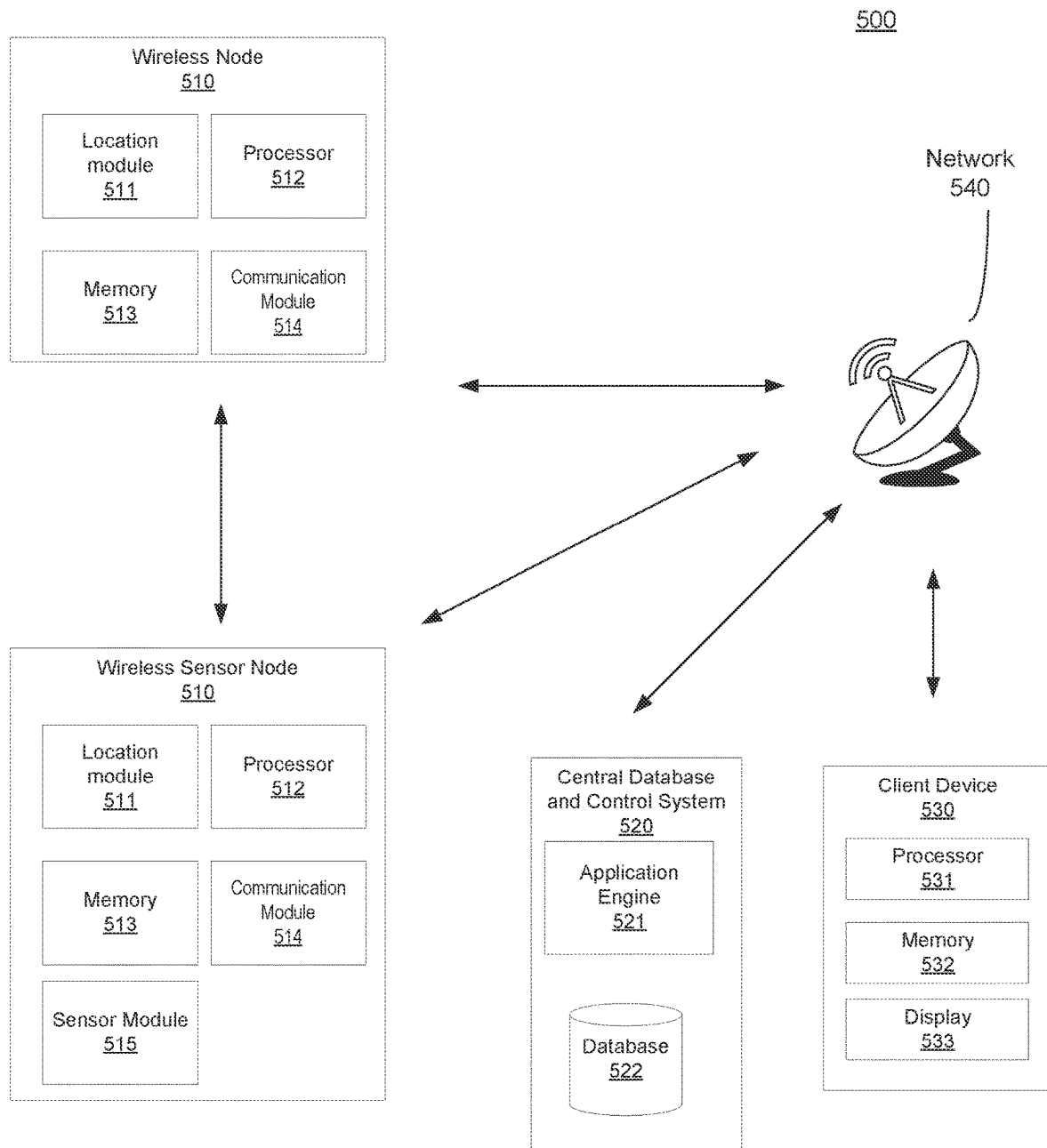


FIG. 6B

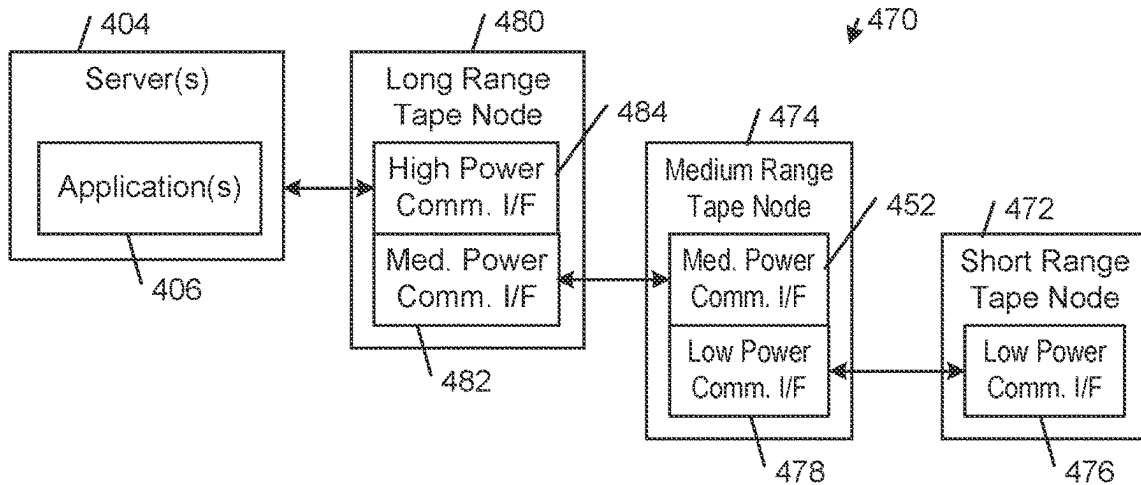


FIG. 7

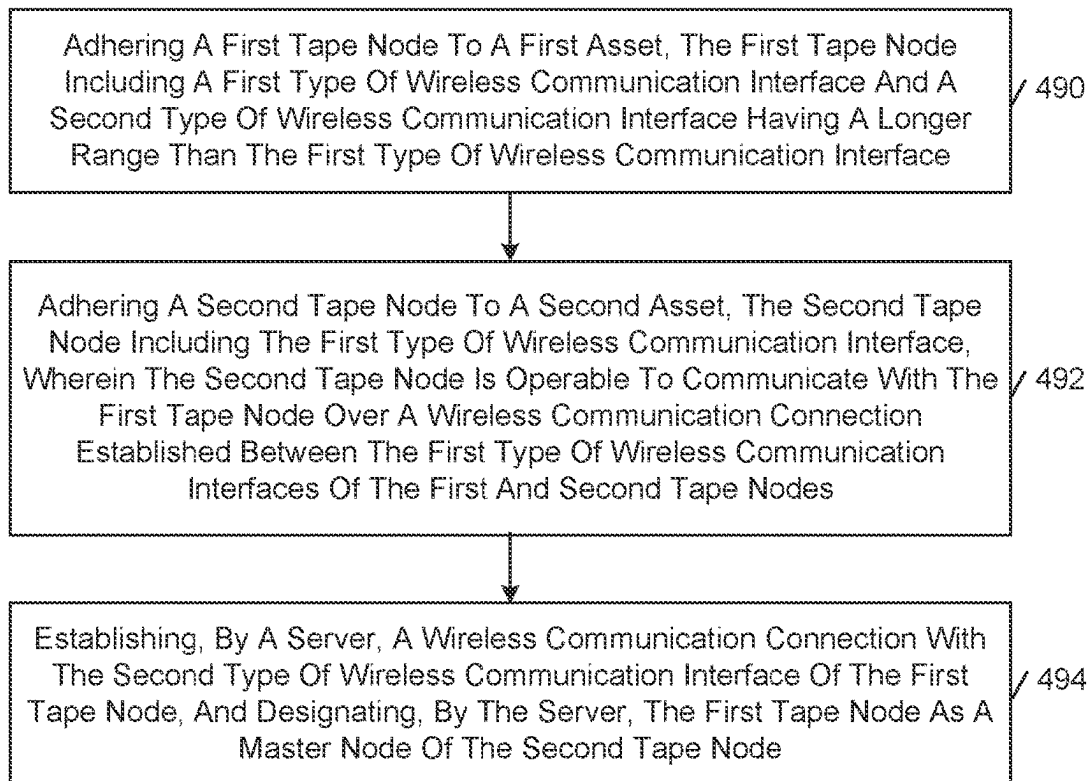


FIG. 8

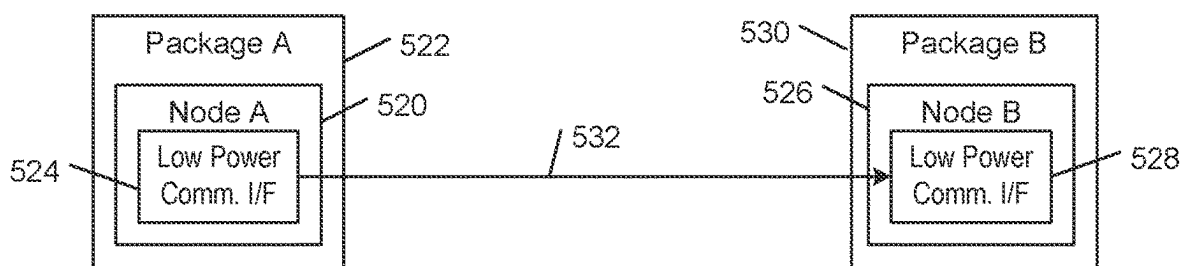


FIG. 9A

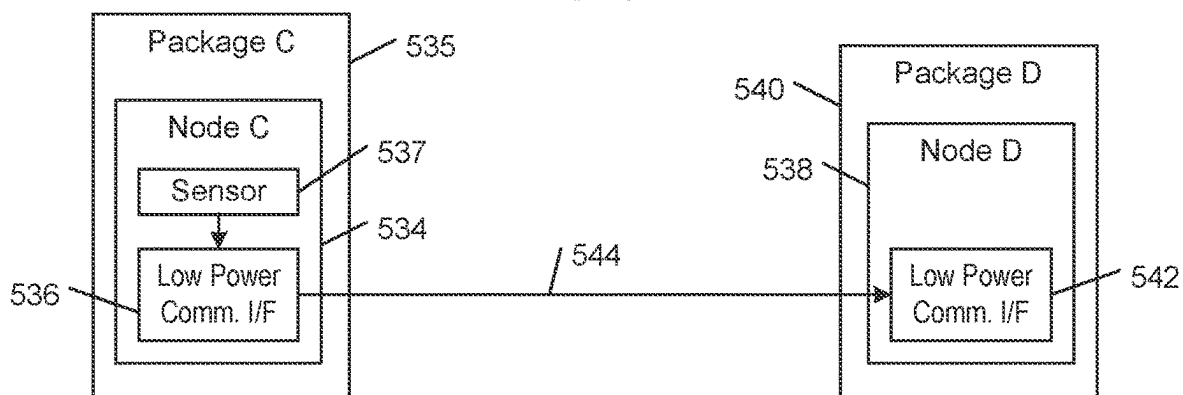


FIG. 9B

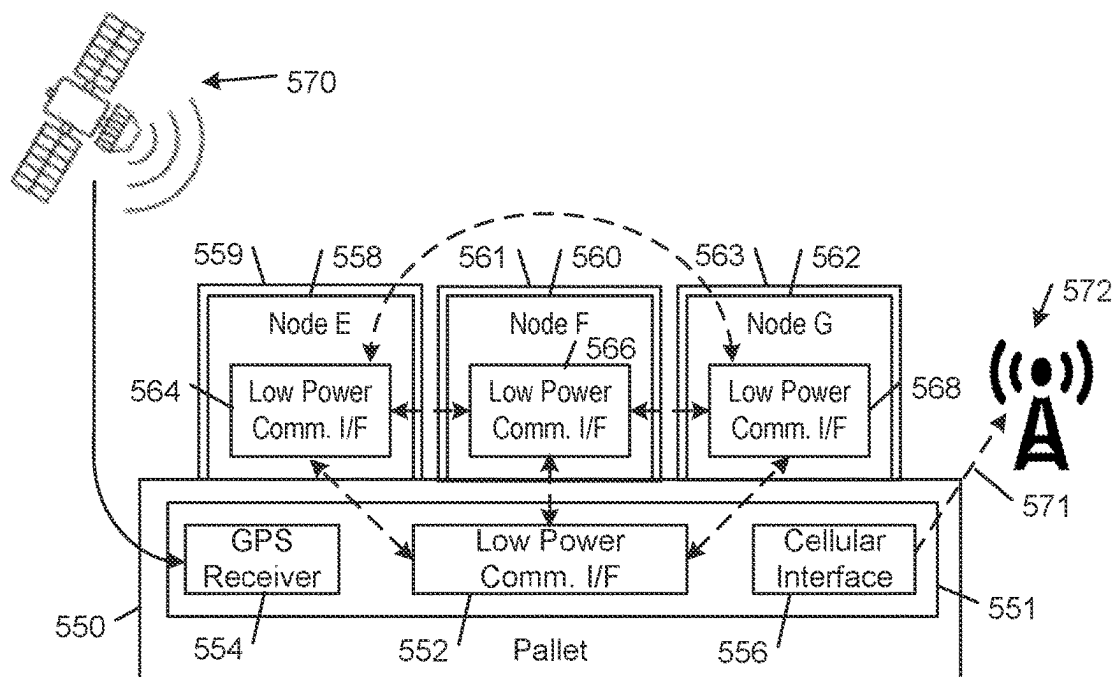


FIG. 9C

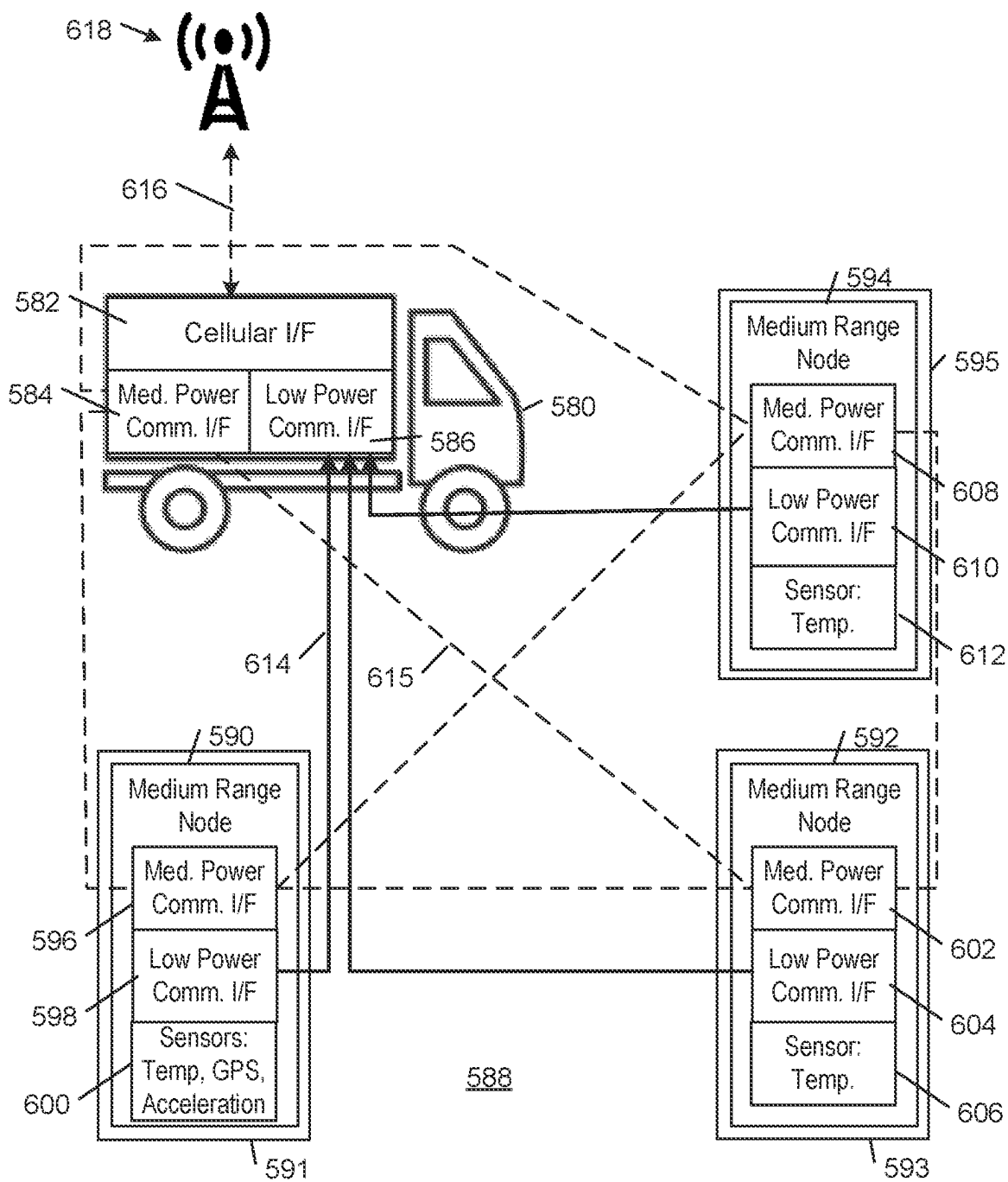


FIG. 9D

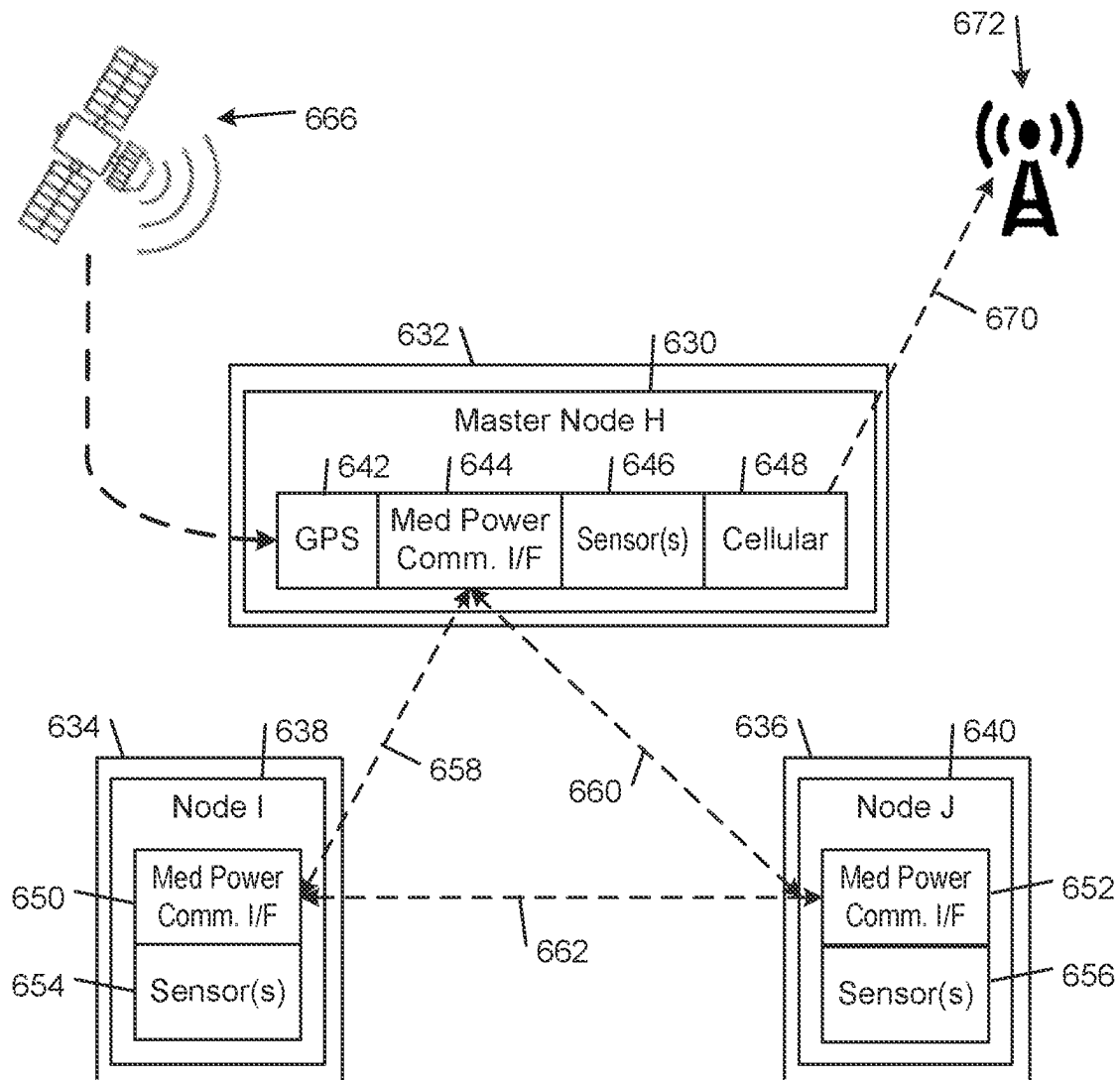


FIG. 9E

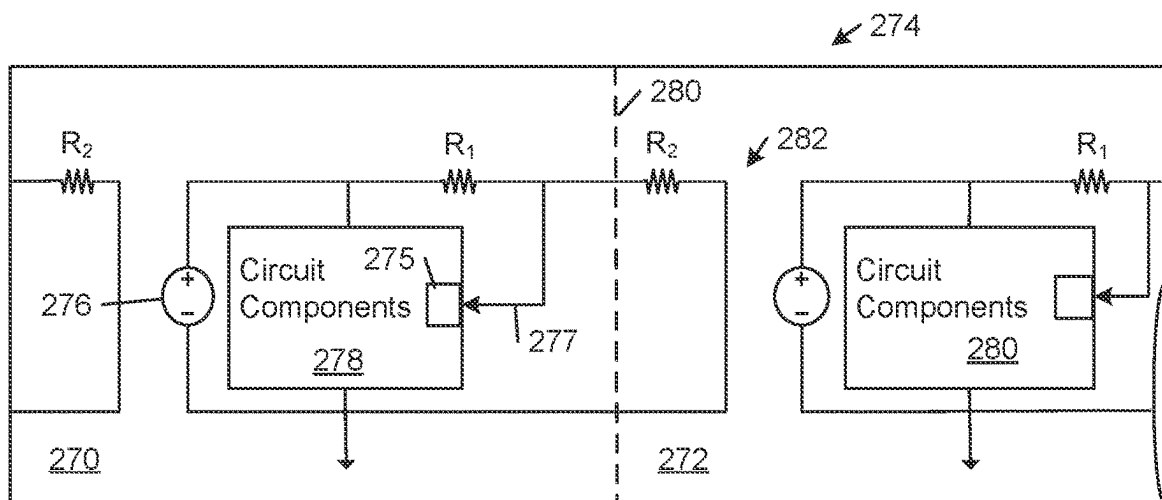


FIG. 10A

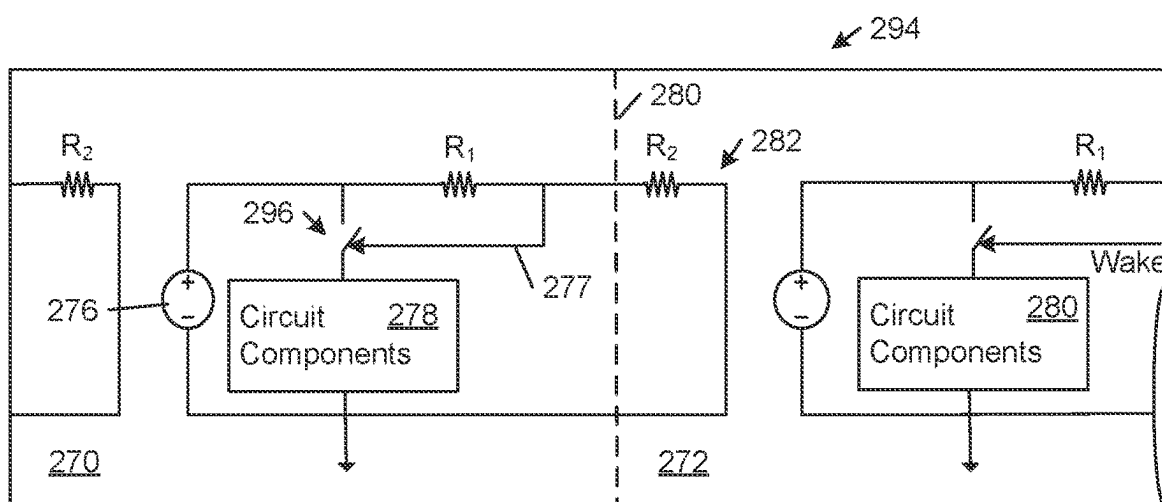


FIG. 10B

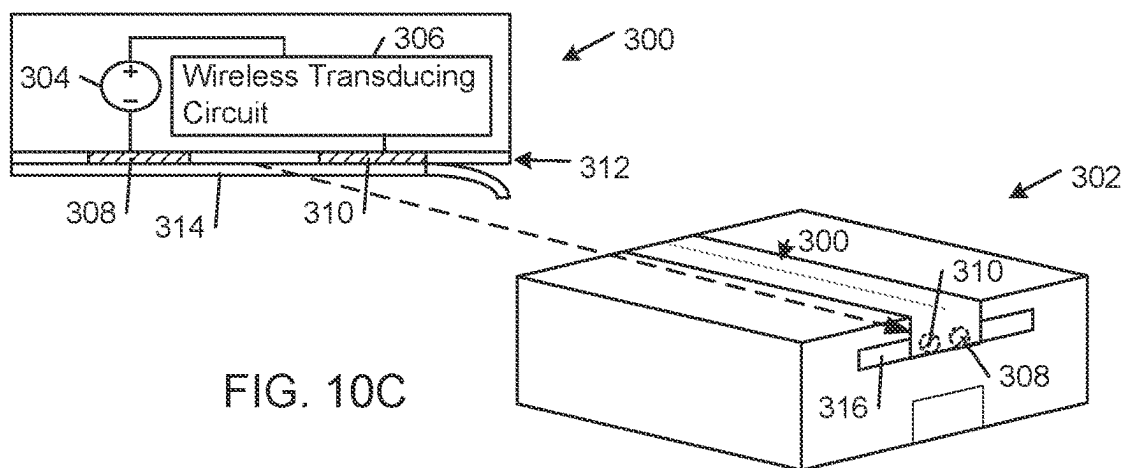
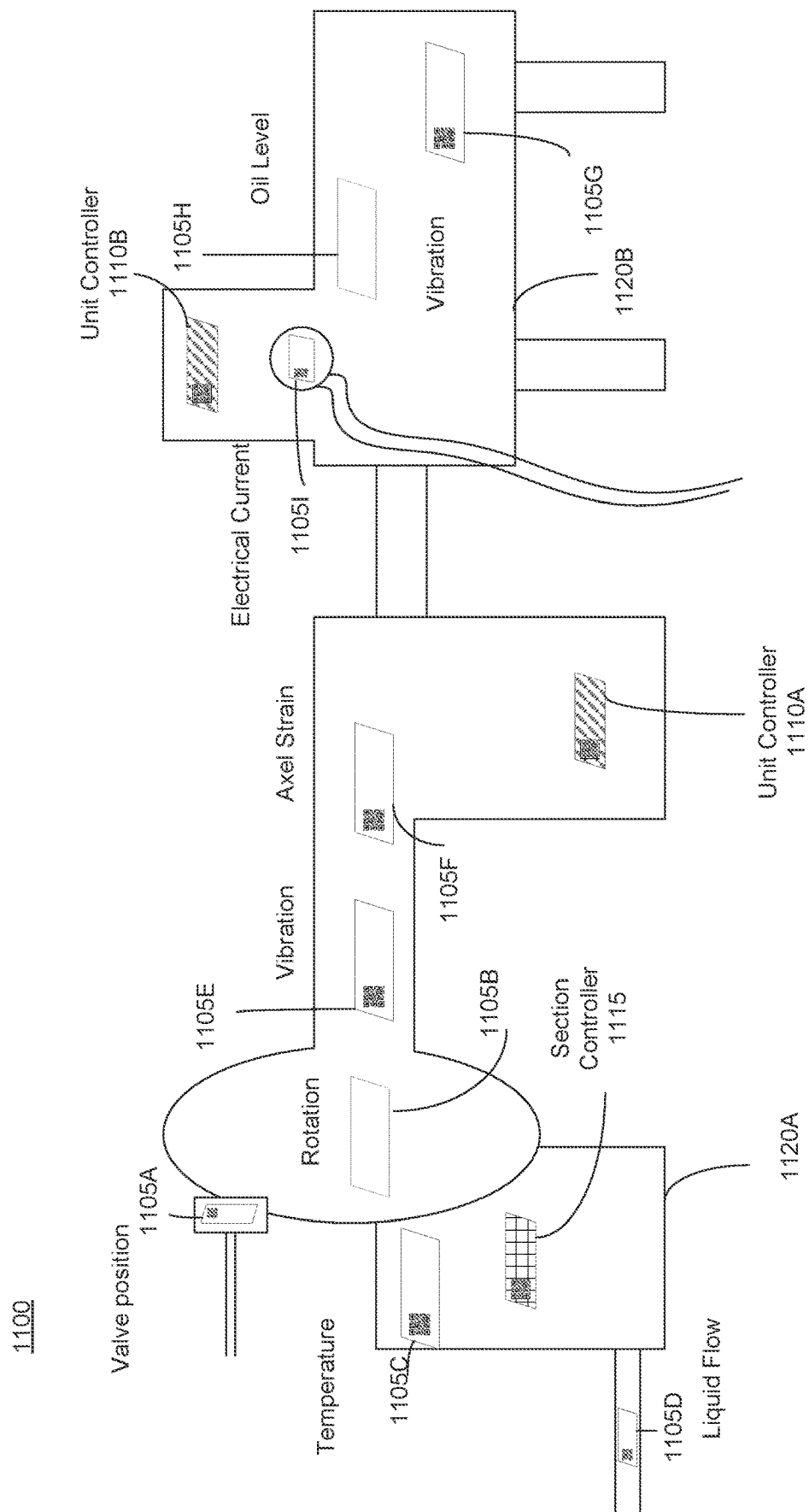


FIG. 10C



11
11
G
E

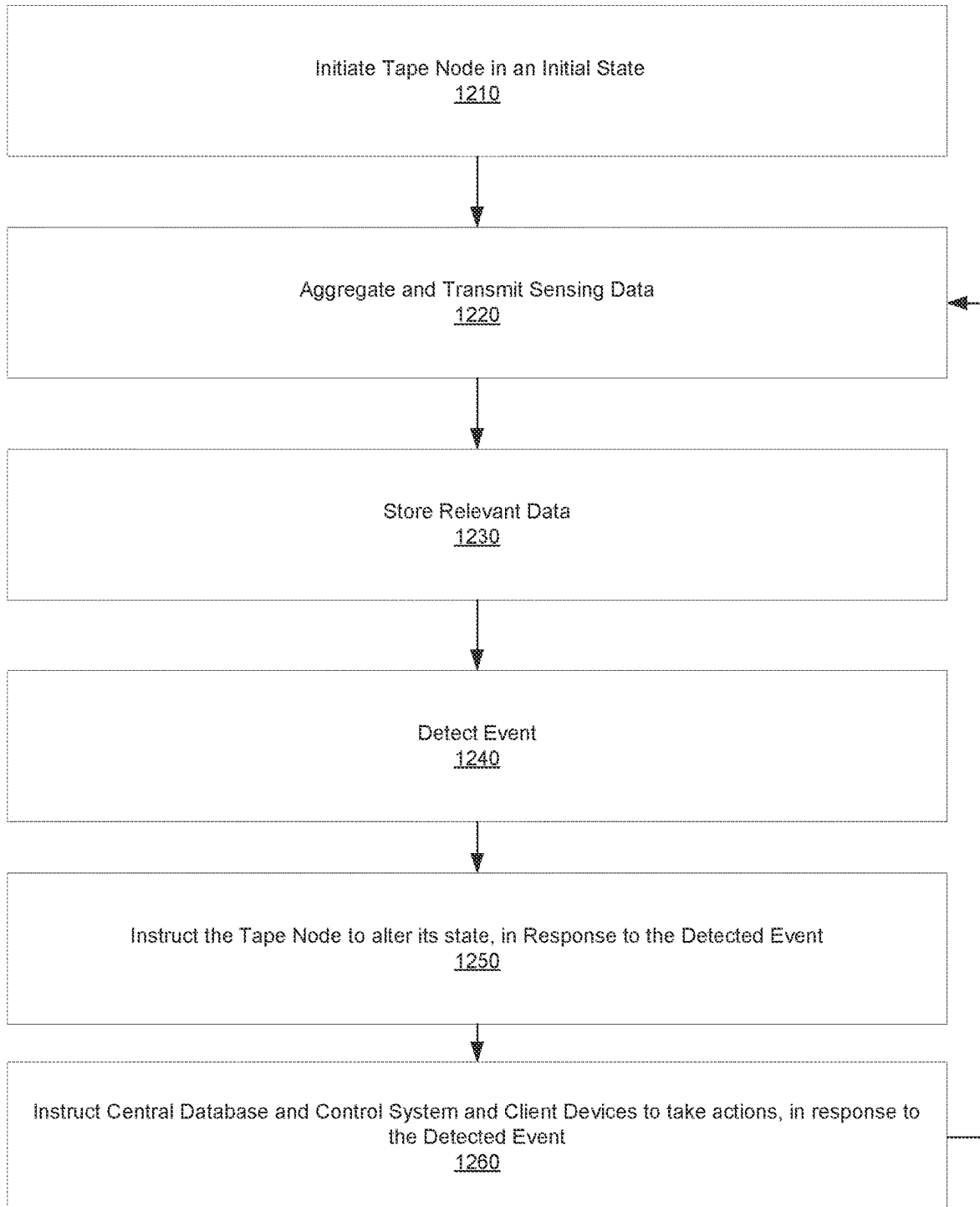
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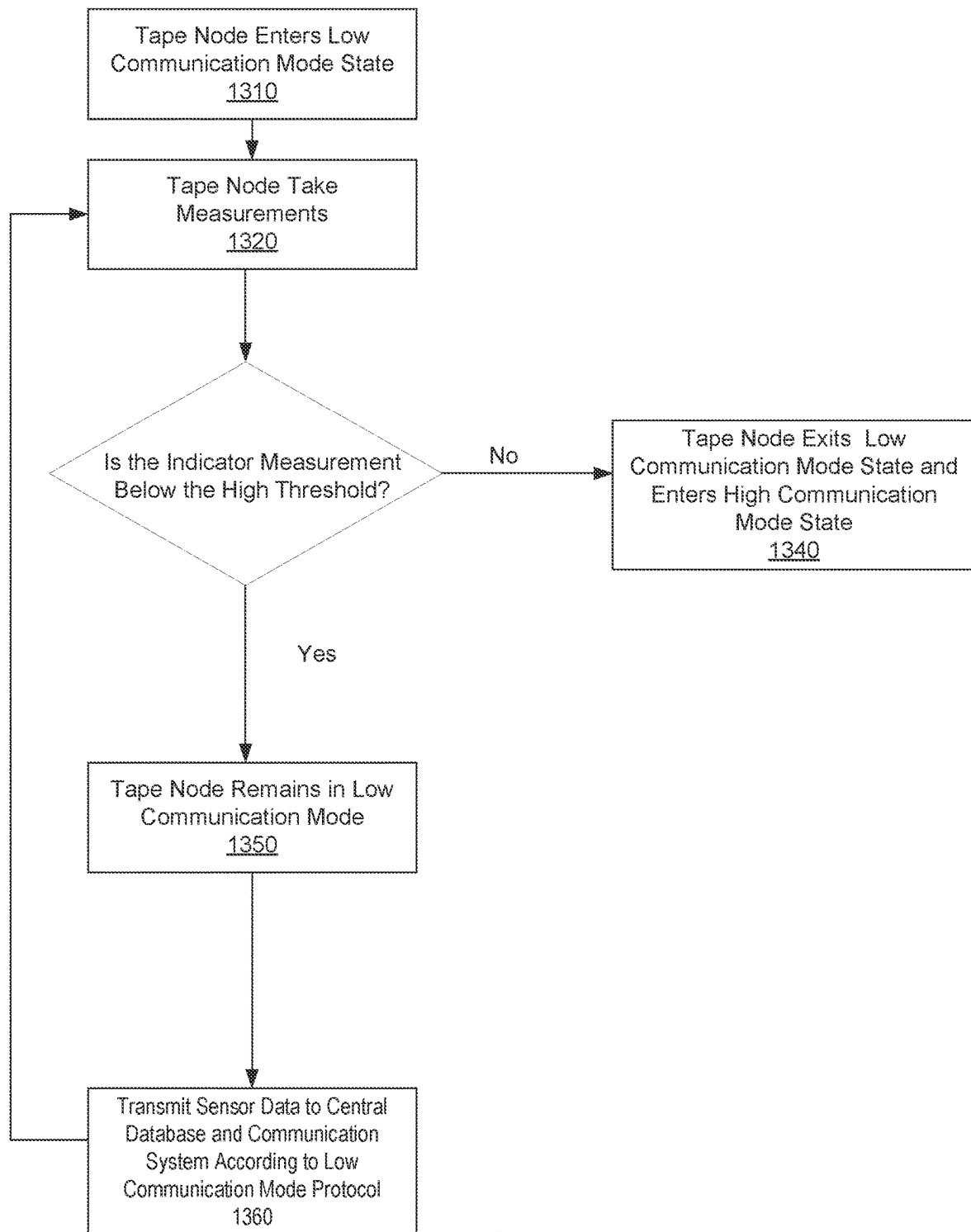


FIG. 13

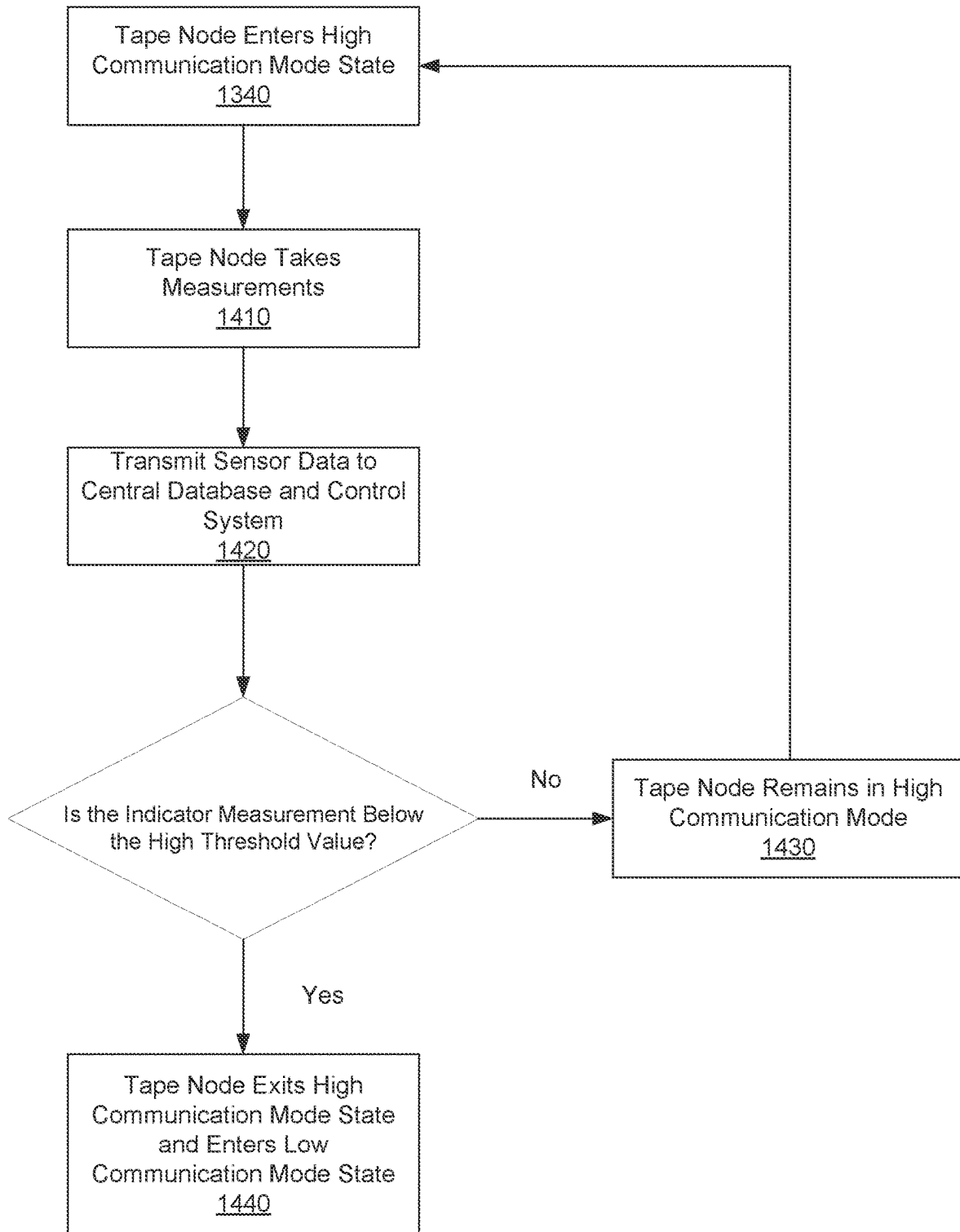


FIG. 14

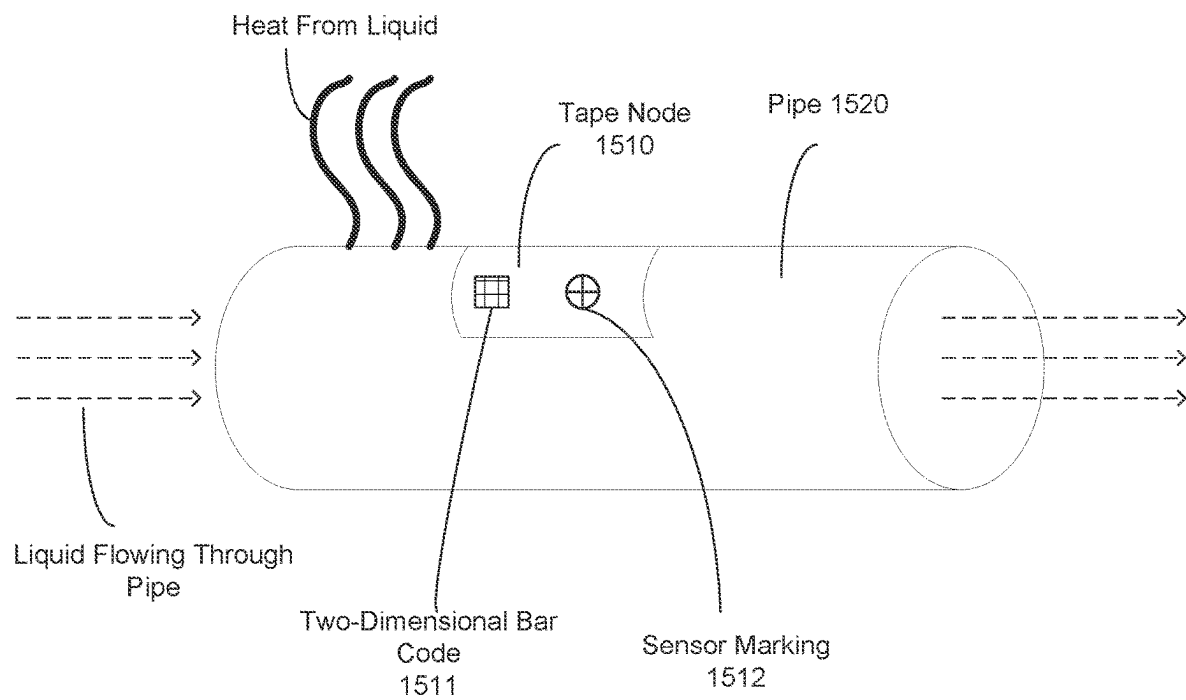


FIG. 15

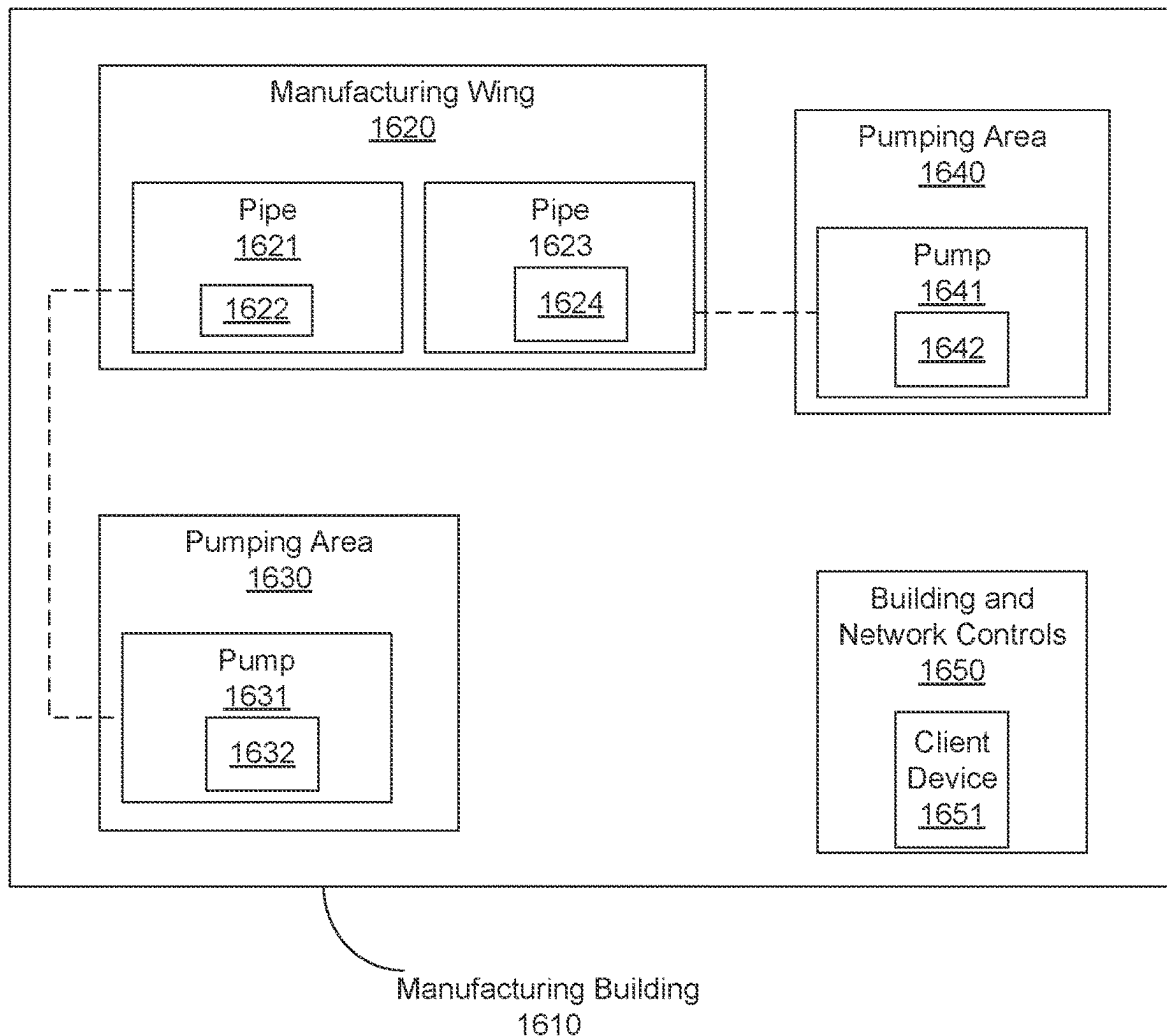
1600

FIG. 16

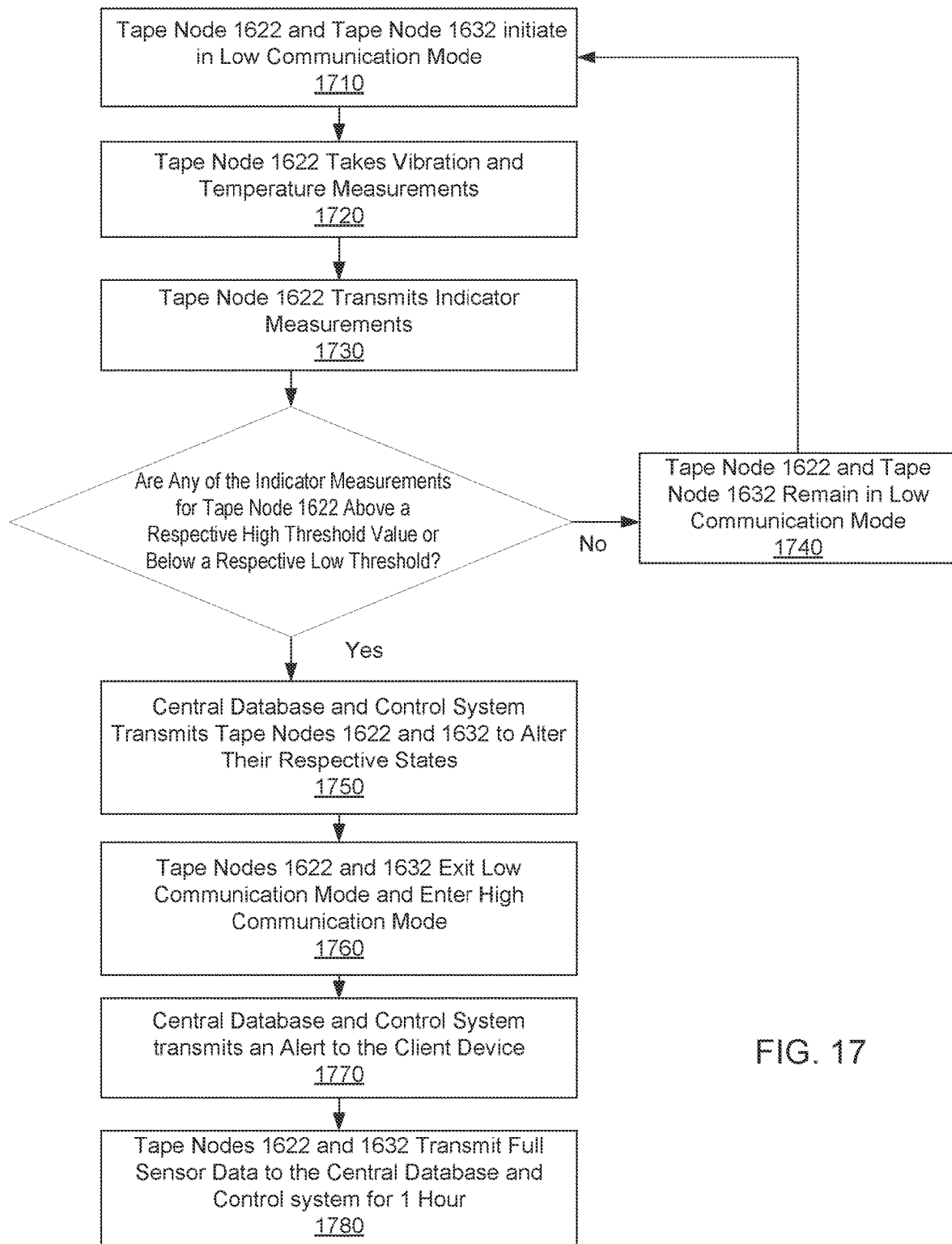


FIG. 17

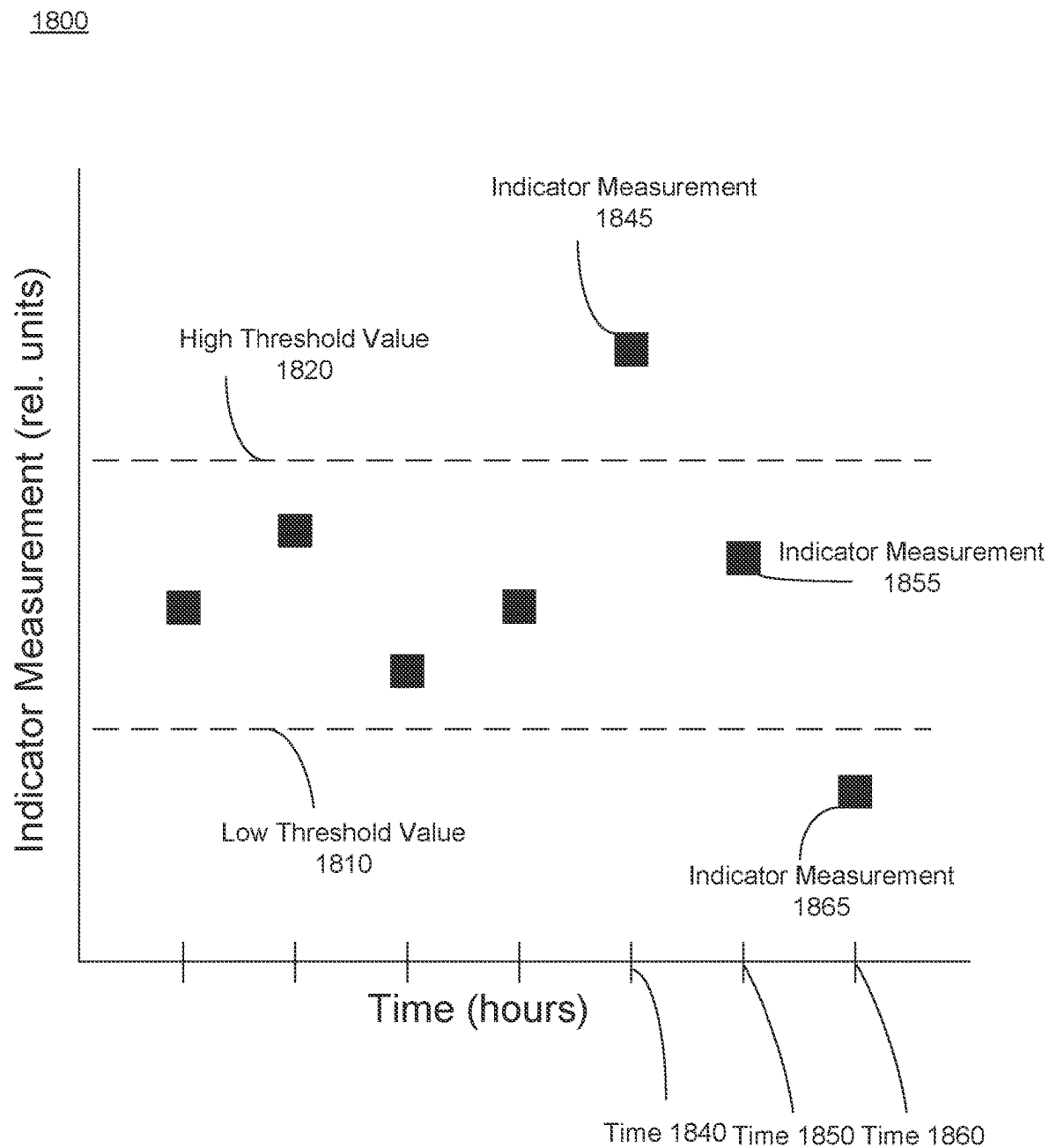


FIG. 18

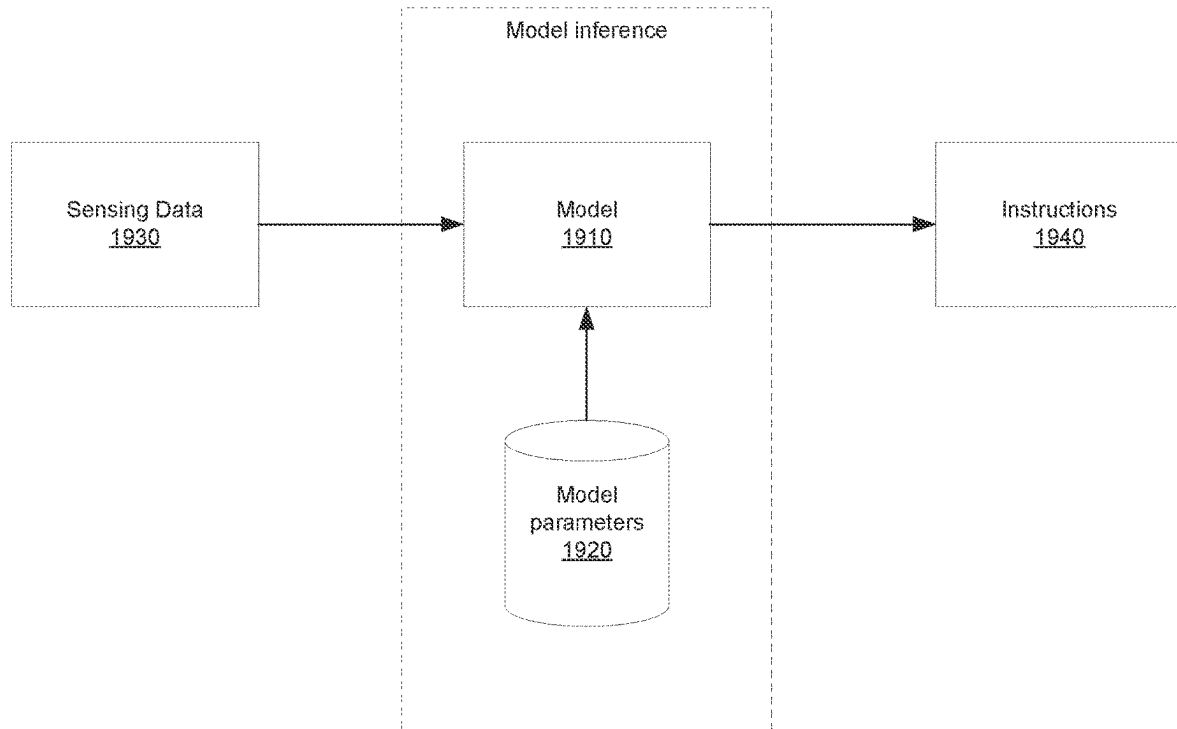


FIG. 19

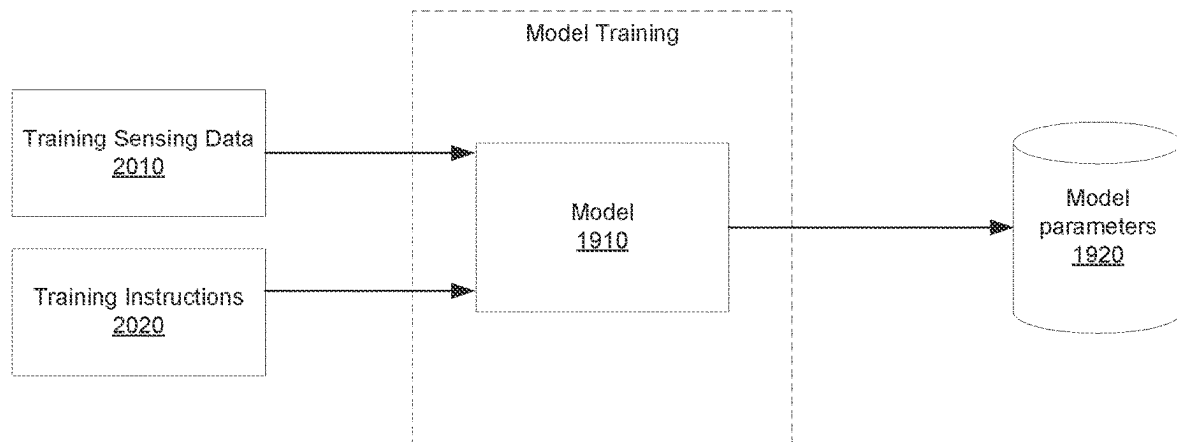
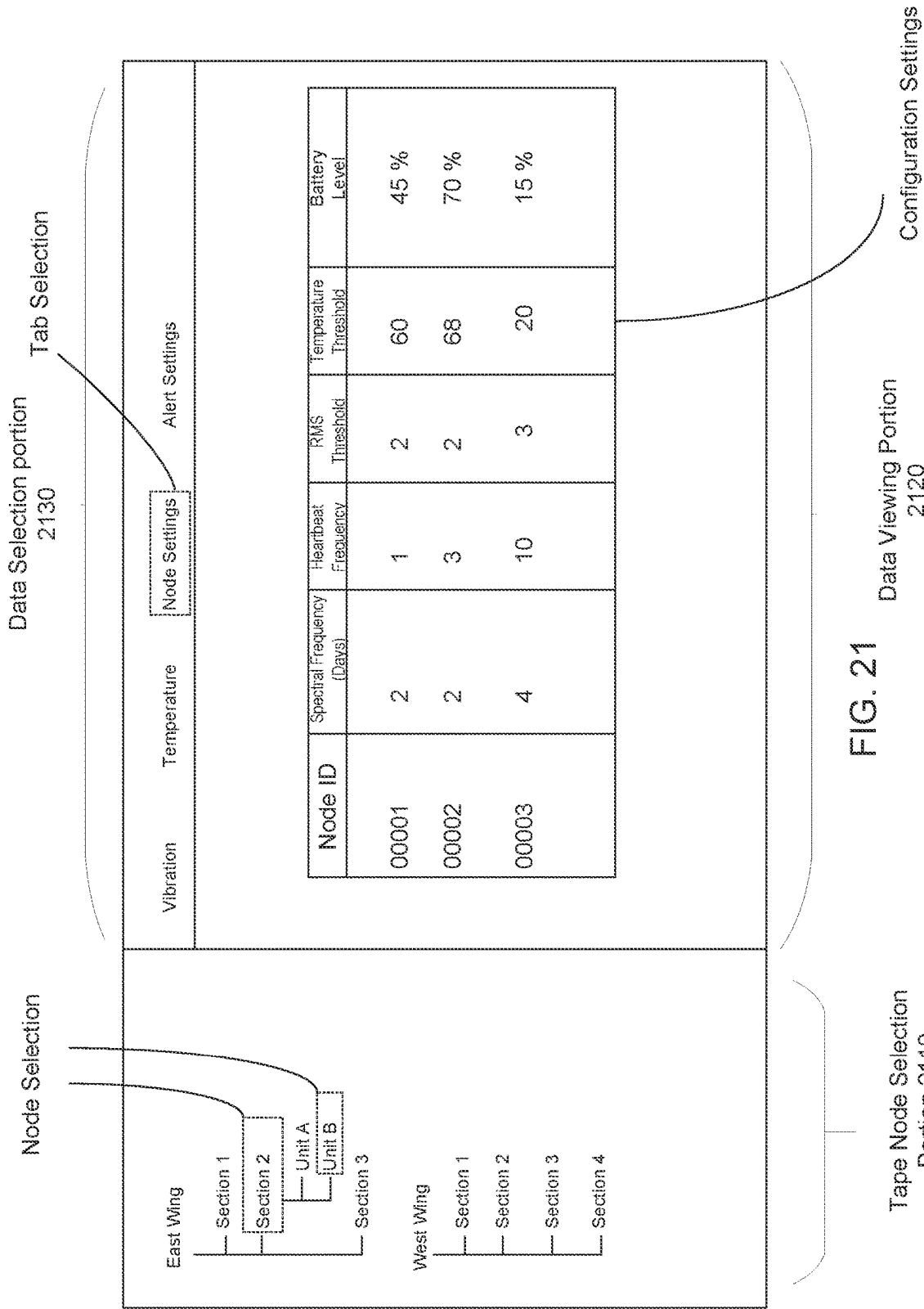


FIG. 20

1500



1500

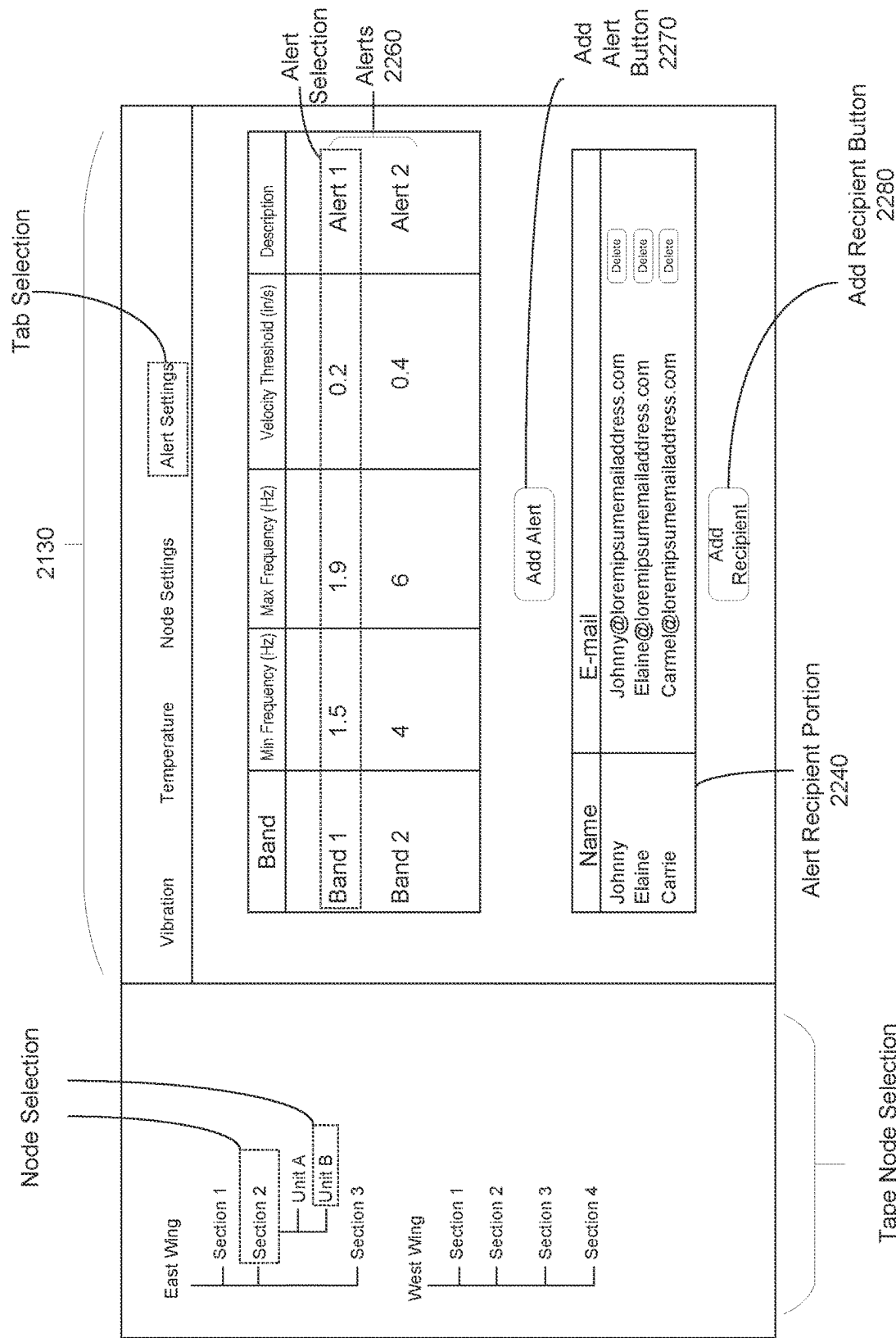
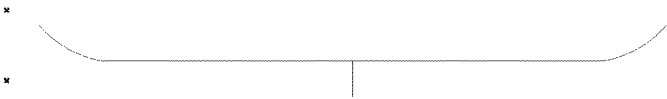


FIG. 22

Time Domain Measurements		Frequency Domain Values		
Acceleration (G)	Time (s)	Acceleration (G)	Frequency (Hz)	Velocity (in/s)
-0.3345	0	0.0033	0.5	0.11
-0.0103	0.001	0.0028	1.0	0.33
0.00001	0.002	0.0100	1.5	0.65
-0.0033	0.003	0.0003	2.0	0.14
-0.2463	0.004	0.0060	2.5	0.18
-0.3013	0.005	0.0078	3.0	0.78
-0.2879	0.006	0.0043	3.5	0.65



Time Domain Acceleration
Data
1810



Frequency Domain Data
1820

FIG. 23

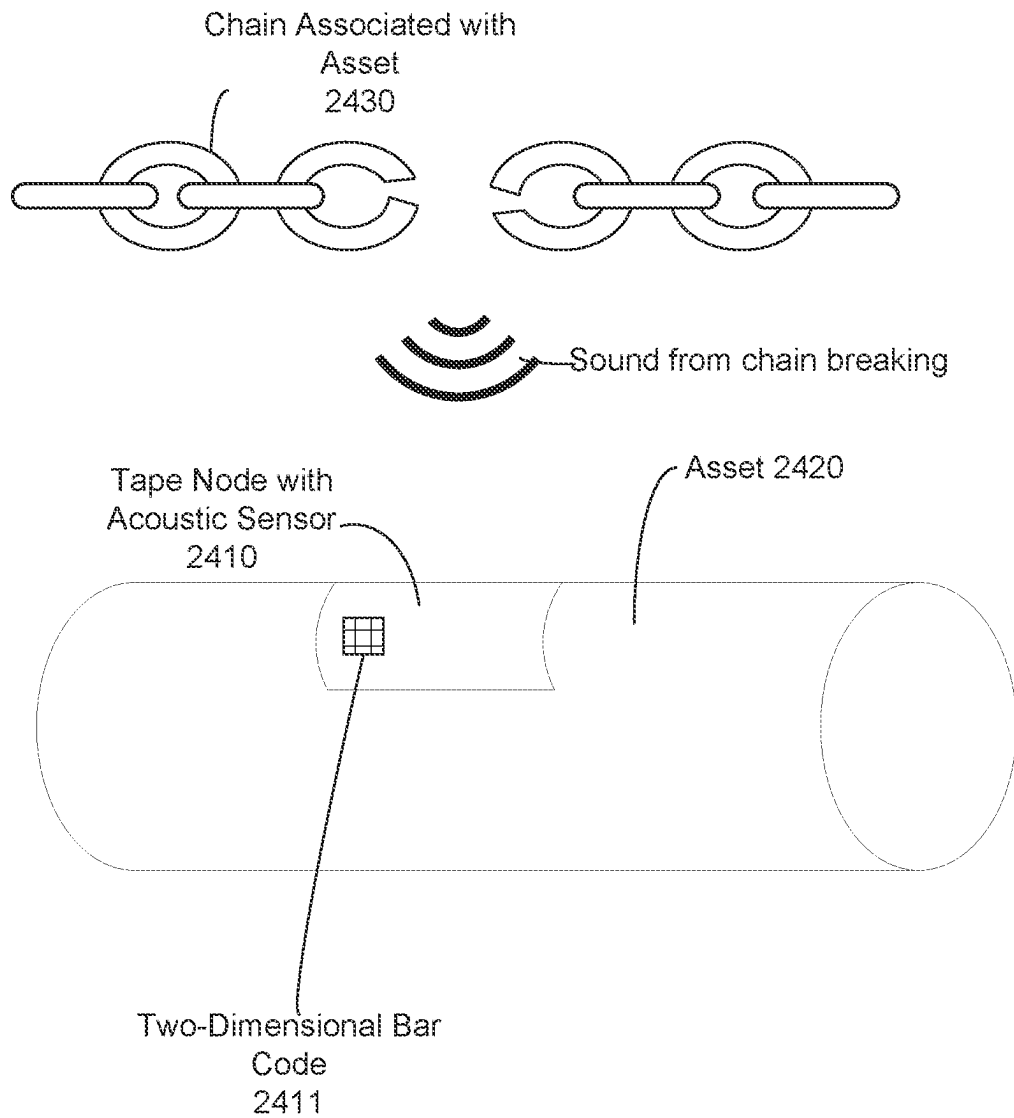


FIG. 24A

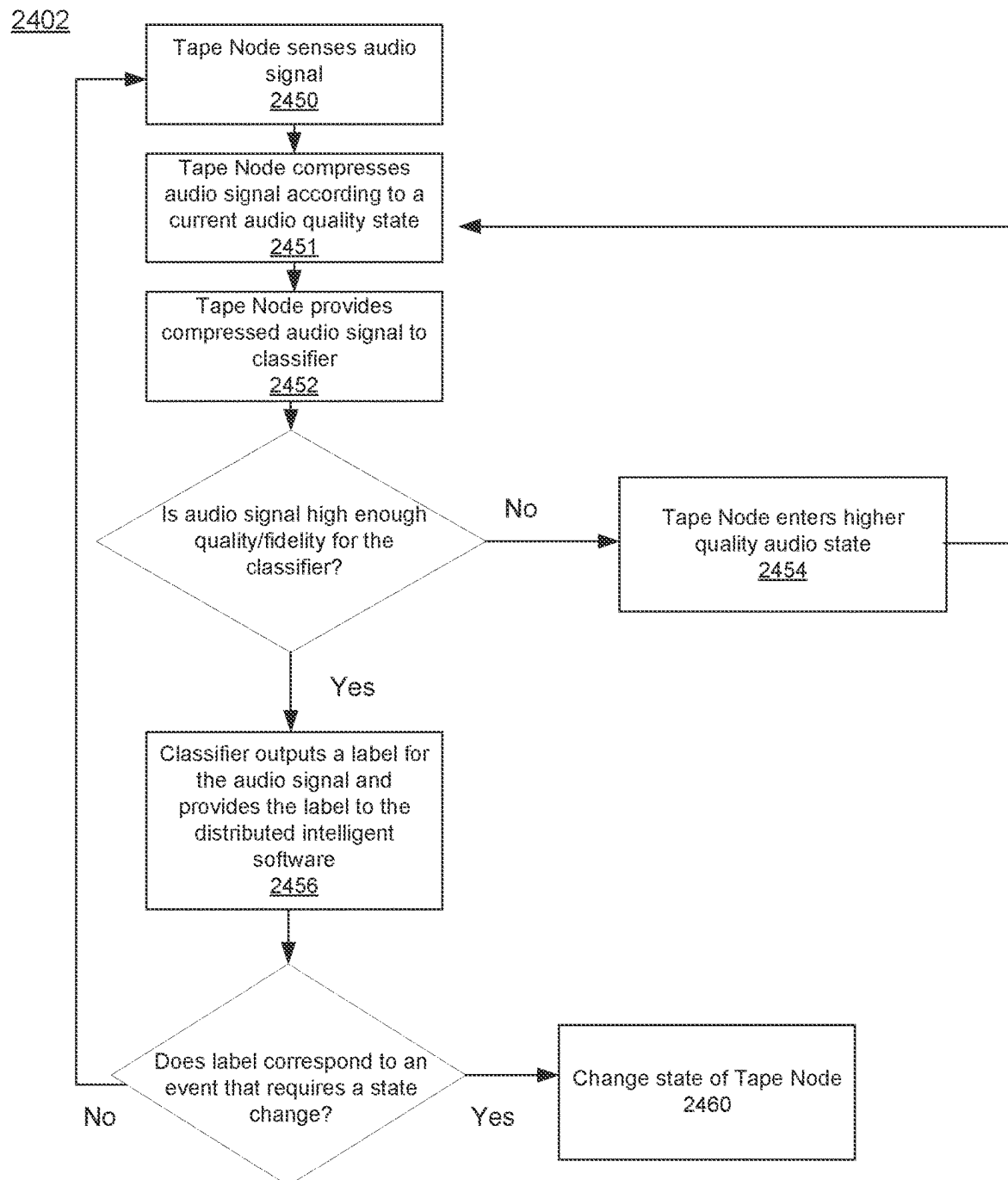


FIG. 24B

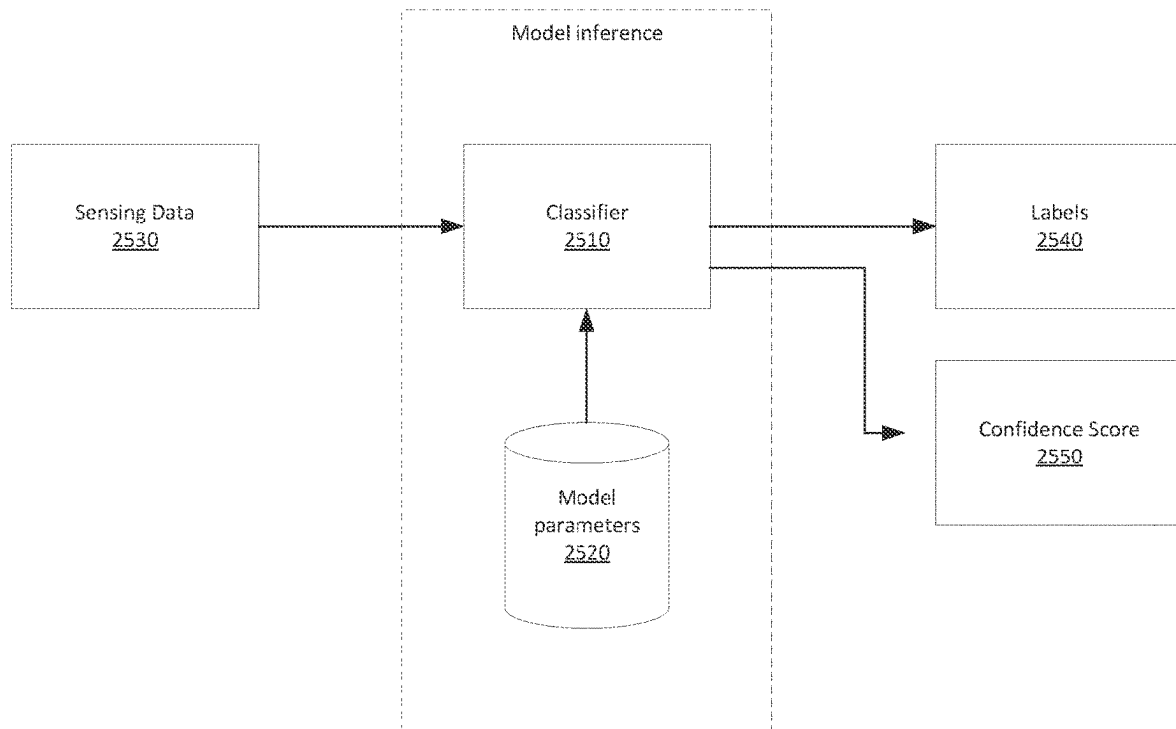


FIG. 25

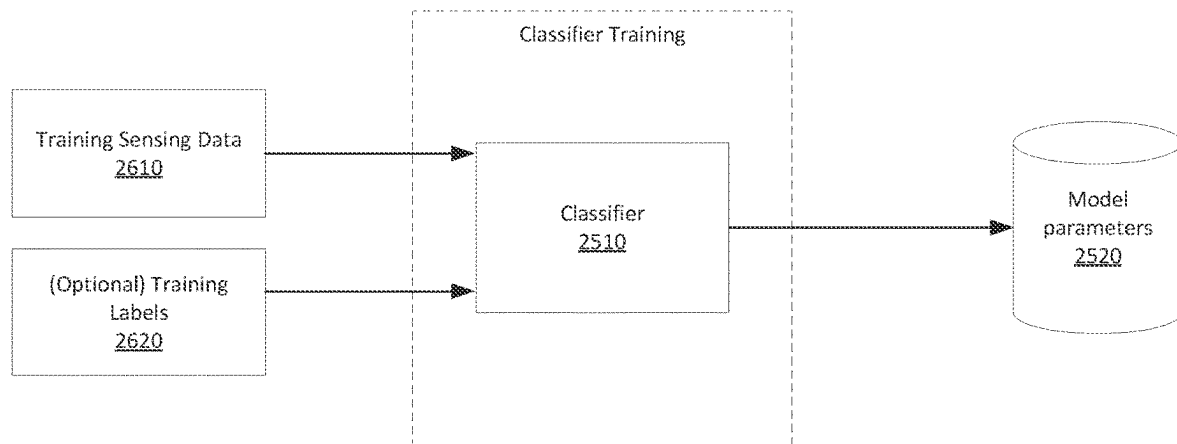


FIG. 26A

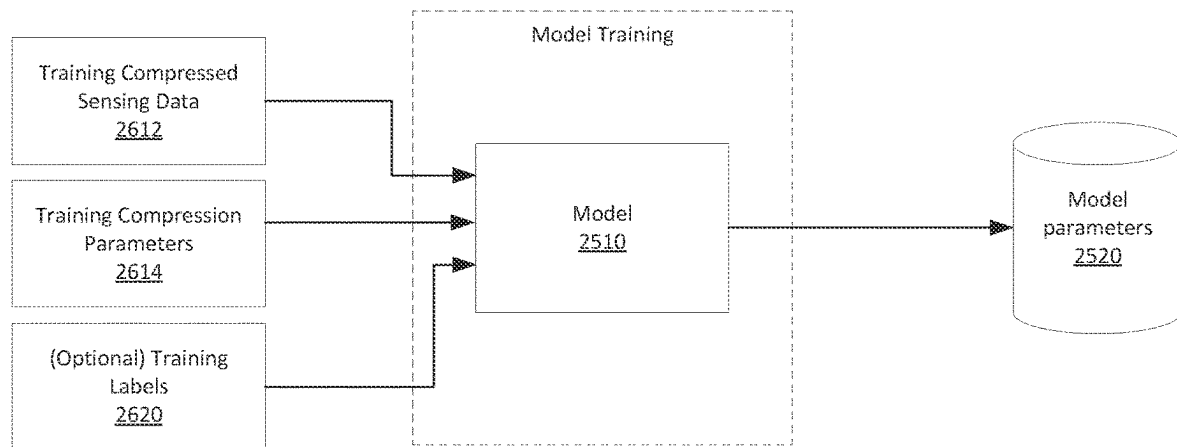


FIG. 26B

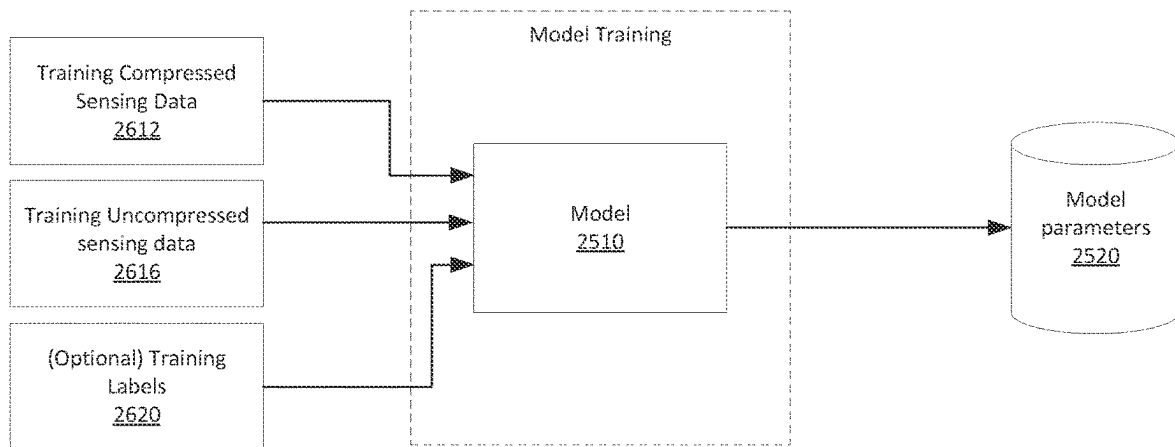


FIG. 26C

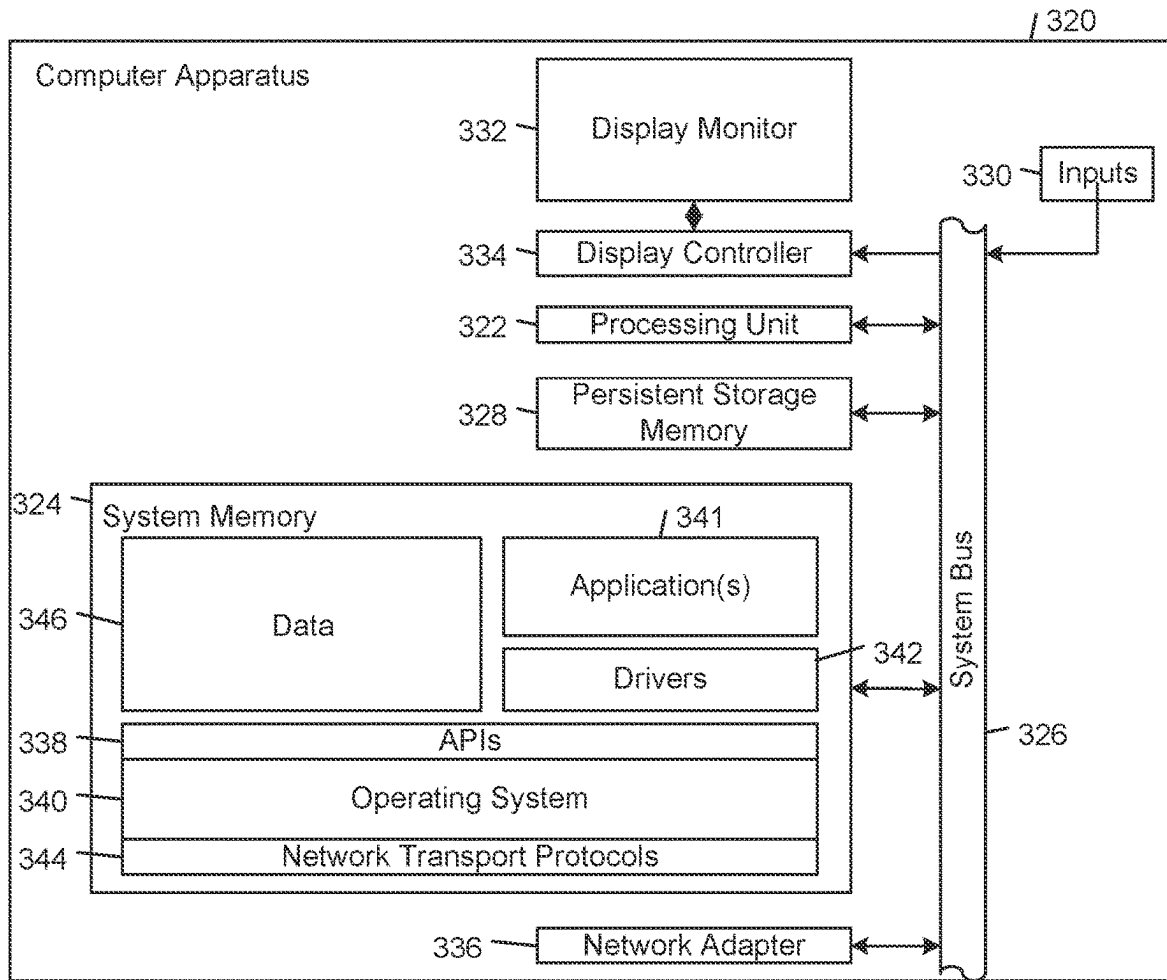


FIG. 27

1

DISTRIBUTED INTELLIGENT SOFTWARE FOR VIBRATION AND ACOUSTIC MONITORING AND SYSTEMS AND METHODS IMPLEMENTING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part of U.S. patent application Ser. No. 17/209,192, filed on Mar. 22, 2021, now U.S. Pat. No. 11,363,427, which claims priority to U.S. Provisional Patent Application No. 62/992,904, filed on Mar. 21, 2020 and to U.S. Provisional Patent Application No. 63/081,887, filed on Sep. 22, 2020, each of which is incorporated herein in its entirety. This application also claims priority to pending U.S. Provisional Patent Application No. 63/087,284, filed on Oct. 4, 2020, which is incorporated herein in its entirety.

FIELD OF THE DISCLOSURE

This disclosure generally relates to sensor devices related to asset management, including tracking, warehousing, inventorying, and monitoring items (e.g., objects, tools, and other equipment).

BACKGROUND

When equipment failure occurs, multiple factors may contribute to the failure or malfunctioning of the equipment. It is conventionally difficult to isolate the cause of failure without continuous monitoring of the various factors. Therefore, it is desirable to be able to monitor a large amount of data on different aspects of equipment or assets in order to determine the condition of the equipment or assets and causes for failure or malfunctioning.

In some applications, an array of sensor devices may be deployed at different physical locations in the field to gather information about various conditions. In some cases, sensor devices at each location communicate with a system that provides sensor data to users and takes further actions based on the sensor data it receives. The sensor devices may continuously communicate the full range of sensor data collected to the central system at all times. However, communicating large volumes of data for extended periods of time or continuously at all times results in high power consumption, particularly when the sensor devices are using wireless communication to send data to the central system. Situations may arise where an object or location being monitored does not require the full functionality of the sensing device or only requires a limited set of functionalities.

SUMMARY

Disclosed herein is a method and sensing system for monitoring equipment using wireless sensor nodes. The wireless sensor nodes are configured to gather sensor data using sensors integrated into the wireless sensor nodes and wirelessly communicate data with other wireless nodes of the sensing system including communication of the gathered sensor data. The wireless sensor nodes are wireless sensor devices that may be attached or coupled to an object of interest for measuring sensor data relevant to the object of interest. In some embodiments, a wireless sensor node is an embodiment of an adhesive tape platform, examples of which are discussed below with respect to FIGS. 1-5C and

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10A-10C. Wireless sensor nodes attached to various components of equipment gather sensor data on the components of equipment. The sensor data may be transmitted and relayed through wireless nodes of the sensing system to a control system stored on the cloud. The sensing system is configured to coordinate communication of sensor data in a way that conserves battery life of wireless sensor nodes without sacrificing the ability to detect events based on the sensor data.

10 According to some embodiments, a distributed intelligent software supports communication to, from, and/or between one or more sensor devices deployed in the field. The term distributed intelligent software herein includes hardware used to implement the logic and collect data associated with the distributed intelligent software. A system for collecting sensor information includes an array of sensor devices, each sensor device located at one of a plurality of locations or coupled to one of a plurality of objects. The sensor devices collect measurements on a relevant property or signal. For example, a sensor device in the array of sensor devices may collect data on temperature measurements, vibration measurements, or some combination thereof. Each sensor device wirelessly communicates with another device, such as another sensor device, and/or a central database and control system. A sensor device may operate in an initial state. The initial state may be a low communication mode when an indicator measurement is within a moderate range, according to some embodiments. In the low communication mode, the sensor device limits the communication with other devices, such as another sensor device, and/or a central database and control system and only transmits a low amount of data periodically. For example, the sensor device may only transmit a root-mean-square (RMS) value periodically, rather than sending a full spectrum of data that includes measurements over multiple frequencies and times. If an event is detected based on the collected data, the state of the sensor device is altered, according to the distributed intelligent software. For example, if an indicator measurement is outside of the moderate range (above a high threshold or below a low threshold), the distributed intelligent software may include instructions located on the sensor device that, when executed in response to the indicator measurement being outside of the moderate range, transitions from the low communication mode to the high communication mode, according to some embodiments. In the high communication mode, the distributed intelligent software may cause the sensor device to transmit a larger volume of data to other devices and/or to the central database and control system than in the low communication mode (e.g., transmitting a full range of data included, such as a full frequency spectrum), according to some embodiments. The sensor device may also transmit data more frequently in the high communication mode than in the low communication mode. The distributed intelligent software that generates the instruction which causes the sensor to transition from low to high communication mode may be located at the sensor device, or at another device, such as another sensor device, a gateway device, and/or the central database and control system.

60 In further embodiments, the sensor device may take additional actions in the high communication mode. For example, the sensor device may send alerts to the central database and control system, send alerts to other sensor devices in the sensor device array, increase the sampling rate of measurements of the sensor device and/or another sensor device, or some combination thereof. In response to the sensor device entering the high communication mode, the

central database and control system and one or more client devices may also take additional actions.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a diagrammatic view of an asset that has been sealed for shipment using a segment of an example adhesive tape platform dispensed from a roll.

FIG. 1B is a diagrammatic top view of a portion of the segment of the example adhesive tape platform shown in FIG. 1A.

FIG. 2 is a diagrammatic view of an example of an envelope carrying a segment of an example adhesive tape platform dispensed from a backing sheet.

FIG. 3 is a schematic view of an example segment of an adhesive tape platform.

FIG. 4 is a diagrammatic top view of a length of an example adhesive tape platform.

FIGS. 5A-5C show diagrammatic cross-sectional side views of portions of different respective adhesive tape platforms.

FIG. 6A is a diagrammatic view of an example of a network environment supporting communications with segments of an adhesive tape platform.

FIG. 6B is a diagram showing an example system environment for the sensing system including the adhesive tape platform, according to some embodiments.

FIG. 7 is a diagrammatic view of a hierarchical communications network.

FIG. 8 is a flow diagram of a method of creating a hierarchical communications network.

FIGS. 9A-9E are diagrammatic views of exemplary use cases for a distributed agent operating system.

FIGS. 10A-10C are diagrammatic top views of a length of an example tracking adhesive product.

FIG. 11 shows an example environment for a fleet of wireless sensor nodes and wireless nodes of the sensing system for generating sensing data on a section of equipment and communicating the sensing data with the sensing system, according to some embodiments.

FIG. 12 is a flow chart depicting steps implemented, via execution by one or more processors of the adhesive tape platform, central database and control system, client device, or any combination thereof, by the distributed intelligent software, according to some embodiments.

FIG. 13 is a flowchart showing example steps executed according to the distributed intelligent software, according to some embodiments.

FIG. 14 is a flowchart showing example steps after the tape node enters the high communication mode, according to the distributed intelligent software, according to some embodiments.

FIG. 15 shows an example of a tape node being used for vibration and temperature sensing.

according to some embodiments.

FIG. 16 is a block diagram illustrating an example where multiple tape nodes are used to monitor various items at different locations in a manufacturing building and collect sensor data on the items in a manufacturing building, according to some embodiments.

FIG. 17 is a flow chart showing example steps executed according to the distributed intelligent software by components of the sensing system deployed in the manufacturing building shown in FIG. 16, according to some embodiments.

FIG. 18 shows an example plot of an indicator measurement collected at various times by the tape node in the example discussed with respect to FIGS. 16 and 17, according to some embodiments.

FIG. 19 is a block diagram illustrating an example of a trained machine learning model being used to generate instructions for the distributed intelligent software, according to some embodiments.

FIG. 20 is a block diagram illustrating an example of training the machine learning model shown in FIG. 19, according to some embodiments.

FIG. 21 is an example user interface for displaying information and sensor data received from a tape node, according to some embodiments.

FIG. 22 is an example user interface for setting parameters relevant to the distributed intelligent software, according to some embodiments.

FIG. 23 is a chart showing example sensor data received by the central database and control system from a tape node and downloaded to a client device associated with the interface, according to some embodiments.

FIGS. 24A-24B are diagrams for an example of a tape node with an acoustic sensor collecting audio data relevant to an asset that the tape node is monitoring, according to some embodiments.

FIG. 25 is a diagram showing an example of using a classifier to detect events based on sensing data collected by a tape node, according to some embodiments.

FIGS. 26A-26C are block diagrams illustrating an example of a classifier, according to some embodiments.

FIG. 27 shows an example embodiment of a computer apparatus 320, according to some embodiments.

DETAILED DESCRIPTION

Disclosed herein is a method and sensing system for monitoring equipment using wireless sensor nodes. The wireless sensor nodes are configured to gather sensor data using sensors integrated into the wireless sensor nodes and wirelessly communicate data with other wireless nodes of the sensing system including communication of the gathered sensor data. The wireless sensor nodes are wireless sensor devices that may be attached or coupled to an object of interest for measuring sensor data relevant to the object of interest. In some embodiments, a wireless sensor node is an embodiment of an adhesive tape platform, examples of which are discussed below with respect to FIGS. 1-5C and 10A-10C. Wireless sensor nodes attached to various components of equipment gather sensor data on the components of equipment. The sensor data may be transmitted and relayed through wireless nodes of the sensing system to a control system stored on the cloud. The sensing system is configured to coordinate communication of sensor data in a way that conserves battery life of wireless sensor nodes without sacrificing the ability to detect events based on the sensor data.

Also disclosed herein is distributed intelligent software for managing a real-time sensing system including one or more of the following members: one or more sensor devices, a central database and control system, and one or more client devices. In some embodiments of the present disclosure, a sensor device (also referred to herein as an “adhesive tape platform”) has a form factor of a flexible adhesive tape. The adhesive tape platform includes a flexible substrate with an adhesive on an outer surface of the flexible substrate, a flexible cover layer, a device layer between the flexible substrate and the flexible cover layer, and (optionally) a

flexible battery. The adhesive tape platform has a dual functionality as both a sensor device for measuring properties and/or signals relevant to an object of interest and an adhesive tape that can be adhered to the objects of interest or used to seal or close items (e.g., a box).

The distributed intelligent software defines one or more rules, algorithms, protocols, logic, and other methods that analyze sensing data and instructs the one or more sensor devices, the central database and control system, and the one or more client devices based on the sensing data. The distributed intelligent software controls the one or more sensing devices, the central database and control system, and/or the one or more client devices to determine that specific events have occurred based on the sensing data collected by the one or more sensor devices and communications between members of the sensing system. Using the distributed intelligent software, the sensing system is able to automatically alter the operation of the sensing system depending on the various conditions and contexts that may occur in real-time.

The distributed intelligent software may be in the form of executable code that may be run on one or more of a sensor device, a client device (e.g., a computer and/or a smart-phone), a server, a central database and control system, or any combination thereof. In some embodiments, the distributed intelligent software may be part of or integrated with an operating system for one or more of a sensor device, a sensing system, a client device, a server, a central database and control system, or any combination thereof.

In the following description, like reference numbers are used to identify like elements. Furthermore, the drawings are intended to illustrate major features of exemplary embodiments in a diagrammatic manner. The drawings are not intended to depict every feature of actual embodiments nor relative dimensions of the depicted elements and are not drawn to scale.

As used herein, the term “or” refers an inclusive “or” rather than an exclusive “or.” In addition, the articles “a” and “an” as used in the specification and claims mean “one or more” unless specified otherwise or clear from the context to refer the singular form.

Each of the one or more sensor devices (also referred to herein as “wireless sensing nodes”) in the sensing system may be a sensor device having an adhesive tape form factor (also referred to herein as an “adhesive tape platform”). The term “tape node” refers to an adhesive tape platform or a segment thereof that is equipped with sensor, processor, memory, energy source/harvesting mechanism, and wireless communications functionality, where the adhesive product has a variety of different form factors, including a multilayer roll or a sheet that includes a plurality of divisible adhesive segments. Once deployed, each tape node can function, for example, as an adhesive tape, label, sticker, decal, or the like, and as a wireless communications device.

The terms “adhesive tape node,” “wireless node,” or “tape node” may be used interchangeably in certain contexts, and refer to an adhesive tape platform or a segment thereof that is equipped with sensor, processor, memory, energy source/harvesting mechanism, and wireless communications functionality, where the adhesive product has a variety of different form factors, including a multilayer roll or a sheet that includes a plurality of divisible adhesive segments. Once deployed, each tape node or wireless node can function, for example, as an adhesive tape, label, sticker, decal, or the like, and as a wireless communications device. A “periph-

eral” tape node or wireless node, also referred to as an outer node, leaf node, or terminal node, refers to a node that does not have any child nodes.

In certain contexts, the terms “package,” “envelope,” “box,” “parcel,” “container,” “pallet,” “carton,” “wrapping,” and the like are used interchangeably herein to refer to an packaged item or items.

In certain contexts, the terms “wireless tracking system,” “hierarchical communications network,” “distributed agent operating system,” and the like are used interchangeably herein to refer to a system or network of wireless nodes.

Adhesive Tape Platform

FIG. 1A shows an example asset **10** that is sealed for shipment using an example adhesive tape platform **12** that includes embedded components of a wireless transducing circuit **14** (collectively referred to herein as a “tape node”). In this example, a length **13** of the adhesive tape platform **12** is dispensed from a roll **16** and affixed to the asset **10**. The adhesive tape platform **12** includes an adhesive side **18** and a non-adhesive side **20**. The adhesive tape platform **12** can be dispensed from the roll **16** in the same way as any conventional packing tape, shipping tape, or duct tape. For example, the adhesive tape platform **12** may be dispensed from the roll **16** by hand, laid across the seam where the two top flaps of the asset **10** meet, and cut to a suitable length either by hand or using a cutting instrument (e.g., scissors or an automated or manual tape dispenser). Examples of such tapes include tapes having non-adhesive sides **20** that carry one or more coatings or layers (e.g., colored, light reflective, light absorbing, and/or light emitting coatings or layers).

Referring to FIG. 1B, in some examples, the non-adhesive side **20** of the length **13** of the adhesive tape platform **12** includes writing or other markings that convey instructions, warnings, or other information to a person or machine (e.g., a bar code reader), or may simply be decorative and/or entertaining. For example, different types of adhesive tape platforms may be marked with distinctive colorations to distinguish one type of adhesive tape platform from another. In the illustrated example, the length **13** of the adhesive tape platform **12** includes a two-dimensional bar code (e.g., a QR Code) **22**, written instructions **24** (i.e., “Cut Here”), and an associated cut line **26** that indicates where the user should cut the adhesive tape platform **12**. The written instructions **24** and the cut line **26** typically are printed or otherwise marked on the top non-adhesive surface **20** of the adhesive tape platform **12** during manufacture. The two-dimensional bar code **22**, on the other hand, may be marked on the non-adhesive surface **20** of the adhesive tape platform **12** during the manufacture of the adhesive product **12** or, alternatively, may be marked on the non-adhesive surface **20** of the adhesive tape platform **12** as needed using, for example, a printer or other marking device.

In order to avoid damage to the functionality of the segments of the adhesive tape platform **12**, the cut lines **26** typically demarcate the boundaries between adjacent segments at locations that are free of any active components of the wireless transducing circuit **14**. The spacing between the wireless transducing circuit components **14** and the cut lines **26** may vary depending on the intended communication, transducing and/or adhesive taping application. In the example illustrated in FIG. 1A, the length of the adhesive tape platform **12** that is dispensed to seal the asset **10** corresponds to a single segment of the adhesive tape platform **12**. In other examples, the length of the adhesive tape platform **12** needed to seal a asset or otherwise serve the adhesive function for which the adhesive tape platform **12** is being applied may include multiple segments **13** of the

adhesive tape platform 12, one or more of which segments 13 may be activated upon cutting the length of the adhesive tape platform 12 from the roll 16 and/or applying the length of the adhesive tape platform to the asset 10.

In some examples, the transducing components 14 that are embedded in one or more segments 13 of the adhesive tape platform 12 are activated when the adhesive tape platform 12 is cut along the cut line 26. In these examples, the adhesive tape platform 12 includes one or more embedded energy sources (e.g., thin film batteries, which may be printed, or conventional cell batteries, such as conventional watch style batteries, rechargeable batteries, or other energy storage device, such as a super capacitor or charge pump) that supply power to the transducing components 14 in one or more segments of the adhesive tape platform 12 in response to being separated from the adhesive tape platform 12 (e.g., along the cut line 26).

In some examples, each segment 13 of the adhesive tape platform 12 includes its own respective energy source including energy harvesting elements that can harvest energy from the environment. In some of these examples, each energy source is configured to only supply power to the components in its respective adhesive tape platform segment regardless of the number of contiguous segments 13 that are in a given length of the adhesive tape platform 12. In other examples, when a given length of the adhesive tape platform 12 includes multiple segments 13, the energy sources in the respective segments 13 are configured to supply power to the transducing components 14 in all of the segments 13 in the given length of the adhesive tape platform 12. In some of these examples, the energy sources are connected in parallel and concurrently activated to power the transducing components 14 in all of the segments 13 at the same time. In other examples, the energy sources are connected in parallel and alternately activated to power the transducing components 14 in respective ones of the adhesive tape platform segments 13 at different time periods, which may or may not overlap.

FIG. 2 shows an example adhesive tape platform 30 that includes a set of adhesive tape platform segments 32 each of which includes a respective set of embedded wireless transducing circuit components 34, and a backing sheet 36 with a release coating that prevents the adhesive segments 32 from adhering strongly to the backing sheet 36. Each adhesive tape platform segment 32 includes an adhesive side facing the backing sheet 36, and an opposing non-adhesive side 40. In this example, a particular segment 32' of the adhesive tape platform 30 has been removed from the backing sheet 36 and affixed to an envelope 44. Each segment 32 of the adhesive tape platform 30 can be removed from the backing sheet 36 in the same way that adhesive labels can be removed from a conventional sheet of adhesive labels (e.g., by manually peeling a segment 32 from the backing sheet 36). In general, the non-adhesive side 40' of the segment 32' may include any type of writing, markings, decorative designs, or other ornamentation. In the illustrated example, the non-adhesive side 40' of the segment 32' includes writing or other markings that correspond to a destination address for the envelope 44. The envelope 44 also includes a return address 46 and, optionally, a postage stamp or mark 48.

In some examples, segments of the adhesive tape platform 12 are deployed by a human operator. The human operator may be equipped with a mobile phone or other device that allows the operator to authenticate and initialize the adhesive tape platform 12. In addition, the operator can take a picture of an asset including the adhesive tape platform and

any barcodes associated with the asset and, thereby, create a persistent record that links the adhesive tape platform 12 to the asset. In addition, the human operator typically will send the picture to a network service and/or transmit the picture to the adhesive tape platform 12 for storage in a memory component of the adhesive tape platform 12.

In some examples, the wireless transducing circuit components 34 that are embedded in a segment 32 of the adhesive tape platform 12 are activated when the segment 32 is removed from the backing sheet 32. In some of these examples, each segment 32 includes an embedded capacitive sensing system that can sense a change in capacitance when the segment 32 is removed from the backing sheet 36. As explained in detail below, a segment 32 of the adhesive tape platform 30 includes one or more embedded energy sources (e.g., thin film batteries, common disk-shaped cell batteries, or rechargeable batteries or other energy storage devices, such as a super capacitor or charge pump) that can be configured to supply power to the wireless transducing circuit components 34 in the segment 32 in response to the detection of a change in capacitance between the segment 32 and the backing sheet 36 as a result of removing the segment 32 from the backing sheet 36.

FIG. 3 shows a block diagram of the components of an example wireless transducing circuit 70 that includes a number of communication systems 72, 74. Example communication systems 72, 74 include a GPS system that includes a GPS receiver circuit 82 (e.g., a receiver integrated circuit) and a GPS antenna 84, and one or more wireless communication systems each of which includes a respective transceiver circuit 86 (e.g., a transceiver integrated circuit) and a respective antenna 88. Example wireless communication systems include a cellular communication system (e.g., GSM/GPRS), a Wi-Fi communication system, an RF communication system (e.g., LoRa), a Bluetooth communication system (e.g., a Bluetooth Low Energy system), a Z-wave communication system, and a ZigBee communication system. The wireless transducing circuit 70 also includes a processor 90 (e.g., a microcontroller or micro-processor), one or more energy storage devices 92 (e.g., non-rechargeable or rechargeable printed flexible battery, conventional single or multiple cell battery, and/or a super capacitor or charge pump), one or more transducers 94 (e.g., sensors and/or actuators, and, optionally, one or more energy harvesting transducer components). In some examples, the conventional single or multiple cell battery may be a watch style disk or button cell battery that is associated electrical connection apparatus (e.g., a metal clip) that electrically connects the electrodes of the battery to contact pads on the flexible circuit 116.

Examples of sensing transducers 94 include a capacitive sensor, an altimeter, a gyroscope, an accelerometer, a temperature sensor, a strain sensor, a pressure sensor, a piezoelectric sensor, a weight sensor, an optical or light sensor (e.g., a photodiode or a camera), an acoustic or sound sensor (e.g., a microphone), a smoke detector, a radioactivity sensor, a chemical sensor (e.g., an explosives detector), a biosensor (e.g., a blood glucose biosensor, odor detectors, antibody based pathogen, food, and water contaminant and toxin detectors, DNA detectors, microbial detectors, pregnancy detectors, and ozone detectors), a magnetic sensor, an electromagnetic field sensor, and a humidity sensor. Examples of actuating (e.g., energy emitting) transducers 94 include light emitting components (e.g., light emitting diodes and displays), electro-acoustic transducers (e.g., audio speakers), electric motors, and thermal radiators (e.g., an electrical resistor or a thermoelectric cooler).

In some examples, the wireless transducing circuit 70 includes a memory 96 for storing data, including, e.g., profile data, state data, event data, sensor data, localization data, security data, and one or more unique identifiers (ID) 98 associated with the wireless transducing circuit 70, such as a product ID, a type ID, and a media access control (MAC) ID, and control code 99. In some examples, the memory 96 may be incorporated into one or more of the processor 90 or transducers 94, or may be a separate component that is integrated in the wireless transducing circuit 70 as shown in FIG. 3. The control code typically is implemented as programmatic functions or program modules that control the operation of the wireless transducing circuit 70, including a tape node communication manager that manages the manner and timing of tape node communications, a tape node power manager that manages power consumption, and a tape node connection manager that controls whether connections with other tape nodes are secure connections or unsecure connections, and a tape node storage manager that securely manages the local data storage on the node. The tape node connection manager ensures the level of security required by the end application and supports various encryption mechanisms. The tape node power manager and tape communication manager work together to optimize the battery consumption for data communication. In some examples, execution of the control code by the different types of tape nodes described herein may result in the performance of similar or different functions.

FIG. 4 is a top view of a portion of an example flexible adhesive tape platform 100 that shows a first segment 102 and a portion of a second segment 104. Each segment 102, 104 of the flexible adhesive tape platform 100 includes a respective set 106, 108 of the components of the wireless transducing circuit 70. The segments 102, 104 and their respective sets of components 106, 108 typically are identical and configured in the same way. In some other embodiments, however, the segments 102, 104 and/or their respective sets of components 106, 108 are different and/or configured in different ways. For example, in some examples, different sets of the segments of the flexible adhesive tape platform 100 have different sets or configurations of tracking and/or transducing components that are designed and/or optimized for different applications, or different sets of segments of the flexible adhesive tape platform may have different ornamentations (e.g., markings on the exterior surface of the platform) and/or different (e.g., alternating) lengths.

An example method of fabricating the adhesive tape platform 100 (see FIG. 4) according to a roll-to-roll fabrication process is described in connection with FIGS. 6, 7A, and 7B of U.S. Pat. No. 10,262,255, issued Apr. 15, 2019, the entirety of which is incorporated herein by reference.

The instant specification describes an example system of adhesive tape platforms (also referred to herein as “tape nodes”) that can be used to implement a low-cost wireless network infrastructure for performing monitoring, tracking, and other asset management functions relating to, for example, parcels, persons, tools, equipment and other physical assets and objects. The example system includes a set of three different types of tape nodes that have different respective functionalities and different respective cover markings that visually distinguish the different tape node types from one another. In one non-limiting example, the covers of the different tape node types are marked with different colors (e.g., white, green, and black). In the illustrated examples, the different tape node types are distinguishable from one

another by their respective wireless communications capabilities and their respective sensing capabilities.

FIG. 5A shows a cross-sectional side view of a portion of an example segment 102 of the flexible adhesive tape platform 100 that includes a respective set of the components of the wireless transducing circuit 106 corresponding to the first tape node type (i.e., white). The flexible adhesive tape platform segment 102 includes an adhesive layer 112, an optional flexible substrate 110, and an optional adhesive layer 114 on the bottom surface of the flexible substrate 110. If the bottom adhesive layer 114 is present, a release liner (not shown) may be (weakly) adhered to the bottom surface of the adhesive layer 114. In some examples, the adhesive layer 114 includes an adhesive (e.g., an acrylic foam adhesive) that has a high bond strength that is sufficient to prevent removal of the adhesive segment 102 from a surface on which the adhesive layer 114 is adhered without destroying the physical or mechanical integrity of the adhesive segment 102 and/or one or more of its constituent components. In some examples, the optional flexible substrate 110 is implemented as a prefabricated adhesive tape that includes the adhesive layers 112, 114 and the optional release liner. In other examples, the adhesive layers 112, 114 are applied to the top and bottom surfaces of the flexible substrate 110 during the fabrication of the adhesive tape platform 100. The adhesive layer 112 bonds the flexible substrate 110 to a bottom surface of a flexible circuit 116, that includes one or more wiring layers (not shown) that connect the processor 90, a low power wireless communication interface 81 (e.g., a Zigbee, Bluetooth® Low Energy (BLE) interface, or other low power communication interface), a timer circuit 83, transducing and/or energy harvesting component(s) 94 (if present), the memory 96, and other components in a device layer 122 to each other and to the energy storage component 92 and, thereby, enable the transducing, tracking and other functionalities of the flexible adhesive tape platform segment 102. The low power wireless communication interface 81 typically includes one or more of the antennas 84, 88 and one or more of the wireless circuits 82, 86.

FIG. 5B shows a cross-sectional side view of a portion of an example segment 103 of the flexible adhesive tape platform 100 that includes a respective set of the components of the wireless transducing circuit 106 corresponding to the second tape node type (i.e., green). In this example, the flexible adhesive tape platform segment 103 differs from the segment 102 shown in FIG. 5A by the inclusion of a medium power communication interface 85 (e.g., a LoRa interface) in addition to the low power communications interface that is present in the first tape node type (i.e., white). The medium power communication interface has longer communication range than the low power communication interface. In some examples, one or more other components of the flexible adhesive tape platform segment 103 differ, for example, in functionality or capacity (e.g., larger energy source).

FIG. 5C shows a cross-sectional side view of a portion of an example segment 105 of the flexible adhesive tape platform 100 that includes a respective set of the components of the wireless transducing circuit 106 corresponding to the third tape node type (i.e., black). In this example, the flexible adhesive tape platform segment 105 includes a high power communications interface 87 (e.g., a cellular interface; e.g., GSM/GPRS) and an optional medium and/or low power communications interface 85. The high power communication range provides global coverage to available infrastructure (e.g. the cellular network). In some examples,

one or more other components of the flexible adhesive tape platform segment **105** differ, for example, in functionality or capacity (e.g., larger energy source).

FIGS. 5A-5C show examples in which the cover layer **128** of the flexible adhesive tape platform **100** includes one or more interfacial regions **129** positioned over one or more of the transducers **94**. In examples, one or more of the interfacial regions **129** have features, properties, compositions, dimensions, and/or characteristics that are designed to improve the operating performance of the platform **100** for specific applications. In some examples, the flexible adhesive tape platform **100** includes multiple interfacial regions **129** over respective transducers **94**, which may be the same or different depending on the target applications. Example interfacial regions include an opening, an optically transparent window, and/or a membrane located in the interfacial region **129** of the cover **128** that is positioned over the one or more transducers and/or energy harvesting components **94**. Additional details regarding the structure and operation of example interfacial regions **129** are described in U.S. Provisional Patent Application No. 62/680,716, filed Jun. 5, 2018, PCT Patent Application No. PCT/US2018/064919, filed Dec. 11, 2018, U.S. Pat. No. 10,885,420, issued Jan. 4, 2021, U.S. Pat. No. 10,902,310 issued Jan. 25, 2021, and U.S. Provisional Patent Application No. 62/670,712, filed May 11, 2018.

In some examples, a flexible polymer layer **124** encapsulates the device layer **122** and thereby reduces the risk of damage that may result from the intrusion of contaminants and/or liquids (e.g., water) into the device layer **122**. The flexible polymer layer **124** also planarizes the device layer **122**. This facilitates optional stacking of additional layers on the device layer **122** and also distributes forces generated in, on, or across the adhesive tape platform segment **102** so as to reduce potentially damaging asymmetric stresses that might be caused by the application of bending, torqueing, pressing, or other forces that may be applied to the flexible adhesive tape platform segment **102** during use. In the illustrated example, a flexible cover **128** is bonded to the planarizing polymer **124** by an adhesive layer (not shown).

The flexible cover **128** and the flexible substrate **110** may have the same or different compositions depending on the intended application. In some examples, one or both of the flexible cover **128** and the flexible substrate **110** include flexible film layers and/or paper substrates, where the film layers may have reflective surfaces or reflective surface coatings. Example compositions for the flexible film layers include polymer films, such as polyester, polyimide, polyethylene terephthalate (PET), and other plastics. The optional adhesive layer on the bottom surface of the flexible cover **128** and the adhesive layers **112**, **114** on the top and bottom surfaces of the flexible substrate **110** typically include a pressure-sensitive adhesive (e.g., a silicon-based adhesive). In some examples, the adhesive layers are applied to the flexible cover **128** and the flexible substrate **110** during manufacture of the adhesive tape platform **100** (e.g., during a roll-to-roll or sheet-to-sheet fabrication process). In other examples, the flexible cover **128** may be implemented by a prefabricated single-sided pressure-sensitive adhesive tape and the flexible substrate **110** may be implemented by a prefabricated double-sided pressure-sensitive adhesive tape; both kinds of tape may be readily incorporated into a roll-to-roll or sheet-to-sheet fabrication process. In some examples, the flexible polymer layer **124** is composed of a flexible epoxy (e.g., silicone).

In some examples, the energy storage device **92** is a flexible battery that includes a printed electrochemical cell,

which includes a planar arrangement of an anode and a cathode and battery contact pads. In some examples, the flexible battery may include lithium-ion cells or nickel-cadmium electro-chemical cells. The flexible battery typically is formed by a process that includes printing or laminating the electro-chemical cells on a flexible substrate (e.g., a polymer film layer). In some examples, other components may be integrated on the same substrate as the flexible battery. For example, the low power wireless communication interface **81** and/or the processor(s) **90** may be integrated on the flexible battery substrate. In some examples, one or more of such components also (e.g., the flexible antennas and the flexible interconnect circuits) may be printed on the flexible battery substrate.

In some examples, the flexible circuit **116** is formed on a flexible substrate by printing, etching, or laminating circuit patterns on the flexible substrate. In some examples, the flexible circuit **116** is implemented by one or more of a single-sided flex circuit, a double access or back bared flex circuit, a sculpted flex circuit, a double-sided flex circuit, a multi-layer flex circuit, a rigid flex circuit, and a polymer thick film flex circuit. A single-sided flexible circuit has a single conductor layer made of, for example, a metal or conductive (e.g., metal filled) polymer on a flexible dielectric film. A double access or back bared flexible circuit has a single conductor layer but is processed so as to allow access to selected features of the conductor pattern from both sides. A sculpted flex circuit is formed using a multi-step etching process that produces a flex circuit that has finished copper conductors that vary in thickness along their respective lengths. A multilayer flex circuit has three or more layers of conductors, where the layers typically are interconnected using plated through holes. Rigid flex circuits are a hybrid construction of flex circuit consisting of rigid and flexible substrates that are laminated together into a single structure, where the layers typically are electrically interconnected via plated through holes. In polymer thick film (PTF) flex circuits, the circuit conductors are printed onto a polymer base film, where there may be a single conductor layer or multiple conductor layers that are insulated from one another by respective printed insulating layers.

In the example flexible adhesive tape platform segments **102** shown in FIGS. 5A-5C, the flexible circuit **116** is a single access flex circuit that interconnects the components of the adhesive tape platform on a single side of the flexible circuit **116**. In other examples, the flexible circuit **116** is a double access flex circuit that includes a front-side conductive pattern that interconnects the low power communications interface **81**, the timer circuit **83**, the processor **90**, the one or more transducers **94** (if present), and the memory **96**, and allows through-hole access (not shown) to a back-side conductive pattern that is connected to the flexible battery (not shown). In these examples, the front-side conductive pattern of the flexible circuit **116** connects the communications circuits **82**, **86** (e.g., receivers, transmitters, and transceivers) to their respective antennas **84**, **88** and to the processor **90**, and also connects the processor **90** to the one or more sensors **94** and the memory **96**. The backside conductive pattern connects the active electronics (e.g., the processor **90**, the communications circuits **82**, **86**, and the transducers) on the front-side of the flexible circuit **116** to the electrodes of the flexible battery **116** via one or more through holes in the substrate of the flexible circuit **116**.

Depending on the target application, the wireless transducing circuits **70** are distributed across the flexible adhesive tape platform **100** according to a specified sampling density, which is the number of wireless transducing circuits **70** for

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a given unit size (e.g., length or area) of the flexible adhesive tape platform **100**. In some examples, a set of multiple flexible adhesive tape platforms **100** are provided that include different respective sampling densities in order to seal different asset sizes with a desired number of wireless transducing circuits **70**. In particular, the number of wireless transducing circuits per asset size is given by the product of the sampling density specified for the adhesive tape platform and the respective size of the adhesive tape platform **100** needed to seal the asset. This allows an automated packaging system to select the appropriate type of flexible adhesive tape platform **100** to use for sealing a given asset with the desired redundancy (if any) in the number of wireless transducer circuits **70**. In some example applications (e.g., shipping low value goods), only one wireless transducing circuit **70** is used per asset, whereas in other applications (e.g., shipping high value goods) multiple wireless transducing circuits **70** are used per asset. Thus, a flexible adhesive tape platform **100** with a lower sampling density of wireless transducing circuits **70** can be used for the former application, and a flexible adhesive tape platform **100** with a higher sampling density of wireless transducing circuits **70** can be used for the latter application. In some examples, the flexible adhesive tape platforms **100** are color-coded or otherwise marked to indicate the respective sampling densities with which the wireless transducing circuits **70** are distributed across the different types of adhesive tape platforms **100**.

In some embodiments, the adhesive tape platforms discussed in conjunction with FIGS. **1A-5C** are dispensed by a handheld tape dispenser, as discussed further in conjunction with FIGS. **11-13**. The handheld tape dispenser is configured to receive a roll of tape, the roll of tape comprising a plurality of adhesive tape platforms distributed throughout the roll, e.g., in uniform length segments, and to dispense the roll of tape for use by a sensing system, e.g., in an environment as illustrated in FIGS. **6A** and **6B**.
Deployment of Tape Nodes

FIG. **6A** shows an example network communications environment **400** that includes a network **402** that supports communications between one or more servers **404** executing one or more applications of a network service **408**, mobile gateways **410**, **412**, a stationary gateway **414**, and various types of tape nodes that are associated with various assets (e.g., parcels, equipment, tools, persons, and other things) The example network communication environment **400** may also be referred to as a sensing system **400** or wireless tracking system **400**. The nodes of the sensing system **400** may refer to the tape nodes, other wireless devices, the gateway devices, client devices, servers, and other components of the wireless sensing system **400**. In some examples, the network **402** includes one or more network communication systems and technologies, including any one or more of wide area networks, local area networks, public networks (e.g., the internet), private networks (e.g., intranets and extranets), wired networks, and wireless networks. For example, the network **402** includes communications infrastructure equipment, such as a geolocation satellite system **416** (e.g., GPS, GLONASS, and NAVSTAR), cellular communication systems (e.g., GSM/GPRS), Wi-Fi communication systems, RF communication systems (e.g., LoRa), Bluetooth communication systems (e.g., a Bluetooth Low Energy system), Z-wave communication systems, and Zig-Bee communication systems.

In some examples, the one or more network service applications **406** leverage the above-mentioned communications technologies to create a hierarchical wireless net-

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work of tape nodes that improves asset management operations by reducing costs and improving efficiency in a wide range of processes, from asset packaging, asset transporting, asset tracking, asset condition monitoring, asset inventorying, and asset security verification. Communication across the network is secured by a variety of different security mechanisms. In the case of existing infrastructure, a communication link the communication uses the infrastructure security mechanisms. In case of communications among tapes nodes, the communication is secured through a custom security mechanism. In certain cases, tape nodes can also be configured to support block chain to protect the transmitted and stored data.

A set of tape nodes can be configured by the network service **408** to create hierarchical communications network. The hierarchy can be defined in terms of one or more factors, including functionality (e.g., wireless transmission range or power), role (e.g., master tape node vs. peripheral tape node), or cost (e.g., a tape node equipped with a cellular transceiver vs. a peripheral tape node equipped with a Bluetooth LE transceiver). Tape nodes can be assigned to different levels of a hierarchical network according to one or more of the above-mentioned factors. For example, the hierarchy can be defined in terms of communication range or power, where tape nodes with higher power or longer communication range transceivers are arranged at a higher level of the hierarchy than tape nodes with lower power or lower range transceivers. In another example, the hierarchy is defined in terms of role, where, e.g., a master tape node is programmed to bridge communications between a designated group of peripheral tape nodes and a gateway node or server node. The problem of finding an optimal hierarchical structure can be formulated as an optimization problem with battery capacity of nodes, power consumption in various modes of operation, desired latency, external environment, etc. and can be solved using modern optimization methods e.g. neural networks, artificial intelligence, and other machine learning computing systems that take expected and historical data to create an optimal solution and can create algorithms for modifying the system's behavior adaptively in the field.

The tape nodes may be deployed by automated equipment or manually. As described in the embodiments of FIGS. **11-14**, the tape nodes are deployed by an automated handheld tape dispenser. In this process, a tape node typically is separated from a roll or sheet and adhered to an asset, or other stationary or mobile object (e.g., a structural element of a warehouse, or a vehicle, such as a delivery truck) or stationary object (e.g., a structural element of a building). This process activates the tape node and causes the tape node to communicate with a server **404** of the network service **408**. In this process, the tape node may communicate through one or more other tape nodes in the communication hierarchy. In this process, the network server **404** executes the network service application **406** to programmatically configure tape nodes that are deployed in the environment **400**. In some examples, there are multiple classes or types of tape nodes, where each tape node class has a different respective set of functionalities and/or capacities.

In some examples, the one or more network service servers **404** communicate over the network **402** with one or more gateways that are configured to send, transmit, forward, or relay messages to the network **402** and activated tape nodes that are associated with respective assets and within communication range. Example gateways include mobile gateways **410**, **412** and a stationary gateway **414**. In some examples, the mobile gateways **410**, **412**, and the

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stationary gateway **414** are able to communicate with the network **402** and with designated sets or groups of tape nodes.

In some examples, the mobile gateway **412** is a vehicle (e.g., a delivery truck or other mobile hub) that includes a wireless communications unit **416** that is configured by the network service **408** to communicate with a designated set of tape nodes, including a peripheral tape node **418** in the form of a label that is adhered to an asset **420** contained within a parcel **421** (e.g., an envelope), and is further configured to communicate with the network service **408** over the network **402**. In some examples, the peripheral tape node **418** includes a lower power wireless communications interface of the type used in, e.g., tape node **102** (shown in FIG. 5A), and the wireless communications unit **416** is implemented by a tape node (e.g., one of tape node **103** or tape node **105**, respectively shown in FIGS. 5B and 5C) that includes a lower power communications interface for communicating with tape nodes within range of the mobile gateway **412** and a higher power communications interface for communicating with the network **402**. In this way, the tape nodes **418** and **416** create a hierarchical wireless network of nodes for transmitting, forwarding, bridging, relaying, or otherwise communicating wireless messages to, between, or on behalf of the peripheral tape node **418** and the network service **408** in a power-efficient and cost-effective way.

In some examples, the mobile gateway **410** is a mobile phone that is operated by a human operator and executes a client application **422** that is configured by the network service **408** to communicate with a designated set of tape nodes, including a master tape node **424** that is adhered to a parcel **426** (e.g., a box), and is further configured to communicate with the network service **408** over the network **402**. In the illustrated example, the parcel **426** contains a first parcel labeled or sealed by a tape node **428** and containing a first asset **430**, and a second parcel labeled or sealed by a tape node **432** and containing a second asset **434**. As explained in detail below, the master tape node **424** communicates with each of the peripheral tape nodes **428**, **432** and communicates with the mobile gateway **408** in accordance with a hierarchical wireless network of tape nodes. In some examples, each of the peripheral tape nodes **428**, **432** includes a lower power wireless communications interface of the type used in, e.g., tape node **102** (shown in FIG. 5A), and the master tape node **424** is implemented by a tape node (e.g., tape node **103**, shown in FIG. 5B) that includes a lower power communications interface for communicating with the peripheral tape nodes **428**, **432** contained within the parcel **426**, and a higher power communications interface for communicating with the mobile gateway **410**. The master tape node **424** is operable to relay wireless communications between the tape nodes **428**, **432** contained within the parcel **426** and the mobile gateway **410**, and the mobile gateway **410** is operable to relay wireless communications between the master tape node **424** and the network service **408** over the wireless network **402**. In this way, the master tape node **424** and the peripheral tape nodes **428** and **432** create a hierarchical wireless network of nodes for transmitting, forwarding, relaying, or otherwise communicating wireless messages to, between, or on behalf of the peripheral tape nodes **428**, **432** and the network service **408** in a power-efficient and cost-effective way.

In some examples, the stationary gateway **414** is implemented by a server executing a server application that is configured by the network service **408** to communicate with a designated set **440** of tape nodes **442**, **444**, **446**, **448** that

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are adhered to respective parcels containing respective assets **450**, **452**, **454**, **456** on a pallet **458**. In other examples, the stationary gateway **414** is implemented by a tape node (e.g., one of tape node **103** or tape node **105**, respectively shown in FIGS. 5B and 5C) that is adhered to, for example, a wall, column or other infrastructure component of the environment **400**, and includes a lower power communications interface for communicating with tape nodes within range of the stationary gateway **414** and a higher power communications interface for communicating with the network **402**. In one embodiment, each of the tape nodes **442-448** is a peripheral tape node and is configured by the network service **408** to communicate individually with the stationary gateway **414**, which relays communications from the tape nodes **442-448** to the network service **408** through the stationary gateway **414** and over the communications network **402**. In another embodiment, one of the tape nodes **442-448** at a time is configured as a master tape node that transmits, forwards, relays, or otherwise communicate wireless messages to, between, or on behalf of the other tape nodes on the pallet **458**. In this embodiment, the master tape node may be determined by the tape nodes **442-448** or designated by the network service **408**. In some examples, the tape node with the longest range or highest remaining power level is determined to be the master tape node. In some examples, when the power level of the current master tape node drops below a certain level (e.g., a fixed power threshold level or a threshold level relative to the power levels of one or more of the other tape nodes), another one of the tape nodes assumes the role of the master tape node. In some examples, a master tape node **459** is adhered to the pallet **458** and is configured to perform the role of a master node for the tape nodes **442-448**. In these ways, the tape nodes **442-448**, **458** are configurable to create different hierarchical wireless networks of nodes for transmitting, forwarding, relaying, bridging, or otherwise communicating wireless messages with the network service **408** through the stationary gateway **414** and over the network **402** in a power-efficient and cost-effective way.

In the illustrated example, the stationary gateway **414** also is configured by the network service **408** to communicate with a designated set of tape nodes, including a master tape node **460** that is adhered to the inside of a door **462** of a shipping container **464**, and is further configured to communicate with the network service **408** over the network **402**. In the illustrated example, the shipping container **464** contains a number of parcels labeled or sealed by respective peripheral tape nodes **466** and containing respective assets. The master tape node **416** communicates with each of the peripheral tape nodes **466** and communicates with the stationary gateway **415** in accordance with a hierarchical wireless network of tape nodes. In some examples, each of the peripheral tape nodes **466** includes a lower power wireless communications interface of the type used in, e.g., tape node **102** (shown in FIG. 5A), and the master tape node **460** is implemented by a tape node (e.g., tape node **103**, shown in FIG. 5B) that includes a lower power communications interface for communicating with the peripheral tape nodes **466** contained within the shipping container **464**, and a higher power communications interface for communicating with the stationary gateway **414**.

In some examples, when the doors of the shipping container **464** are closed, the master tape node **460** is operable to communicate wirelessly with the peripheral tape nodes **466** contained within the shipping container **464**. In an example, the master tape node **460** is configured to collect sensor data from the peripheral tape nodes and, in some

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embodiments, process the collected data to generate, for example, one or more histograms from the collected data. When the doors of the shipping container **464** are open, the master tape node **460** is programmed to detect the door opening (e.g., with an accelerometer component of the master tape node **460**) and, in addition to reporting the door opening event to the network service **408**, the master tape node **460** is further programmed to transmit the collected data and/or the processed data in one or more wireless messages to the stationary gateway **414**. The stationary gateway **414**, in turn, is operable to transmit the wireless messages received from the master tape node **460** to the network service **408** over the wireless network **402**. Alternatively, in some examples, the stationary gateway **414** also is operable to perform operations on the data received from the master tape node **460** with the same type of data produced by the master node **459** based on sensor data collected from the tape nodes **442-448**. In this way, the master tape node **460** and the peripheral tape nodes **466** create a hierarchical wireless network of nodes for transmitting, forwarding, relaying, or otherwise communicating wireless messages to, between, or on behalf of the peripheral tape nodes **466** and the network service **408** in a power-efficient and cost-effective way.

In an example of the embodiment shown in FIG. 6A, there are three classes of tape nodes: a short range tape node, a medium range tape node, and a long range tape node, as respectively shown in FIGS. 5A-5C. The short range tape nodes typically are adhered directly to parcels containing assets. In the illustrated example, the tape nodes **418**, **428**, **432**, **442-448**, **466** are short range tape nodes. The short range tape nodes typically communicate with a low power wireless communication protocol (e.g., Bluetooth LE, Zigbee, or Z-wave). The medium range tape nodes typically are adhered to objects (e.g., a box **426** and a shipping container **460**) that are associated with multiple parcels or assets that are separated from the medium range tape nodes by a barrier or a large distance. In the illustrated example, the tape nodes **424** and **460** are medium range tape nodes. The medium range tape nodes typically communicate with a medium power wireless communication protocol (e.g., LoRa or Wi-Fi). The long-range tape nodes typically are adhered to mobile or stationary infrastructure of the wireless communication environment **400**. In the illustrated example, the mobile gateway tape node **412** and the stationary gateway tape node **414** are long range tape nodes. The long range tape nodes typically communicate with other nodes using a high power wireless communication protocol (e.g., a cellular data communication protocol). In some examples, the mobile gateway tape node **436** is adhered to a mobile vehicle (e.g., a truck). In these examples, the mobile gateway **412** may be moved to different locations in the environment **400** to assist in connecting other tape nodes to the server **404**. In some examples, the stationary gateway tape node **414** may be attached to a stationary structure (e.g., a wall) in the environment **400** with a known geographic location. In these examples, other tape nodes in the environment can determine their geographic location by querying the gateway tape node **414**.

System Environment

FIG. 6B is a diagram showing an example system environment for the sensing system **500** including the adhesive tape platform **510**. The sensing system **500** includes one or more sections of an adhesive tape platform **510**, a central database and control system **520**, one or more client devices **530**, and a network **540**.

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Embodiments of the adhesive tape platform **510** are also described above with respect to FIGS. 1A-1B, 2A-2C, 3, and 4A-4C. Each section of the adhesive tape platform **510** may include a location module **511**, a processor **512**, memory **513**, a communication module **514**, and a sensor module **515**, according to some embodiments. A section of the adhesive tape platform **510** may also include additional and/or different components not shown in FIG. 6B, according to some embodiments. The location module **511** collects data relevant to the location of the corresponding section of the adhesive tape platform **510**. The location data collected by the location module may be stored in the memory **513**. The location data may also have computations performed on it by the processor **512** and may be transmitted by the communication module **514** to the central database and control system **520** and/or one or more of the client devices **530** via the network **540**. The location data may include geographic locations ascertained from systems including GPS, cellular network systems (e.g., GSM), wireless local area networks (e.g., a system of Wi-Fi access points), a dead-reckoning system, some other location system, or some combination thereof.

The sensing system **500** may also include sensing devices and components deployed in the field other than devices with the flexible adhesive tape form factor. For example, an embodiment of the tape node **510** may include a non-flexible sensing device that may be used to track assets, interact with other tape nodes **510**, communicate with the central database and control system **520** and client devices **530**, perform other functions, or some combination thereof. The tape node **510** may also include gateway devices or other communication devices that perform functions in conjunction with the adhesive tape platform and the sensing system **500**. In some embodiments, multiple tape nodes **510** may have multiple different functionalities, such as performing different types of communication (e.g., long-range, medium-range, short-range), and may be deployed and operate together in the sensing system **500**.

The location module **511** may include the one or more antennas and one or more wireless communication interface circuits of the communication interface **81**, according to some embodiments. The location module **511** may include, for example, a GPS system that includes a GPS receiver circuit (e.g., a receiver integrated circuit) and a GPS antenna. In some embodiments, the location module **511** also includes one or more wireless communication systems each of which includes a respective transceiver circuit (e.g., a transceiver integrated circuit) and a respective antenna. Example wireless communication systems include a cellular communication system (e.g., GSM/GPRS), a Wi-Fi communication system, an RF communication system (e.g., LoRa), a Bluetooth communication system (e.g., a Bluetooth Low Energy system), a Z-wave communication system, and a ZigBee communication system.

The processor **512** may be a microcontroller or micro-processor, according to some embodiments. The processor **512** may be an embodiment of the processor **90**. In some embodiments, each section of the adhesive tape platform includes more than one processor **512**. The memory **513** stores sensing data, location data, and other data necessary for the functioning of the adhesive tape platform **510**. The memory **513** may be incorporated into the one or more of the processor **512** or may be a separate component. The memory **513** may be an embodiment of the memory **96**.

The communication module **514** enables communication between the adhesive tape platform **510** and the other members of the sensing system **500** via the network **540**. The

communication module may include embodiments of the communication interface **81**. In some embodiments, the communication module **514** enables a segment of the adhesive tape platform **510** to communicate with another segment of the adhesive tape platform **510**. The communication module includes one or more wireless communication systems each of which includes a respective transceiver circuit (e.g., a transceiver integrated circuit) and a respective antenna. Example wireless communication systems include a cellular communication system (e.g., GSM/GPRS), a Wi-Fi communication system, an RF communication system (e.g., LoRa), a Bluetooth communication system (e.g., a Bluetooth Low Energy system), a Z-wave communication system, and a ZigBee communication system. The one or more wireless communication systems in the communication module **514** may be shared with the location module **511**.

The sensor module **515** includes one or more sensors and/or sensor devices. The one or more sensors may include the examples of sensing transducers **94** described above with respect to FIG. **3**. The sensor module **515** collects sensor data which may be stored in the memory **513**, have computations performed on it by the processor **512**, be transmitted to one or more of the central database and control system **520** and the one or more client devices **530**, or some combination thereof. The sensor data may include acceleration data, velocity data, vibration data, capacitive sensing data, humidity data, audio recording data, optical sensor data, infrared sensor data, temperature data, other sensor data, or some combination thereof. The sensor data may include examples of data collected by sensors not described herein.

In some embodiments, different segments of the adhesive tape platform **510** include different communication modules that have corresponding communication ranges. For example, a first tape node **510** may only include short-range communication capabilities (e.g., Bluetooth communications), while a second tape node **510** may include the short-range communication capabilities and longer range communication capabilities (e.g., LoRa, cellular communications, or WiFi). The second tape node **510** may be positioned in a location within the communication range of the first tape node **510**. In this case, the first tape node **510** may communicate with other members of the sensing system **500** by communicating data to the second tape node **510** via the short-range communication. The second tape node **510**, in turn, relays the data to and/or from the other members of the sensing system **500**. In some embodiments, the communication is relayed from and to the first tape node **510** by a gateway device (stationary or mobile) that may have a form factor different than that of an adhesive tape.

The distributed intelligent software may define how one or more devices of the system process and operate in response to data (also referred to herein as sensor data) collected by the adhesive tape platform **510**. The sensing data includes the location data collected by the location module **511**, the sensor data collected by the sensor module **515**, data received by the communication module **514** from the central database and control system **520** and/or the one or more client devices **530**, or some combination thereof. Based on the sensing data, the distributed intelligent software may alter the state of the adhesive tape platform **510**. Altering the state of the adhesive tape platform **510** alters the actions, functions, and behavior of the location module **511**, processor **512**, communication module **513**, and the sensor module **515** of the adhesive tape platform **510**. In some embodiments, altering the state of the adhesive tape plat-

form **510** alters other aspects of the adhesive tape platform. The processor **512** of the adhesive tape platform executes computations and functions based on instructions of the distributed intelligent software to alter the state of the adhesive tape platform **510**. Examples of various states are discussed below, with respect to FIG. **6**. Instructions to alter the state of the tape node **510** may be in the form of executable programmatic code, according to some embodiments.

In some embodiments, logic relevant to the distributed intelligent software is stored in the memory **513** of the adhesive tape platform **510**, or multiple segments thereof. The processor **512** may then generate and execute the instructions for altering the state of the adhesive tape platform **510** based on the stored logic and corresponding sensing data. In other embodiments, the communication module **514** receives the instructions from the central database and control center **520**, the client devices **530**, or some combination thereof. In other embodiments, the logic is distributed between some combination of any one or more of the adhesive tape platform **510**, other adhesive tape platforms, the central database and control system **520**, and the client devices **530**.

The central database and control system **520** is a system for storing data (including sensing data), running applications, transmitting/receiving data to the adhesive tape platform **510** and the one or more client devices **530**, and communicating with the adhesive tape platform **510** and the one or more client devices **530**. According to some embodiments, the central database and control system **520** is hosted on one or more servers. The central database and control system **520** includes an application engine **521** and a database **522**, according to some embodiments. The central database and control system **520** may include additional and/or different components than are shown in FIG. **6B**.

The application engine **521** executes applications associated with the sensing system **500**. For example, the application engine **521** may receive communications and data from the adhesive tape platform **510** and update the database **522** based on the data received from the adhesive tape platform **510**. In another example, the application engine **521** may provide data from the database **522** to one or more of the client devices **530** and communicate instructions to display the data on the display **533** of the client device.

In some embodiments, the application engine **521** performs tasks according to the distributed intelligent software. For example, the application engine **521** detects that an event has occurred based on sensing data received from the adhesive tape platform **510**. In response to the detected event, the application engine **521** generates instructions to alter the state of the adhesive tape platform **510** according to the distributed intelligent software and transmits the instructions to the adhesive tape platform **510**. In some embodiments, the application engine **521** also generates instructions for the central database and control system **520** and executes the instructions, in response to the detected event. The application engine **521** may also generate instructions for the client device **530**, in response to the detected event, according to some embodiments.

The database **522** stores data and logs relevant to the adhesive tape platform **510**. The database **522** stores sensing data that it receives from the adhesive tape platform **510** via the network **540**. The sensing data may include location data (e.g., GPS coordinates, geographic coordinates, etc.), sensor data, other data relevant to monitoring an item with the adhesive tape platform **510**, or some combination thereof. The database **522** may also store data received from one or

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more client devices **530**. For example, a client device **530** may scan a barcode on the adhesive tape node **510** or on an asset being tracked by the adhesive tape node **510**. The client device **530** then transmits a notification regarding the scanning of the barcode to the central database and control system **520** which the application engine **521** logs on the database **522**, creating a comprehensive log of data relevant to the tracking and monitoring of items using the sensing system **500**.

The application engine **521** may perform calculations on the sensing data stored on the database **522** and store calculated values on the database **522**. For example, if the central database and control system receives time-domain sensor data, the application engine **521** may calculate a frequency spectrum for the sensor data by performing a fast Fourier transform (FFT) on the time-domain sensor data and store the frequency spectrum on the database **522**. In some embodiments, the application engine **521** uses a trained machine learning model to perform computations relevant to the adhesive tape platform **510**. A trained machine learning model may be used to detect events in the stored sensing data and determine rules that are part of the distributed intelligent software, according to some embodiments. For example, the application engine **521** may input sensing data from the adhesive tape platform **510** to a trained machine learning model which outputs instructions for altering the state of the adhesive tape platform **510** in response. The use of a machine learning model to generate instructions according to the distributed intelligent software is described in further detail below, with respect to FIGS. **13** and **14**.

The one or more client devices **530** includes computing devices used by users, human operators, and/or administrators of the sensing system **500**. Examples of the client devices **530** include personal computers, smartphones, barcode scanning devices, and other computational devices. A client device **530** may be a dedicated computing device for interacting with the adhesive tape platform **510** and the central database and control system **520**. Each client device **530** includes a processor **531**, memory **532**, and optionally a display **533**, according to some embodiments. A client device **530** may also include a camera, a sensor, a barcode scanning apparatus, communication systems, other components, or some combination thereof. Each client device may execute one or more applications for interacting with the adhesive tape platform **510** and the central database and control system **520**. For example, a client device may run an application that receives sensing data collected by the adhesive tape platform and provided from the central database and control system **520**, store the sensing data in the memory **532**, and display the sensing data on the display **533**. A client device **530** may include additional and/or different components than are shown in FIG. **6B**, according to some embodiments.

In some examples, a human operator uses one or more client devices **530** to interact directly with the adhesive tape platform. The human operator may be equipped with a client device **530** (e.g., mobile phone or other device) that allows the operator to authenticate and initialize the adhesive tape platform **510**. In addition, the operator can take a picture of an asset using the client device **530** including the adhesive tape platform **510** and any barcodes associated with the asset and, thereby, create a persistent record that links the adhesive tape platform **510** to the asset. In addition, the human operator typically sends the picture to a network service and/or transmit the picture to the adhesive tape platform **510** for storage in the memory **513** and/or to the central database and control system **520** for storage in the database **424**. The

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display **533** may display sensing data, notifications, instructions, a user interface, or some combination thereof. In some embodiments, a client device **530** may communicate directly with the adhesive tape platform **510**, for example, using Bluetooth communications, near-field communications (NFC), Wi-Fi, some other communication method, or some combination thereof. In further embodiments, the adhesive tape platform **510** transmits sensing data directly to one or more client devices **530**.

In some embodiments, the client device **530** performs tasks based on the distributed intelligent software. An application running on the processor **531** detects that an event has occurred based on sensing data received from the adhesive tape platform **510** and/or from the central database and control system **520**. Alternatively, the processor **431** may receive an indication from the adhesive tape platform **510** or the central database and control system **420** indicating such event has occurred, without requiring the processor **531** to process the sensing data and expressly detect the event therein. In response to the detected event, the processor **531** generates instructions to alter the state of the adhesive tape platform **510** based on the distributed intelligent software (part of which may be stored on the memory **532**) and transmits the instructions to the adhesive tape platform **510**. In some embodiments, the processor **531** also generates instructions for the central database and control system **520** and the client device **530** transmits the instructions to the central database and control system **520**, in response to the detected event. The processor **531** may also generate instructions for the client device **530** and execute the generated instructions, in response to the detected event, according to some embodiments.

The adhesive tape platform **510**, the central database and control system **520**, and the client devices **530** are configured to communicate via the network **540**, which may comprise any combination of local area networks, wide area networks, public network (e.g., the internet), private networks (e.g., intranets and extranets), using wired and/or wireless communication system. In one embodiment, the network **540** uses standard communications technologies and/or protocols. For example, the network **540** includes communication links using technologies such as Ethernet, 802.11, worldwide interoperability for microwave access (WiMAX), 3G, 4G, code division multiple access (CDMA), digital subscriber line (DSL), geolocation satellite systems (e.g., GPS, GLONASS, and NAVSTAR), cellular communication systems (e.g., GSM/GPRS), Wi-Fi communication systems, RF communication systems (e.g., LoRa), Bluetooth communication systems (e.g., a Bluetooth Low Energy system), Z-wave communication systems, ZigBee communication systems, etc. Examples of networking protocols used for communicating via the network **540** include multiprotocol label switching (MPLS), transmission control protocol/Internet protocol (TCP/IP), hypertext transport protocol (HTTP), simple mail transfer protocol (SMTP), and file transfer protocol (FTP). Data exchanged over the network **120** may be represented using any suitable format, such as hypertext markup language (HTML) or extensible markup language (XML). In some embodiments, all or some of the communication links of the network **540** may be encrypted using any suitable technique or techniques.

The distributed intelligent software includes logic for managing each of the adhesive tape platform segments **510**, the central database and control system **520**, and the client devices **530**. In some embodiments, computation related to the distributed intelligent software is distributed among one or more of the adhesive tape platform **510**, other adhesive

tape platforms, the central database and control system **520**, and the client devices **530**, and combinations thereof. For example, logic relevant to the behavior of the tape nodes **510** may be stored locally on the memory **513** of each of the tape nodes **510**. The processor **512** of the tape node **510** may then access the stored logic and execute the logic based on sensing data that the sensor module **515**, the location module **511**, and the communication module **515** has collected, altering the state of the tape node **510** without receiving any commands from the central database and control system **520** and/or a client device **530**.

In other embodiments, the logic relevant to the behavior of the tape nodes **510** may be stored and executed on the central database and control system **520**. For example, the distributed intelligence engine **521** may execute the logic in response to receiving sensing data from one of the tape nodes **510** and transmit instructions to the tape node **510** to alter its state. In further embodiments, the processor **512** of the tape nodes **510** does not execute any of the intelligent software logic and only alters the state of the tape node **510** in response to commands received from the central database and control system **520** or the client devices **530**.

In some embodiments, the logic relevant to the behavior of the tape nodes **510** may be stored on the memory **532** of the client device **530** and executed by the processor **531**. For example, the processor **531** may execute the logic in response to receiving sensing data from one of the tape nodes **510**, resulting in the client device **530** transmitting one or more instruction to the tape node **510** to alter its state.

Hierarchical Wireless Communications Network

FIG. 7 shows an example hierarchical wireless communications network of tape nodes **470**. In this example, the short range tape node **472** and the medium range tape node **474** communicate with one another over their respective low power wireless communication interfaces **476**, **478**. The medium range tape node **474** and the long range tape node **480** communicate with one another over their respective medium power wireless communication interfaces **478**, **482**. The long range tape node **480** and the network server **404** communicate with one another over the high power wireless communication interface **484**. In some examples, the low power communication interfaces **476**, **478** establish wireless communications with one another in accordance with the Bluetooth LE protocol, the medium power communication interfaces **452**, **482** establish wireless communications with one another in accordance with the LoRa communications protocol, and the high power communication interface **484** establishes wireless communications with the server **404** in accordance with a cellular communications protocol.

In some examples, the different types of tape nodes are deployed at different levels in the communications hierarchy according to their respective communications ranges, with the long range tape nodes generally at the top of the hierarchy, the medium range tape nodes generally in the middle of the hierarchy, and the short range tape nodes generally at the bottom of the hierarchy. In some examples, the different types of tape nodes are implemented with different feature sets that are associated with component costs and operational costs that vary according to their respective levels in the hierarchy. This allows system administrators flexibility to optimize the deployment of the tape nodes to achieve various objectives, including cost minimization, asset tracking, asset localization, and power conservation.

In some examples, a server **404** of the network service **408** designates a tape node at a higher level in a hierarchical communications network as a master node of a designated

set of tape nodes at a lower level in the hierarchical communications network. For example, the designated master tape node may be adhered to a parcel (e.g., a box, pallet, or shipping container) that contains one or more tape nodes that are adhered to one or more packages containing respective assets. In order to conserve power, the tape nodes typically communicate according to a schedule promulgated by the server **404** of the network service **408**. The schedule usually dictates all aspects of the communication, including the times when particular tape nodes should communicate, the mode of communication, and the contents of the communication. In one example, the server **404** transmits programmatic Global Scheduling Description Language (GSDL) code to the master tape node and each of the lower-level tape nodes in the designated set. In this example, execution of the GSDL code causes each of the tape nodes in the designated set to connect to the master tape node at a different respective time that is specified in the GSDL code, and to communicate a respective set of one or more data packets of one or more specified types of information over the respective connection. In some examples, the master tape node simply forwards the data packets to the server network node **404**, either directly or indirectly through a gateway tape node (e.g., the long range tape node **416** adhered to the mobile vehicle **412** or the long range tape node **414** adhered to an infrastructure component of the environment **400**). In other examples, the master tape node processes the information contained in the received data packets and transmits the processed information to the server network node **404**.

FIG. 8 shows an example method of creating a hierarchical communications network. In accordance with this method, a first tape node is adhered to a first asset in a set of associated assets, the first tape node including a first type of wireless communication interface and a second type of wireless communication interface having a longer range than the first type of wireless communication interface (FIG. 8, block **490**). A second tape node is adhered to a second asset in the set, the second tape node including the first type of wireless communication interface, wherein the second tape node is operable to communicate with the first tape node over a wireless communication connection established between the first type of wireless communication interfaces of the first and second tape nodes (FIG. 8, block **492**). An application executing on a computer system (e.g., a server **404** of a network service **408**) establishes a wireless communication connection with the second type of wireless communication interface of the first tape node, and the application transmits programmatic code executable by the first tape node to function as a master tape node with respect to the second tape node (FIG. 8, block **494**).

In some embodiments, the second tape node is assigned the role of the master tape node with respect to the first tape node.

Distributed Agent Operating System

As used herein, the term “node” refers to both a tape node and a non-tape node unless the node is explicitly designated as a “tape node” or a “non-tape node.” In some embodiments, a non-tape node may have the same or similar communication, sensing, processing and other functionalities and capabilities as the tape nodes described herein, except without being integrated into a tape platform. In some embodiments, non-tape nodes can interact seamlessly with tape nodes. Each node is assigned a respective unique identifier.

The following disclosure describes a distributed software operating system that is implemented by distributed hard-

ware nodes executing intelligent agent software to perform various tasks or algorithms. In some embodiments, the operating system distributes functionalities (e.g., performing analytics on data or statistics collected or generated by nodes) geographically across multiple intelligent agents that are bound to items (e.g., parcels, containers, packages, boxes, pallets, a loading dock, a door, a light switch, a vehicle such as a delivery truck, a shipping facility, a port, a hub, etc.). In addition, the operating system dynamically allocates the hierarchical roles (e.g., master and slave roles) that nodes perform over time in order to improve system performance, such as optimizing battery life across nodes, improving responsiveness, and achieving overall objectives. In some embodiments, optimization is achieved using a simulation environment for optimizing key performance indicators (PKIs).

In some embodiments, the nodes are programmed to operate individually or collectively as autonomous intelligent agents. In some embodiments, nodes are configured to communicate and coordinate actions and respond to events. In some embodiments, a node is characterized by its identity, its mission, and the services that it can provide to other nodes. A node's identity is defined by its capabilities (e.g., battery life, sensing capabilities, and communications interfaces). A node's mission (or objective) is defined by the respective program code, instructions, or directives it receives from another node (e.g., a server or a master node) and the actions or tasks that it performs in accordance with that program code, instructions, or directives (e.g., sense temperature every hour and send temperature data to a master node to upload to a server). A node's services define the functions or tasks that it is permitted to perform for other nodes (e.g., retrieve temperature data from a peripheral node and send the received temperature data to the server). At least for certain tasks, once programmed and configured with their identities, missions, and services, nodes can communicate with one another and request services from and provide services to one another independently of the server.

Thus, in accordance with the runtime operating system every agent knows its objectives (programmed). Every agent knows which capabilities/resources it needs to fulfill objective. Every agent communicates with every other node in proximity to see if it can offer the capability. Examples include communicate data to the server, authorize going to lower power level, temperature reading, send an alert to local hub, send location data, triangulate location, any boxes in same group that already completed group objectives.

Nodes can be associated with items. Examples of an item includes, but are not limited to for example, a package, a box, pallet, a container, a truck or other conveyance, infrastructure such as a door, a conveyor belt, a light switch, a road, or any other thing that can be tracked, monitored, sensed, etc. or that can transmit data concerning its state or environment. In some examples, a server or a master node may associate the unique node identifiers with the items.

Communication paths between tape and/or non-tape nodes may be represented by a graph of edges between the corresponding assets (e.g., a storage unit, truck, or hub). In some embodiments, each node in the graph has a unique identifier. A set of connected edges between nodes is represented by a sequence of the node identifiers that defines a communication path between a set of nodes.

Referring to FIG. 9A, a node **520** (Node A) is associated with a package **522** (Package A). In some embodiments, the node **520** may be implemented as a tape node that is used to seal the package **522** or it may be implemented as a label

node that is used to label the package **522**; alternatively, the node **520** may be implemented as a non-tape node that is inserted within the package **522** or embedded in or otherwise attached to the interior or exterior of the package **522**. In the illustrated embodiment, the node **520** includes a low power communications interface **524** (e.g., a Bluetooth Low Energy communications interface). Another node **526** (Node B), which is associated with another package **530** (Package B), is similarly equipped with a compatible low power communications interface **528** (e.g., a Bluetooth Low Energy communications interface).

In an example scenario, in accordance with the programmatic code stored in its memory, node **526** (Node B) requires a connection to node **520** (Node A) to perform a task that involves checking the battery life of Node A. Initially, Node B is unconnected to any other nodes. In accordance with the programmatic code stored in its memory, Node B periodically broadcasts advertising packets into the surrounding area. When the other node **520** (Node A) is within range of Node B and is operating in a listening mode, Node A will extract the address of Node B and potentially other information (e.g., security information) from an advertising packet. If, according to its programmatic code, Node A determines that it is authorized to connect to Node B, Node A will attempt to pair with Node B. In this process, Node A and Node B determine each other's identities, capabilities, and services. For example, after successfully establishing a communication path **532** with Node A (e.g., a Bluetooth Low Energy formatted communication path), Node B determines Node A's identity information (e.g., master node), Node A's capabilities include reporting its current battery life, and Node A's services include transmitting its current battery life to other nodes. In response to a request from Node B, Node A transmits an indication of its current battery life to Node B.

Referring to FIG. 9B, a node **534** (Node C) is associated with a package **535** (Package C). In the illustrated embodiment, the Node C includes a low power communications interface **536** (e.g., a Bluetooth Low Energy communications interface), and a sensor **537** (e.g., a temperature sensor). Another node **538** (Node D), which is associated with another package **540** (Package D), is similarly equipped with a compatible low power communications interface **542** (e.g., a Bluetooth Low Energy communications interface).

In an example scenario, in accordance with the programmatic code stored in its memory, Node D requires a connection to Node C to perform a task that involves checking the temperature in the vicinity of Node C. Initially, Node D is unconnected to any other nodes. In accordance with the programmatic code stored in its memory, Node D periodically broadcasts advertising packets in the surrounding area. When Node C is within range of Node D and is operating in a listening mode, Node C will extract the address of Node D and potentially other information (e.g., security information) from the advertising packet. If, according to its programmatic code, Node C determines that it is authorized to connect to Node D, Node C will attempt to pair with Node D. In this process, Node C and Node D determine each other's identities, capabilities, and services. For example, after successfully establishing a communication path **544** with Node C (e.g., a Bluetooth Low Energy formatted communication path), Node D determines Node C's identity information (e.g., a peripheral node), Node C's capabilities include retrieving temperature data, and Node C's services include transmitting temperature data to other nodes. In

response to a request from Node D, Node C transmits its measured and/or locally processed temperature data to Node D.

Referring to FIG. 9C, a pallet **550** is associated with a master node **551** that includes a low power communications interface **552**, a GPS receiver **554**, and a cellular communications interface **556**. In some embodiments, the master node **551** may be implemented as a tape node or a label node that is adhered to the pallet **550**. In other embodiments, the master node **551** may be implemented as a non-tape node that is inserted within the body of the pallet **550** or embedded in or otherwise attached to the interior or exterior of the pallet **550**.

The pallet **550** provides a structure for grouping and containing packages **559**, **561**, **563** each of which is associated with a respective peripheral node **558**, **560**, **562** (Node E, Node F, and Node G). Each of the peripheral nodes **558**, **560**, **562** includes a respective low power communications interface **564**, **566**, **568** (e.g., Bluetooth Low Energy communications interface). In the illustrated embodiment, each of the nodes E, F, G and the master node **551** are connected to each of the other nodes over a respective low power communications path (shown by dashed lines).

In some embodiments, the packages **559**, **561**, **563** are grouped together because they are related. For example, the packages **559**, **561**, **563** may share the same shipping itinerary or a portion thereof. In an example scenario, the master pallet node **550** scans for advertising packets that are broadcasted from the peripheral nodes **558**, **560**, **562**. In some examples, the peripheral nodes broadcast advertising packets during respective scheduled broadcast intervals. The master node **551** can determine the presence of the packages **559**, **561**, **563** in the vicinity of the pallet **550** based on receipt of one or more advertising packets from each of the nodes E, F, and G. In some embodiments, in response to receipt of advertising packets broadcasted by the peripheral nodes **558**, **560**, **562**, the master node **551** transmits respective requests to the server to associate the master node **551** and the respective peripheral nodes **558**, **560**, **562**. In some examples, the master tape node requests authorization from the server to associate the master tape node and the peripheral tape nodes. If the corresponding packages **559**, **561**, **563** are intended to be grouped together (e.g., they share the same itinerary or certain segments of the same itinerary), the server authorizes the master node **551** to associate the peripheral nodes **558**, **560**, **562** with one another as a grouped set of packages. In some embodiments, the server registers the master node and peripheral tape node identifiers with a group identifier. The server also may associate each node ID with a respective physical label ID that is affixed to the respective package.

In some embodiments, after an initial set of packages is assigned to a multi-package group, the master node **551** may identify another package arrives in the vicinity of the multi-package group. The master node may request authorization from the server to associate the other package with the existing multi-package group. If the server determines that the other package is intended to ship with the multi-package group, the server instructs the master node to merge one or more other packages with currently grouped set of packages. After all packages are grouped together, the server authorizes the multi-package group to ship. In some embodiments, this process may involve releasing the multi-package group from a containment area (e.g., customs holding area) in a shipment facility.

In some embodiments, the peripheral nodes **558**, **560**, **562** include environmental sensors for obtaining information

regarding environmental conditions in the vicinity of the associated packages **559**, **561**, **563**. Examples of such environmental sensors include temperature sensors, humidity sensors, acceleration sensors, vibration sensors, shock sensors, pressure sensors, altitude sensors, light sensors, and orientation sensors.

In the illustrated embodiment, the master node **551** can determine its own location based on geolocation data transmitted by a satellite-based radio navigation system **570** (e.g., GPS, GLONASS, and NAVSTAR) and received by the GPS receiver **554** component of the master node **551**. In an alternative embodiment, the location of the master pallet node **551** can be determined using cellular based navigation techniques that use mobile communication technologies (e.g., GSM, GPRS, CDMA, etc.) to implement one or more cell-based localization techniques. After the master node **551** has ascertained its location, the distance of each of the packages **559**, **561**, **563** from the master node **551** can be estimated based on the average signal strength of the advertising packets that the master node **551** receives from the respective peripheral node. The master node **551** can then transmit its own location and the locations of the package nodes E, F, and G to a server over a cellular interface connection with a cell tower **572**. Other methods of determining the distance of each of the packages **559**, **561**, **563** from the master node **551**, such as Received Signal-Strength Index (RSSI) based indoor localization techniques, also may be used.

In some embodiments, after determining its own location and the locations of the peripheral nodes, the master node **551** reports the location data and the collected and optionally processed (e.g., either by the peripheral nodes peripheral nodes **558**, **560**, **562** or the master node **551**) sensor data to a server over a cellular communication path **571** on a cellular network **572**.

In some examples, nodes are able to autonomously detect logistics execution errors if packages that are supposed to travel together no longer travel together, and raise an alert. For example, a node (e.g., the master node **551** or one of the peripheral nodes **558**, **560**, **562**) alerts the server when the node determines that a particular package **559** is being or has already been improperly separated from the group of packages. The node may determine that there has been an improper separation of the particular package **559** in a variety of ways. For example, the associated node **558** that is bound to the particular package **559** may include an accelerometer that generates a signal in response to movement of the package from the pallet. In accordance with its intelligent agent program code, the associated node **558** determines that the master node **551** has not disassociated the particular package **559** from the group and therefore broadcasts advertising packets to the master node, which causes the master node **551** to monitor the average signal strength of the advertising packets and, if the master node **551** determines that the signal strength is decreasing over time, the master node **551** will issue an alert either locally (e.g., through a speaker component of the master node **551**) or to the server.

Referring to FIG. 9D, a truck **580** is configured as a mobile node or mobile hub that includes a cellular communications interface **582**, a medium power communications interface **584**, and a low power communications interface **586**. The communications interfaces **580-586** may be implemented on one or more tape and non-tape nodes. In an illustrative scenario, the truck **580** visits a storage facility, such as a warehouse **588**, to wirelessly obtain temperature data generated by temperature sensors in the medium range

nodes **590, 592, 594**. The warehouse **588** contains nodes **590, 592**, and **594** that are associated with respective packages **591, 593, 595**. In the illustrated embodiment, each node **590-594** is a medium range node that includes a respective medium power communications interface **596, 602, 608**, a respective low power communications interface **598, 604, 610** and one or more respective sensors **600, 606, 612**. In the illustrated embodiment, each of the package nodes **590, 592, 594** and the truck **580** is connected to each of the other ones of the package nodes through a respective medium power communications path (shown by dashed lines). In some embodiments, the medium power communications paths are LoRa formatted communication paths.

In some embodiments, the communications interfaces **584** and **586** (e.g., a LoRa communications interface and a Bluetooth Low Energy communications interface) on the node on the truck **580** is programmed to broadcast advertisement packets to establish connections with other network nodes within range of the truck node. A warehouse **588** includes medium range nodes **590, 592, 594** that are associated with respective containers **591, 593, 595** (e.g., packages, boxes, pallets, and the like). When the truck node's low power interface **586** is within range of any of the medium range nodes **590, 592, 594** and one or more of the medium range nodes is operating in a listening mode, the medium range node will extract the address of truck node and potentially other information (e.g., security information) from the advertising packet. If, according to its programmatic code, the truck node determines that it is authorized to connect to one of the medium range nodes **590, 592, 594**, the truck node will attempt to pair with the medium range node. In this process, the truck node and the medium range node determine each other's identities, capabilities, and services. For example, after successfully establishing a communication path with the truck node (e.g., a Bluetooth Low Energy formatted communication path **614** or a LoRa formatted communication path **615**), the truck node determines the identity information for the medium range node **590** (e.g., a peripheral node), the medium range node's capabilities include retrieving temperature data, and the medium range node's services include transmitting temperature data to other nodes. Depending of the size of the warehouse **588**, the truck **580** initially may communicate with the nodes **590, 592, 594** using a low power communications interface (e.g., Bluetooth Low Energy interface). If any of the anticipated nodes fails to respond to repeated broadcasts of advertising packets by the truck **580**, the truck **580** will try to communicate with the non-responsive nodes using a medium power communications interface (e.g., LoRa interface). In response to a request from the truck node **584**, the medium range node **590** transmits an indication of its measured temperature data to the truck node. The truck node repeats the process for each of the other medium range nodes **592, 594** that generate temperature measurement data in the warehouse **588**. The truck node reports the collected (and optionally processed, either by the medium range nodes **590, 592, 594** or the truck node) temperature data to a server over a cellular communication path **616** with a cellular network **618**.

Referring to FIG. 9E, a master node **630** is associated with an item **632** (e.g., a package) and grouped together with other items **634, 636** (e.g., packages) that are associated with respective peripheral nodes **638, 640**. The master node **630** includes a GPS receiver **642**, a medium power communications interface **644**, one or more sensors **646**, and a cellular communications interface **648**. Each of the peripheral nodes **638, 640** includes a respective medium power communications interface **650, 652** and one or more respective sensors

654, 656. In the illustrated embodiment, the peripheral and master nodes are connected to one another over respective pairwise communications paths (shown by dashed lines). In some embodiments, the nodes **630, 638, 640** communicate through respective LoRa communications interfaces over LoRa formatted communications paths **658, 660, 662**.

In the illustrated embodiment, the master and peripheral nodes **638, 638, 640** include environmental sensors for obtaining information regarding environmental conditions in the vicinity of the associated packages **632, 634, 636**. Examples of such environmental sensors include temperature sensors, humidity sensors, acceleration sensors, vibration sensors, shock sensors, pressure sensors, altitude sensors, light sensors, and orientation sensors.

In accordance with the programmatic code stored in its memory, the master node **630** periodically broadcasts advertising packets in the surrounding area. When the peripheral nodes **638, 640** are within range of master node **630**, and are operating in a listening mode, the peripheral nodes **638, 640** will extract the address of master node **630** and potentially other information (e.g., security information) from the advertising packets. If, according to their respective programmatic code, the peripheral nodes **638, 640** determine that they are authorized to connect to the master node **630**, the peripheral nodes **638, 640** will attempt to pair with the master node **630**. In this process, the peripheral nodes **638, 640** and the master node and the peripheral nodes determine each other's identities, capabilities, and services. For example, after successfully establishing a respective communication path **658, 660** with each of the peripheral nodes **638, 640** (e.g., a LoRa formatted communication path), the master node **630** determines certain information about the peripheral nodes **638, 640**, such as their identity information (e.g., peripheral nodes), their capabilities (e.g., measuring temperature data), and their services include transmitting temperature data to other nodes.

After establishing LoRa formatted communications paths **658, 660** with the peripheral nodes **638, 640**, the master node **630** transmits requests for the peripheral nodes **638, 640** to transmit their measured and/or locally processed temperature data to the master node **630**.

In the illustrated embodiment, the master node **630** can determine its own location based on geolocation data transmitted by a satellite-based radio navigation system **666** (e.g., GPS, GLONASS, and NAVSTAR) and received by the GPS receiver **642** component of the master node **630**. In an alternative embodiment, the location of the master node **630** can be determined using cellular based navigation techniques that use mobile communication technologies (e.g., GSM, GPRS, CDMA, etc.) to implement one or more cell-based localization techniques. After the master node **630** has ascertained its location, the distance of each of the packages **634, 636** from the master node **630** can be estimated based on the average signal strength of the advertising packets that the master node **630** receives from the respective peripheral node. The master node **630** can then transmit its own location and the locations of the package nodes E, F, and G to a server over a cellular interface connection with a cell tower **672**. Other methods of determining the distance of each of the packages **634, 636** from the master node **630**, such as Received Signal-Strength Index (RSSI) based indoor localization techniques, also may be used.

In some embodiments, after determining its own location and the locations of the peripheral nodes, the master node **630** reports the location data the collected and optionally processed (e.g., either by the peripheral nodes peripheral

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nodes 634, 636 or the master node 630) sensor data to a server over a cellular communication path 670 on a cellular network 672.

Referring to FIG. 10A, in some examples, each of one or more of the segments 270, 272 of a tracking adhesive product 274 includes a respective circuit 275 that delivers power from the respective energy source 276 to the respective tracking circuit 278 (e.g., a processor and one or more wireless communications circuits) in response to an event. In some of these examples, the wake circuit 275 is configured to transition from an off state to an on state when the voltage on the wake node 277 exceeds a threshold level, at which point the wake circuit transitions to an on state to power-on the segment 270. In the illustrated example, this occurs when the user separates the segment from the tracking adhesive product 274, for example, by cutting across the tracking adhesive product 274 at a designated location (e.g., along a designated cut-line 280). In particular, in its initial, un-cut state, a minimal amount of current flows through the resistors R1 and R2. As a result, the voltage on the wake node 270 remains below the threshold turn-on level. After the user cuts across the tracking adhesive product 274 along the designated cut-line 280, the user creates an open circuit in the loop 282, which pulls the voltage of the wake node above the threshold level and turns on the wake circuit 275. As a result, the voltage across the energy source 276 will appear across the tracking circuit 278 and, thereby, turn on the segment 270. In particular embodiments, the resistance value of resistor R1 is greater than the resistance value of R2. In some examples, the resistance values of resistors R1 and R2 are selected based on the overall design of the adhesive product system (e.g., the target wake voltage level and a target leakage current).

In some examples, each of one or more of the segments of a tracking adhesive product includes a respective sensor and a respective wake circuit that delivers power from the respective energy source to the respective one or more of the respective tracking components 278 in response to an output of the sensor. In some examples, the respective sensor is a strain sensor that produces a wake signal based on a change in strain in the respective segment. In some of these examples, the strain sensor is affixed to a tracking adhesive product and configured to detect the stretching of the tracking adhesive product segment as the segment is being peeled off a roll or a sheet of the tracking adhesive product. In some examples, the respective sensor is a capacitive sensor that produces a wake signal based on a change in capacitance in the respective segment. In some of these examples, the capacitive sensor is affixed to a tracking adhesive product and configured to detect the separation of the tracking adhesive product segment from a roll or a sheet of the tracking adhesive product. In some examples, the respective sensor is a flex sensor that produces a wake signal based on a change in curvature in the respective segment. In some of these examples, the flex sensor is affixed to a tracking adhesive product and configured to detect bending of the tracking adhesive product segment as the segment is being peeled off a roll or a sheet of the tracking adhesive product. In some examples, the respective sensor is a near field communications sensor that produces a wake signal based on a change in inductance in the respective segment.

FIG. 10B shows another example of a tracking adhesive product 294 that delivers power from the respective energy source 276 to the respective tracking circuit 278 (e.g., a processor and one or more wireless communications circuits) in response to an event. This example is similar in structure and operation as the tracking adhesive product 294

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shown in FIG. 10A, except that the wake circuit 275 is replaced by a switch 296 that is configured to transition from an open state to a closed state when the voltage on the switch node 277 exceeds a threshold level. In the initial state of the tracking adhesive product 294, the voltage on the switch node is below the threshold level as a result of the low current level flowing through the resistors R1 and R2. After the user cuts across the tracking adhesive product 294 along the designated cut-line 280, the user creates an open circuit in the loop 282, which pulls up the voltage on the switch node above the threshold level to close the switch 296 and turn on the tracking circuit 278.

FIG. 10C shows a diagrammatic cross-sectional front view of an example adhesive tape platform 300 and a perspective view of an example asset 302. Instead of activating the adhesive tape platform in response to separating a segment of the adhesive tape platform from a roll or a sheet of the adhesive tape platform, this example is configured to supply power from the energy source 302 to turn on the wireless transducing circuit 306 in response to establishing an electrical connection between two power terminals 308, 310 that are integrated into the adhesive tape platform. In particular, each segment of the adhesive tape platform 300 includes a respective set of embedded tracking components, an adhesive layer 312, and an optional backing sheet 314 with a release coating that prevents the segments from adhering strongly to the backing sheet 314. In some examples, the power terminals 308, 310 are composed of an electrically conductive material (e.g., a metal, such as copper) that may be printed or otherwise patterned and/or deposited on the backside of the adhesive tape platform 300. In operation, the adhesive tape platform can be activated by removing the backing sheet 314 and applying the exposed adhesive layer 312 to a surface that includes an electrically conductive region 316. In the illustrated embodiment, the electrically conductive region 316 is disposed on a portion of the asset 302. When the adhesive backside of the adhesive tape platform 300 is adhered to the asset with the exposed terminals 308, 310 aligned and in contact with the electrically conductive region 316 on the asset 302, an electrical connection is created through the electrically conductive region 316 between the exposed terminals 308, 310 that completes the circuit and turns on the wireless transducing circuit 306. In particular embodiments, the power terminals 308, 310 are electrically connected to any respective nodes of the wireless transducing circuit 306 that would result in the activation of the tracking circuit 306 in response to the creation of an electrical connection between the power terminals 308, 310.

In some examples, after a tape node is turned on, it will communicate with the network service to confirm that the user/operator who is associated with the tape node is an authorized user who has authenticated himself or herself to the network service 54. In these examples, if the tape node cannot confirm that the user/operator is an authorized user, the tape node will turn itself off.

Monitoring Equipment with Wireless Sensor Nodes

Wireless sensor nodes 510, each wireless sensor node 510 including a sensor, are deployed in the sensing system 500 to monitor various components of equipment and objects of interest, according to some embodiments. Some of the wireless sensor nodes 510 may be directly attached or adhered to components of equipment and objects of interest in order to measure properties of the components of equipment and objects of interest.

FIG. 11 shows an example environment 1100 for a fleet of wireless sensor nodes 1105 and wireless nodes 1110A,

1110B, 1115 of the sensing system 500 for generating sensing data on a section of equipment and communicating the sensing data with the sensing system 500, according to some embodiments. The wireless sensor nodes 1105 includes the wireless sensor nodes 1105A-1105H that are each attached to a different component of an equipment 1120. Each of the wireless sensor nodes 1110 includes a sensor incorporated into the wireless sensor node that generates sensing data on the respective component of the equipment 1120. In the example of FIG. 11, the equipment 1120 is a pumping system for moving liquid in a facility. The equipment 1120 includes a first unit 1120A and a second unit 1120B which are attached to each other.

The wireless sensor node 1105A measures a position of a valve of the pumping system 1120 using an accelerometer or position sensor to measure when the valve moves. The wireless sensor node 1105B is attached to a rotational component of the pumping system 1120 internal to the pumping system 1120 and measures the speed of rotation of the rotational component using an accelerometer. The wireless sensor node 1105C includes a temperature sensor and measures the temperature of the pumping system 1120. The wireless sensor node 1105D includes a flow sensor and detects the flow of liquid through a pipe using the flow sensor. The wireless sensor node 1105E includes a vibration sensor and measure vibrations of the first unit 1120A of the pumping system 1120. The wireless sensor node 1105F includes a strain sensor and measures the amount of axle strain that an axle of the pumping system 1120 is experiencing. The wireless sensor node 1105G includes a vibration sensor and measures vibration of the second unit 1120B of the pumping system 1120. The wireless sensor node 1105H includes a sensor for measuring the oil level in an oil tank of the pumping system 1120. The wireless sensing node 1105I includes an electrical current sensor (e.g., a hall effect sensor) for measuring the electrical current flowing through a plug or wire of the pumping system 1120.

The sensing data of each of the wireless sensor nodes 1105 may be used in combination or separately to detect whether or not the equipment 1120 is functioning properly. The wireless sensor nodes 1105 may generate sensing data to determine and learn baseline data when the equipment 1120 is functioning normally. Sensing data detected by the wireless sensor nodes 1105 that deviates from the baseline may signify a malfunctioning of the equipment 1120. The sensing data may be used in combination to detect specific types of malfunctioning or to confirm that the equipment is indeed malfunctioning. In cases where one of the wireless sensor nodes 1105 is generating data that is deviating from the baseline, the other wireless sensor nodes 1105 can be used to confirm whether the equipment 1120 is functioning. For example, the wireless sensor node 1105B may detect a change in the rotational speed of the rotational component of the pumping system 1120. This may signify that the pumping system 1120 is malfunctioning. To confirm whether this is the case, the sensing system 500 compares the sensing data from other wireless nodes 1105. For example, the sensing system 500 may check the wireless sensor node 1105D to see if there is any detected change in liquid flow within a threshold amount of time from the detected change in rotational speed. If not, the sensing system 500 may determine that the data from the wireless sensing node 1105B is an outlier and that the equipment 1120 is still functioning properly. In further embodiments, the sensing system 500 may also determine that the wireless sensor node 1105D is malfunctioning and requires inspection or replacement. The wireless sensing nodes 1105 wirelessly transmit

sensing data to a respective unit controller 1110 via a wireless communication interface. The wireless sensing nodes 1105 may selectively transmit sensing data, in response to detecting an event, according to some embodiments. For example, the wireless sensing nodes 1105 may only transmit sensing data to a respective unit controller 1110 when the detected sensing data deviates beyond a threshold amount from the baseline data. Each of the wireless sensing nodes 1105 is associated with a unit controller that is located within a close distance from the respective wireless sensing nodes 1105. In the example of FIG. 11, the wireless sensor nodes 1105A-1105F are associated with and communicate with the unit controller 1110A, and the wireless sensor nodes 1105G-1105I are associated with and communicate with the unit controller 1110B.

The unit controllers 1110 (including unit controller 1110A and 1110B) are wireless nodes attached to and associated with respective units of the equipment 1120 and are configured to provide intelligence (e.g., decision making on when to transmit information to other wireless nodes) for making determinations on whether relevant events have occurred based on sensing data received from the wireless sensor nodes 1105. In case of detecting certain events based on the received sensing data, the unit controllers 1110 may communicate selected sensing data and the detected events to the section controller 1115 which is associated with the entirety of the equipment 1120. The wireless communication channel between the wireless sensing nodes 1105 and the unit controllers 1110 may be a low latency communication channel, according to some embodiments. In further embodiments, the low latency communication channel may be a short distance communication channel, such as the ones described with respect to the low power communication interface 476 in FIG. 7. This provides the ability for a unit controller to synchronously capture sensing data from multiple wireless sensing nodes and detect events in real time based on sensing data from multiple wireless sensing nodes. This is important for capturing events that occur at multiple wireless sensing nodes relatively at the same time. Detecting events occurring within a threshold amount of time from one another allows for accurate reporting of equipment failure or events that may reveal a need for maintenance on the equipment 1120. For example, the threshold amount of time may be 10 ms. If a first anomaly is detected in the sensing data of a first wireless sensing node and a second anomaly is detected in the sensing data of a second wireless sensing node within the threshold amount of time from the first event, the respective unit controller 1110 that receives the sensing data from both the first and the second sensing node may determine that an equipment failure is likely. In response, the respective unit controller 1110 may transmit a report including parts of the sensing data to the section controller 1115. Using conventional systems, where the intelligence for determining events occurs at a server or an edge node requiring longer distance higher latency communication channels, it may be difficult to detect events across multiple wireless sensor nodes with high enough resolution in time to accurately determine equipment failure or other conditions of the equipment 1120. By using low latency communications between the wireless sensor nodes 1105 and the unit controllers 1110 which are placed within a short distance (e.g., less than 10 feet) from the wireless sensor nodes, events and conditions relevant to the equipment 1120 may be detected with higher accuracy.

The section controller 1115 is a wireless node associated with the equipment 1120 and configured to receive communications from the unit controllers 1110. In some embodiments,

ments, the section controller is also configured to receive communications directly from one or more of the wireless sensing nodes **1105**. The section controller **1115** may provide further intelligence (e.g., decision making on when to transmit information to the sensing system **500**) and may transmit sensing data to the sensing system **500** in response to receiving data from one or more of the unit controllers **1110** and detecting certain events based on the received data. In some embodiments, the wireless sensor nodes **1105** are embodiments of the short range tape node **472**, the unit controllers **1110** are embodiments of the medium range tape node **474**, and the section controller **1115** is an embodiment of the long range tape node **480**. In this case, the sensing data may be relayed from the wireless sensor nodes **1105** to servers or other nodes of the sensing system **500** (not shown in FIG. **11**) via the section controller **1115**. The section controller **1115** may issue instructions to the unit controller, according to some embodiments. The instructions may be relayed from the server, client devices, or other nodes of the sensing system **500**. Each unit controller **1110** may also issue instructions to respective wireless sensor nodes that are associated with the unit controller **1110**.

FIG. **11** only shows one section of equipment **1120** in a facility. The facility may include multiple instances of the section of equipment **1120** including their own respective wireless sensor nodes **1105**, unit controllers **1110**, and section controllers **1115**. The sensing system allows for simultaneous monitoring of each section of equipment **1120** in the facility. By distributing parts of the computation and intelligence for detecting events based off of the sensing data generated by the wireless sensing nodes **1105** at the wireless sensing nodes **1105** and the wireless nodes **1110** and **1115**, the sensing system **500** creates a mesh network of nodes that can detect events with high accuracy and low latency. Distributing the intelligence also aids in conserving battery life for the wireless sensor nodes **1105** and the wireless nodes **1110** and **1115**. Compared to a system that continuously transmits data from its nodes, the wireless sensor nodes are configured to selectively transmit data in response to detecting certain events and anomalies based on the generated sensing data. By reducing the amount of transmissions, energy consumption and congestion of the wireless communication airwaves is reduced.

Distributed Intelligent Software

The distributed intelligent software may include rules, protocols, logic, and/or instructions for one or more of the nodes (including wireless sensor nodes and wireless nodes), the central database and control system, and the client devices in various scenarios. The distributed intelligent software instructs the wireless sensor nodes to enter different states based on the rules and based on the sensing data collected by the wireless sensor nodes.

The states may include, but are not limited to the following examples: a low power mode where the tape node operates with minimal power consumption; a low communication mode where the tape node limits the amount of transmitted/received data and/or frequency of transmitting and receiving data; a high communication mode where the tape node increases the amount of transmitted/received data and/or frequency of transmitting and receiving data; a sensing mode in which sensors included in the tape node collect sensor data, and the sensor data is transmitted to members of the sensing system **500**; a no-sensing mode in which sensors included in the tape node are deactivated and do not collect sensor data; a low sensing mode which limits the amount of sensor data collected and transmitted (in some embodiments, this includes decreasing the sampling frequency of

the sensors and frequency of transmitting the sensor data); a high sensing mode which increases the amount of sensor data collected and transmitted. (in some embodiments, this includes increasing the sampling frequency of the sensors and frequency of transmitting the sensor data); a sensor configuration mode where a configuration or property of a sensor in the tape node is changed; a sensor activation mode where a specific set of sensors in the tape node are activated (e.g., if a tape node has an acoustic sensor, an accelerometer, and an optical sensor, activating the operation of the accelerometer and the acoustic sensor (e.g., in response to the optical sensor detecting an above threshold value)); a search mode where the tape node searches for a client device in proximity of the tape node to communicate with, a heartbeat mode where the tape node intermittently transmits a signal to the central database and control system to indicate normal functionality of the tape node; an alert mode where the tape node transmits an alert to the central database and control system, a client device of a delivery employee (handler), a client device of a customer, a client device of a final recipient, a client device of an administrator, or some combination thereof; a data processing mode where the tape node calculates values (RMS values, peak values, spectrum analysis, fast Fourier transform (FFT) of data, peak frequency, a time stamp, a relative time a value is reached for a measurement, or other calculated values) based on collected sensing data and only transmits the calculated values a spectral band mode where the tape node collects measurements (e.g., vibration data) and/or calculates values in the form of a spectrum (e.g., a frequency spectrum) but only transmits a portion of the spectrum (e.g., data in a frequency band that is smaller than the full range of frequency-domain data that is collected); a full spectrum mode where the tape node collects measurements (e.g., vibration data) and/or calculates values in the form of a spectrum (e.g., a range of frequencies, a range of times, etc. . . .) and transmits the entire spectrum; a full data mode where the tape node transmits all the sensing data that it has collected; a data history mode where the tape node transmits historical sensing data that it has stored in the memory of the tape node; a high fidelity location mode which increases the resolution and accuracy of location data that is collected and transmitted to the central database and control system (in some embodiments, this includes increasing the sampling frequency of location data and/or the frequency of transmitting the location data, and in other embodiments, this includes activating a GPS module on the tape node and collecting GPS-based location data); a low fidelity location mode which reduces the resolution and accuracy of location data that is collected and transmitted to the central database and control system (in some embodiments, this includes decreasing the sampling frequency of location data and/or the frequency of transmitting the location data and in other embodiments, this includes deactivating a GPS module on the tape node and omitting GPS data in the sensing data, while the tape node is in this mode); and an airplane mode where some of the wireless communication is deactivated based on air travel regulation. The states that the tape node can enter may include additional and/or alternate states not listed above. The tape node may be in multiple states simultaneously, according to some embodiments. For example, the tape node may be in both a high sensing mode and a high communication mode, as described above.

FIG. **12** is a flow chart depicting steps implemented, via execution by one or more processors of the adhesive tape platform, central database and control system, client device, or any combination thereof, by the distributed intelligent

software, according to some embodiments. The steps **1200** include the adhesive tape platform initiating **1210** in an initial state. The initial **1200** state may be any of the states described above or it may be another state. For example, the adhesive tape platform may initiate in the low communication state. The adhesive tape platform then aggregates and transmits **1220** sensing data according to the protocols of the initial state. The adhesive tape platform may transmit some or all of the sensing data to the central database and control system and/or client devices, based on the protocols of the initial state. Relevant data aggregated by the adhesive tape platform during the aggregation and transmitting step **1220** are stored **1230**. The adhesive tape platform may store **1230** the relevant data in its own memory, according to some embodiments. The central database and or control system may store **1230** the relevant data in its database if aggregated data was received from the adhesive tape platform in step **1220**, according to some embodiments. Similarly, a client device may store **1230** the aggregated data in its memory if aggregated data was received from the adhesive tape platform in step **1220** and/or if data is transmitted from the central database and control system.

Based on the relevant data and based on the logic defined in the distributed intelligent software, an event is detected **1240**. The event indicates that the relevant data satisfies one or more of the rules and/or conditions included in the distributed intelligent software. For example, if the relevant data includes sensor data that vibrations on a tracked item exceed a high threshold, “a high vibration” event may be detected. If no event is detected, the process repeats, starting at step **1220**.

The events may include, but are not limited to the following examples: a sensor in the tape node has taken a measurement that is above a threshold value; a sensor in the tape node has taken a measurement that is below a threshold value; a sensor in the tape node has taken a measurement that is below or equal to a high threshold value and above or below a low threshold value; values of sensing data within a frequency band are higher than a threshold value associated with the frequency band; values of sensing data within a frequency band are lower than a threshold value associated with the frequency band, values of sensing data within a frequency band are higher than or equal to a low threshold value and lower than or equal to a high threshold value associated with the frequency band; a sensor in the tape node is unable to take a measurement (e.g., the sensor is malfunctioning); a two-dimensional bar code on the tape node is scanned by a client device; a client device has initiated communication with the tape node; after a period of time has elapsed after a sensor in the tape node has taken a measurement that is above a threshold value the sensor takes another measurement that is also above the threshold value; after a period of time has elapsed after a sensor in the tape node has taken a measurement that is above a threshold value, the sensor takes another measurement that is now below the threshold value; after a period of time has elapsed after a sensor in the tape node has taken a measurement that is below a threshold value, the sensor takes another measurement that is also below the threshold value; after a period of time has elapsed after a sensor in the tape node has taken a measurement that is below a threshold value, the sensor takes another measurement that is now above the threshold value; after a period of time has elapsed after a sensor in the tape node has taken a measurement that is above a first threshold value and below a second threshold value; the sensor takes another measurement that is also above the first threshold value and below the second threshold value; after

a period of time has elapsed after a sensor in the tape node has taken a measurement that is above a first threshold value and below a second threshold value the sensor takes another measurement that is now below the first threshold value or above the second threshold value; a location of the tape node is within a threshold proximity of a target location; a specific duration of time has elapsed since a preceding event (e.g., 5 days have passed since the two-dimensional bar code was scanned); stored energy on an energy storage device (e.g., a battery) on the tape node is below a threshold value or above a threshold value; a specific type of sensor (e.g., light sensor) on the tape node detects a measured signal that is above a threshold value or below a threshold value (e.g., a light sensor detects an above threshold presence of light); the tape node receives a communication from another tape node; the tape node receives a configuration file from another tape node, a gateway device, a client device, the central database and control system, or some combination thereof; the tape node receives data indicating that another tape node in proximity to itself has a battery level below a threshold value or above a threshold value; and the tape node detects another tape node in proximity to the tape node.

In response to detecting **1240** the event, execution of the distributed intelligent software causes **1250** the tape node to alter its state. As discussed above, the instructions may be generated by one or more of the tape nodes, generated and transmitted to the tape node by the central database and control system, generated and transmitted to the tape node by the one or more client devices, or some combination thereof. The tape node then enters a state based on the instruction **1250** from the distributed intelligent software. For example, the tape node may enter a high sensing mode, as described above. In some embodiments, the tape node may also exit the initial state based on the instruction **1250** according to the distributed intelligent software.

The distributed intelligent software also instructs **1260** the central database and control system and the one or more client devices to take corresponding actions, in response to detecting **1240** the event. In some embodiments, execution of the distributed intelligent software causes (e.g., instructs) **1260** the central database and control center to take one or more of the following actions, based on the detected event: transmit a notification to a client device, for example an alert; generate and transmit instructions to the tape node (e.g., instructions to alter the state of the tape node); store a log of the detected event; store a log indicating that the tape node has altered its state; store data received from the tape node and/or client devices; transmit sensor data to a client device; and transmit instructions to a client device (e.g., instructions to update a display on the client device). The instructions for the central database and control system may include actions not listed above. The distributed intelligent software may issue multiple instructions simultaneously or sequentially. For example, the central database and control system may receive instructions to both store a log of the detected event and transmit a notification to a client device.

In some embodiments, the distributed intelligent software causes **1260** a client device to take one or more of the following actions, based on the detected event: display a notification on the display of the client device (e.g., an alert); transmit instructions to the tape node (e.g., instructions to alter the state of the tape node); store a log of the detected event in the client device’s memory; store a log indicating that the tape node has altered its state in the client device’s memory; store data received from the tape node and/or the central database and control system in the client device’s memory; transmit data to the central database and control

system; transmit instructions to the central database and control system. The instructions for client devices may include actions not listed above. The distributed intelligent software may issue multiple instructions simultaneously or sequentially. For example, the client device may receive instructions to both store a log of the detected event and display a notification on the display of the client device.

FIGS. 13 and 14 are flowcharts showing example steps executed according to the distributed intelligent software. In the example in FIGS. 13 and 14, an indicator measurement is included in the sensor data collected using one or more sensors of a tape node (e.g., tape node 510). For example, a tape node 510 including a pressure sensor may be adhered to an asset, and the indicator measurement is the pressure sensed by the pressure sensor. The event being detected in the example of FIGS. 13 and 14 is the indicator measurement being above a high threshold value. The high threshold value is a parameter of the distributed intelligent software that is relevant to the object being monitored. In some cases, the high threshold value is set by a human operator or administrator of the sensing system 500. In other cases, the high threshold value may be set by a machine learning model. For example, the high threshold value may be a parameter that is determined during training of the machine learning model.

FIG. 13 is a flowchart showing example steps executed according to the distributed intelligent software. The tape node 510 enters 1310 a low communication mode state. The tape node may enter 1310 a low communication mode state when the tape node is initiated based on the distributed intelligent software, according to some embodiments. The tape node takes measurements 1320 using one or more sensors on the sensor device. The measurements are aggregated as sensor data. The sensor data includes an indicator measurement, where the indicator measurement is relevant to an object that the sensor device is monitoring. For example, the indicator measurement may be a peak frequency of vibrations monitored by a vibration sensor in the tape node 510. Some or all of the sensor data may be stored on the memory 513 of the tape node 510. In some embodiments, some or all of the sensor data is transmitted to the central database and communication system 520 and/or one or more client devices 530 each of which may store the sensor data in the database 522 or the memory 432, respectively.

In the low communication mode, the tape node 510 has limited communications with the central database and control system 520. In this example, the tape node 510 does not transmit all of the sensor data collected. Of the sensor data, the tape node 510 only transmits the indicator measurement to the central database and control system 520. The tape node 510 transmits data to the central database and control system at a low frequency (e.g., the tape node 510 transmits the indicator measurement every 5 hours). In other embodiments, the tape node enters additional and/or different state in step 1310.

After step 1320, if the indicator measurement is higher than or equal to a high threshold value, the tape node 510 executes instructions to alter its state, according to the distributed intelligent software. In this example, the tape node 510 exits the low communication mode state and enters a high communication mode state 1340 according to the distributed intelligent software. In some embodiments, the logic of the distributed intelligent software including the high threshold value is stored on the memory 513 of the tape node 510, and the tape node 510 determines if the indicator measurement is above the high threshold value. In this case,

the tape node 510 generates the instructions to alter its state according to the distributed intelligent software. In other embodiments, the tape node 510 transmits the indicator measurement to the central database and control system 520, and the central database and control system 520 determines if the indicator measurement is above the high threshold value. In response to determining that the indicator measurement is above the high threshold, the central database and control system 520 generates and transmits the instructions to the tape node 510 to exit the low communication mode state and enter the high communication mode state 1340, according to the distributed intelligent software. Implementing and generating instructions at the tape node, the distributed intelligent software provides the advantage that fewer communication transmissions are necessary. This reduces battery drain on the tape node because it is not transmitting data to, and receiving data from, the central database and control system. Furthermore, this increases the scalability of many tape nodes because it reduces bandwidth used by the central database and control system. In other embodiments, the tape node 510 transmits the indicator measurement to a client device 530, and the client device 530 determines if the indicator measurement is above the high threshold value. In response to determining that the indicator measurement is above the high threshold value, the client device 530 then transmits instructions to the tape node 510 to exit the low communication mode state and enter the high communication mode state 1340, according to the distributed intelligent software. Additional steps executed after the tape node 510 enters the high communication mode 1340 are discussed below, with respect to FIG. 14.

In the high communication mode, the tape node 510 transmits a higher volume of data than in the low communication mode. The tape node 510 may also transmit data more frequently in the high communication mode than in the low communication mode (e.g., the tape node transmits data every 5 seconds). In this example, the tape node 510 transmits all of the sensor data it is collecting to the central database and control system 520, rather than just the indicator measurement. In other embodiments, the tape node enters additional and/or different states than the high communication mode in step 1340.

After step 1320, if the indicator measurement is below the high threshold value, the tape node 510 remains 1350 in the low communication mode state. The tape node 510 proceeds to transmit 1360 sensor data to the central database and communication system 520 and the client device 530 according to protocols of the low communication mode. The process is then repeated starting at step 1320.

As discussed above, in the example described with respect to FIGS. 6 and 13, battery life may be extended without compromising the ability to detect certain events or violations by using the distributed intelligent software to control aspects of the tape nodes' functions. In other embodiments, the distributed intelligent software provides the benefit of improved detection of the violations. For example, entering the high communication mode and a high sensing mode (where the sampling rate of the sensors in the tape nodes is increased) may improve the ability for the sensing system to detect anomalous behavior in the assets being monitored by providing more granular data to users and/or the central database and control system.

FIG. 14 is a flowchart showing example steps executed continuing from step 1340 in FIG. 13 after the tape node 510 enters the high communication mode, according to the distributed intelligent software. The steps of FIG. 14 may also be implemented without implementing the steps of FIG.

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13. In response to the indicator measurement being above the high threshold value, the tape node is instructed to enter 1340 the high communication mode state, according to the distributed intelligent software. The tape node 510 proceeds to take measurements 1410 using its one or more sensors. In some embodiments, the tape node 510 also enters a high sensing mode state at step 1340, and the tape node 510 takes measurements 1410 at a higher sampling rate than in its initial state.

Following the high communication mode protocol, the tape node 510 transmits 1420 the sensor data that it collects in step 1410 to the central database and control system 520. The indicator measurement is included in the sensor data. If the indicator measurement included in the measurements collected in step 1320 is higher than or equal to the high threshold value, the tape node remains 1430 in the high communication mode. The process then repeats starting from step 1340.

If the indicator measurement is below the high threshold value, the central database and control system 520 generates instructions to alter the state of the tape node 510 and transmits the instructions to the tape node 510, according to the distributed intelligent software. In other embodiments, the tape node 510 generates the instructions and executes them without needing to communicate with the central database and control system 520. In other embodiments, a client device 530 generates and transmits the instructions. In this example, the tape node exits the high communication mode and enters the low communication mode state 1440. The tape node then begins the process shown in FIG. 13, starting at step 1320.

Although FIGS. 13 and 14 only show steps in relation to a single high threshold, the distributed intelligent software may consider multiple thresholds when instructing the adhesive tape platform 510, the central database and control system 520, and the one or more client devices 530. For example, in the process shown in FIG. 14, the distributed intelligent software may include instructions to alter the state of the tape node 510 in the high communication mode in response to the indicator measurement being higher than a second high threshold.

Vibration and Temperature Monitoring

The adhesive tape platform 510 may be used in some applications for vibration and temperature monitoring, according to some embodiments. In this case, a tape node may include at least one of a vibration sensor and a temperature sensor. In some embodiments, each tape node includes both a vibration sensor and a temperature sensor. In other embodiments, a first tape node including a vibration sensor is used to measure vibrations, and a second tape node including a temperature sensor is used to measure temperature. The first tape node and second tape node may be installed on an item of interest in proximity to each other, in some examples. In other examples, the first tape node and the second tape node may be installed at different positions on the item of interest.

FIG. 15 shows an example of a tape node being used for vibration and temperature sensing. In the example of FIG. 15, the tape node 1510 is adhered to a section of a pipe 1520 that has a liquid flowing through it. The embodiment of FIG. 15 may also apply equally for gas flowing through the pipe 1520. The liquid may be hot or may transfer heat to the pipe, in some cases. The vibration of the section of the pipe 1520 is monitored in order to determine a flow rate of the liquid flowing through the pipe 810, in some embodiments. In some cases, the vibration monitoring may be used to deter-

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mine additional and/or alternate properties besides flow rate of a liquid through the pipe 610.

The tape node 1510 is an embodiment of the segment 102 of the adhesive tape platform 100. A non-adhesive side of the tape node 1510 is shown in FIG. 15, including a two-dimensional bar code 1511 that appears on the adhesive side of the tape node 1510 and a sensor marking 1512 that also appears on the non-adhesive side of the tape node 1510. The sensor marking 1512 indicates the position of one or more sensors that are included in the tape node 1510, according to some embodiments. In other embodiments, the tape node 1510 does not include the sensor marking 1512.

In the example shown in FIG. 15, the tape node 1510 includes at least one vibration sensor and at least one temperature sensor. In other embodiments, the tape node 1510 includes different configurations of sensors, types of sensors, number of sensors, or some combination thereof. For example, the tape node 1510 may only include a vibration sensor for applications where only vibration sensing is relevant. In another example, the tape node 1510 includes a vibration sensor for vibration sensing and another tape node including a temperature sensor is adhered at a different position on the section of the pipe 1520 for temperature sensing. In other embodiments, other configurations of tape nodes may be used, including a different number of tape nodes, different placements of the tape nodes, different types of tape nodes, or some combinations thereof.

The tape node 1510 wirelessly transmits sensing data collected by its sensors, including data on the vibrations and temperature of the pipe 1520, to an associated embodiment of the central database and control system 520. In some embodiments, the tape node 1510 also transmits sensing data to one or more client devices. An embodiment of a client device 530 may be used to scan the two-dimensional bar code 1511. This may be done to register the tape node 1510 in a log or database, initialize the tape node 1510, pair the tape node 1510 with the client device, for other functions, or some combination thereof.

The tape node 1510 measures vibrations of the section of the pipe 1520 using a vibration sensor, according to some embodiments. The vibration sensor may use one or more of the following sensors to measure vibrations: an accelerometer (piezoelectric-based accelerometer, capacitive-based accelerometer, piezoresistive accelerometer, MEMS-based accelerometer, MEMS resonator-based accelerometer); a strain sensor; a velocity sensor; a microphone or acoustic pressure sensor; an Optical or laser-based sensor; a capacitive sensor or another sensor configured to detect vibrations. Vibration may be monitored using vibration sensing data including acceleration data, velocity data, displacement data, acoustic pressure (e.g., audio) data, other data relevant to calculating the vibration, or some combination thereof.

FIG. 16 is a block diagram 1600 illustrating an example where multiple tape nodes are used to monitor various items at different locations in a manufacturing building 1610 and collect sensor data on the items in a manufacturing building, according to some embodiments. The locations in the manufacturing building 1610 include a manufacturing wing 1620, a pumping area 1630, a pumping Area 1640, and a building and network controls area 1650. In this example, the manufacturing wing 1620 is an area where the flow rate (using vibration measurements) of liquid through pipes and the temperature of the liquid is being monitored using multiple tape nodes, as shown in FIG. 21. In some cases, the vibration monitoring may be used to determine other properties than flow rate of a liquid.

The manufacturing wing **1620** includes a section of a pipe **1621** that has its vibrations and temperature monitored with a tape node **1622**. The manufacturing wing **1620** includes another section of a pipe **1623** that is being monitored with another tape node **1624**. The liquid flowing through the pipe **1621** may be the same or different from the liquid flowing through the pipe **1623**.

The adhesive tape platform is used for safety reasons, according to some embodiments. For example, if the flow rate or the temperature of a liquid through pipe **1621** or pipe **1623** is too high, it may create an unsafe condition in the manufacturing wing **1620**. In other cases, the adhesive tape platform is used to measure performance. For example, if the flow rate or the temperature of a liquid through pipe **1621** or **1623** is too low, it may cause lower efficiency of tools in the manufacturing wing.

The pumping area **1630** includes a pump **1631** connected to the pipe **1621**. The pump **1631** controls the flow of the liquid flowing through the pipe **1621**. For example, the pump **1631** may control a pressure in the pipe **1621** that moves liquid into the pipe **1621**. The pumping area **1630** is in a separate location from the manufacturing wing **1620**. A tape node **1632** is used to monitor vibrations and temperature of the pump **1631**. For example, the tape node **1632** may be adhered to a surface of the pump **1631**.

The pumping area **1640** includes a pump **1641** connected to the pipe **1623**. The pump **1641** controls the flow of the liquid flowing through the pipe **1623**. For example, the pump **1641** may control a pressure in the pipe **1623** that moves liquid into the pipe **1623**. The pumping area **1630** is in a separate location from the manufacturing wing **1620** and the pumping area **1630**. A tape node **1642** is used to monitor vibrations and temperature of the pump **1641**. For example, the tape node **1642** may be adhered to a surface of the pump **1641**.

The building and network controls area **1650** is an area used for the management of tools, operations, personnel, processes, and other functions of the manufacturing building **1610**. The building and network controls may be an area where a floor manager or some other human operator oversees the operations of the manufacturing building **1610**. The building and network controls area **1650** includes at least one client device **1651** that is used to communicate with the tape nodes, **1622**, **1624**, **1632**, **1642**, controls for the pump **1631**, controls for the pump **1641**, and the associated central database and control system **520**. The client device **1651** may also communicate with other client devices, tools, machinery, or other devices in the manufacturing building **1610**. The client device **1651** is an embodiment of the client device **530**.

The tape nodes **1621**, **1624**, **1632**, **1642** are embodiments of the tape node **1510**, and each of the tape nodes **1621**, **1624**, **1632**, **1642** includes at least one vibration sensor and at least one temperature sensor, according to some embodiments.

The distributed intelligent software includes logic and instructions for the operations of the tape nodes **1621**, **1624**, **1632**, **1642**, the client device **1651**, and the associated central database and control system **520**. The distributed intelligent software may be customized with parameters and protocols (e.g., threshold values, emergency protocols) that are specific to the applications where the sensing system **500** is being used. For example, the distributed intelligent software may be customized by a user inputting parameters and protocols relevant to the sensing system to the central database and control system using a client device. The central database and control system may then distribute

configuration settings (e.g., a configuration file) to the tape nodes which include the customized parameters and protocols. In some examples, the central database and control system does not need to provide the configuration settings to all of the tape nodes, and the tape nodes that already have the configuration settings can transmit configuration settings relevant to the distributed intelligent software to other tape nodes. In other embodiments, the tape nodes may receive configuration settings from a client device. For example, the client device may transmit configuration settings to a tape node when the client device scans the two-dimensional barcode on the tape node. An example of a procedure executed by the sensing system **500** deployed in the manufacturing building **1610** according to the distributed intelligent software is discussed below with respect to FIG. **17**.

FIG. **17** is a flow chart showing example steps executed according to the distributed intelligent software by components of the sensing system deployed in the manufacturing building **1610** shown in FIG. **16**. The example of FIG. **17** involves the tape node **1622** which monitors the vibrations and temperature of pipe **1621** and the tape node **1632** which monitors the vibrations and temperature of pipe **1631**. The steps shown in FIG. **17** may similarly apply to the tape node **1624** which monitors the vibrations and temperature of pipe **1623** and the tape node **1642** which monitors the vibrations and temperature of the pump **1641**. The steps in FIG. **17** illustrate one example of the logic and protocols included in the distributed intelligent software for the example application of monitoring in the manufacturing building **1610**, but the distributed intelligent software and its applications are not limited thereto.

The tape node **1622** and the tape node **1632** initiate in the low communication mode state. In one example of FIG. **17**, a client device scans a bar code on each of the tape nodes **1622** and **1632** upon installation of the tape nodes on the respective pipes **1621**, **1622**. The client device, another tape node, a gateway device, an associated central database and control system, another source, or some combination thereof transmits configuration settings to the tape node **1622** and **1632** to provision the tape nodes to a given task. The provisioning may include downloading certain instructions and logic to the tape node, the client device, and/or the central database and control system to implement a given distributed intelligent software logic for the associated task of the provisioned tape node. For example, the configuration settings may be part of a configuration file that is transmitted to the tape nodes **1622** and **1632**. The configuration file may include programs and software instructions, including those part of the distributed intelligent software. After provisioning, the tape nodes may operate according to the intelligent software to control data capture and communication steps as discussed herein.

In the low communication mode, the tape node **1622** and the tape node **1632** limit the communication they perform with the associated central database and control system **520**. The tape nodes **1622**, **1632** each collect a full range of vibration and temperature sensor data. In this example, the full range of vibration and temperature sensor data include a frequency spectrum (e.g., velocity/acceleration/energy/amplitude values over a range of frequencies) calculated for the vibration measurements by performing a fast Fourier transform (FFT) on the vibration sensor data. The tape nodes **1622**, **1632** only transmit a set of indicator measurements in the low communication mode to the central database and control system **520**, instead of the full range of vibration and temperature sensor data. The indicator measurements include select FFT frequency-domain values at three indi-

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cator frequencies (100 Hz, 1 kHz, and 12 kHz). Each FFT frequency-domain value indicates a component of the vibrations measured that occurs at the corresponding frequency. In other examples, the indicator measurements include additional or alternate FFT frequency-domain values at other frequencies. The indicator measurements also include the current temperature sensed by the respective temperature sensor in the sensor node. In the low communication mode, the indicator measurements are transmitted to the central database and control system 520 at a low frequency (e.g., each tape node transmits the indicator measurements every 1 hour).

The tape node 1622 takes vibration and temperature measurements 1720. Some or all of the sensor data collected by the tape nodes may be stored on the memory of the tape node 1622. The tape node 1622 calculates a frequency spectrum of the vibrations by performing a FFT of the vibration data collected. In some embodiments, the tape node 1622 stores the full frequency spectrum in its memory. In other embodiments, the tape node 1622 only stores the select FFT frequency-domain values at the three indicator frequencies.

The tape node 1622 transmits the indicator measurements to the central database and control system according to the protocol of the low communication mode. If each of the indicator measurements received by the central database and control system 520 from the tape node 1622 are above or equal to a respective low threshold value and below or equal to a respective high threshold value (i.e., in a moderate range), the tape node 1622 and the tape node 1632 remain 1740 in the low communication mode, according to the distributed intelligent software. The process then repeats, starting from step 1710. Each indicator measurement may have a corresponding high threshold and low threshold, according to some embodiments. In some embodiments, some of the indicator measurements only have a high threshold or a low threshold. For example, the select FFT frequency-domain values at the three frequencies may each have a corresponding high threshold and low threshold, whereas the temperature measurements included in the indicator measurements only has a high threshold. In this case, the tape node 1622 and the tape node 1632 only remain in the low communication mode 1740 if each of the select FFT frequency-domain values are below or equal to the corresponding high threshold and above or equal to the corresponding low threshold and if the temperature measurement is below or equal to the corresponding high threshold.

If any of the indicator measurements received by the central database and control system 520 from tape node 1622 is above a respective high threshold value or below a respective low threshold value, the central database and control system 520 transmits 1750 respective instructions to the tape node 1622 and the tape node 1632 to alter their respective states. The tape node 1622 executes the received instructions, exiting the low communication mode and entering a high communication mode, and the tape node 1632 likewise executes the received instructions, exiting the low communication mode and entering the high communication mode 1760. In this example, the tape node 1622 and the tape node 1632 receive the same instructions and both enter the high communication mode. In other embodiments, the tape node 1632 receives different instructions from the tape node 1622 and enters a different state from the tape node 1622. For example, the tape node 1632 may enter a high sensing mode, while the tape node 1622 enters the high communication mode.

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In certain embodiments, steps 1730-1760 are all implemented on a tape node, a gateway device, or some combination thereof, and any instructions discussed above sent from the central database and control system are sent from the tape node, instead. Accordingly, this allows for the tape node to act autonomously, without requiring additional data communication to and from the central database.

The central database (or the tape node or gateway) then transmits 1770 an alert to the client device 1651. The alert to the client device 1651 may be displayed on a display associated with the client device 1651. The alert may notify a human operator of the indicator measurements that are above a respective high threshold or below a respective low threshold. For example, the alert may be displayed on the client device 1651 that indicates that the magnitude of vibrations (according to the selected FFT value) at 1 kHz is above a high threshold for 1 kHz vibrations.

In the high communication mode, the tape node 1622 and the tape node 1632 each transmits 1780 its full range of sensor data (also referred to herein as the “full sensor data”) collected to the central database and control system 520 for an hour. The full sensor data may include all of the sensor data collected by the vibration sensor and the temperature sensor in each of the tape node 1622 and the tape node 1632. The full sensor data includes the frequency spectrum of the vibrations for each of the tape nodes 1622, 1623, according to some embodiments. While in the high communication mode, the tape node 1622 and the tape node 1632 each transmits the full sensor data more frequently (e.g., every 5 seconds) than the low frequency of transmission (e.g., every 1 hour) of the indication measurements in the low communication mode.

The full sensor data received by the central database and control system 520 may be stored on the database 522. Further calculations may be performed on the stored data by the application engine 521. The full sensor data is transmitted from the central database and control system 520 to the client device 1651, and the client device 1651 stores the full sensor data in its memory. The client device 1651 displays the full sensor data on its display, for example, for a user to view. The full sensor data may be displayed to a human operator in the building and network controls area 1650, who may make a decision based on the full sensor data. For example, if the vibration and temperature data collected by tape node 1632 indicates that the pump 1631 is functioning incorrectly, the human operator may deactivate the pump using an emergency shutoff. In some embodiments, the emergency shutoff is engaged using the client device 1651. In other embodiments, the client device 1651 automatically engages the emergency shutoff, in response to detecting that the vibration and temperature data collected by the tape node 1632 correspond to a malfunctioning of the pump 1631. In some embodiments, if any of the indicator measurements that the central database and control system 520 from the tape node 1622 are above a second high threshold (value greater than the high threshold value), the client device 1651 is instructed to deactivate the pump using an emergency shutoff.

After an hour of the tape nodes 1622, 1632 transmitting 1780 the full sensor data in the high communication mode, if any of the indicator measurements received from the tape node 1622 or by the tape node 1632 are still above the respective high threshold or below the respective low threshold, the tape nodes 1622, 1632 remain in the high communication mode. This is repeated until the indicator measurements are all below or equal to their respective high threshold and above or equal to their respective low thresh-

old, according to some embodiments. If all of the indicator measurements received from the tape node **1622** and by the tape node **1632** are below their respective high threshold value and above their respective low threshold after step **1780**, the tape node **1622** and the tape node **1632** each exit the high communication mode and return to the low communication mode. The process then repeats, starting at step **1710**.

Although only the indicator measurements for the tape node **1622** are used to decide whether to proceed from step **1730** to step **1740** or step **1750** in the example of FIG. **17**, in other examples, the indicator measurements for the tape node **1632** may be considered.

FIG. **18** shows an example plot **1800** of an indicator measurement collected at various times by the tape node **1622** in the example discussed with respect to FIGS. **16** and **17**. Each indicator measurement shown in the example plot **1800** is a measurement collected by a sensor in the tape node **1622** or a data value calculated based on the collected measurement. For example, each of the indicator measurements shown in the example plot **1800** may be a temperature measurement or a selected FFT frequency-domain values corresponding to the 1 kHz frequency.

The example plot **1800** shows 7 indicator measurements taken over a time period of 7 hours, with a one hour interval between each measurement. Although only 7 indicator measurements are shown, the tape node **1622** may take more than 7 indicator measurements in the time period shown. The example plot **1800** shows the relative values of the low threshold value **1810** and the high threshold value **1820** as corresponding to the indicator measurements. Each threshold value is indicated by a respective dotted line. The first four indicator measurements shown in the example plot **1800** (corresponding to the first four hours shown in the example plot **1800**) are in the moderate range, each having a value below or equal to the high threshold value **1820** and above or equal to the low threshold. During the period (before time **1840**) when the first four measurements are taken, the tape node **1622** is in the low communication mode according to the distributed intelligent software, corresponding to steps **1710** to **1740** of FIG. **17**.

At time **1840**, the tape node **1622** collects the indicator measurement **1845** which has a value higher than the high threshold value **1820**. Execution of the instructions of the distributed intelligent software causes the tape node to enter the high communication mode, and execute steps **1750** to **1780** in FIG. **17**. At time **1850** after an hour has passed from the time **1840** when the tape node **1622** enters the high communication mode, the tape node **1622** collects the indicator measurement **1855** which is in the moderate range. Further execution of the instructions of the distributed intelligent software causes the tape node **1622** to exit the high communication mode and re-enter the low communication mode at time **1850**.

At time **1860**, the tape node **1622** collects the indicator measurement **1865** which has a value lower than the low threshold value **1810**. Further execution of instructions of the distributed intelligent software causes the tape node **1622** to exit the low communication mode and enter the high communication mode at time **1860**. The steps **1750** to **1780** in FIG. **17** are then executed.

FIG. **19** is a block diagram illustrating an example of a trained machine learning model being used to generate instructions for the distributed intelligent software. Aspects of the trained model **1910** and associated model parameters **1920** may be stored on an embodiment of the central database and control system **520**, an embodiment of the

client device **530**, an embodiment of the adhesive tape platform **510**, or any combination thereof. The trained model **1910** receives sensing data **1930** as an input. The sensing data **1930** may be received from a tape node, a client device, the central database and control system, or some combination thereof. The trained model **1910** accesses model parameters **1920** relevant to the distributed intelligent software and generates instructions **1940** to alter the state of one or more tape nodes based on the received sensing data **1930** and the accessed model parameters **1920**. In some embodiments, instructions **1940** also include instructions for the central database control center **530**, one or more client devices **520**, or any combination thereof. The model parameters **1920** may include values, parameters, conditions, and logic relevant to the distributed intelligent software. For example, the model parameters **1920** may include one or more threshold values for the sensing data **1330** used in detecting if an event has occurred.

FIG. **20** is a block diagram illustrating an example of training the machine learning model shown in FIG. **19**. During training, the model **1910** is trained using training sensing data **2010** as an input. The model **2010** may also optionally receive training instructions **2020** that correspond to the training sensing data **2010** as an input. For example, this may be the case when the model **1910** is trained using supervised training methods. The training instructions **2020** may include instructions to alter the state of one or more tape nodes according to the distributed intelligent software and corresponding to the training sensing data **2020**. The model **1910** is trained by determining model parameters **1920**, so as to best represent the relationship between the training sensing data **2010** and associated instructions that will be generated by the trained model **1910**. For example, the model **1910** may be a neural network trained using a stochastic gradient descent technique or some other type of machine learning model. Once the model parameter **1920** are known, the trained model **1910** may be used to generate instructions for one or more tape nodes, the central database and control system, one or more client devices, or some combination thereof. As discussed with respect to FIG. **19**, the output of the trained model **1910** is used to generate instructions according to the distributed intelligent software.

Example Interface

FIGS. **21-23** relate to example user interfaces that may be shown on one or more client devices **530** for displaying sensing data, managing the adhesive tape platform and sensing system **500**, and setting parameters of the distributed intelligent software, according to some embodiments. The example user interfaces **2100**, **2200** in FIG. **21** and FIG. **22**, respectively, may be displayed on one or more embodiments of the client device **530**. In some embodiments, the example user interfaces may be displayed on a personal computer display as part of an application, web app, website, or some combination thereof. In other embodiments, the example user interfaces are displayed on a smart phone display, as part of a smartphone app.

FIG. **21** is an example user interface **2100** for displaying information (e.g., properties and settings of the tape node) and sensor data received from a tape node. On the left side of FIG. **21** in the tape node selection portion, two tape nodes are shown in the user interface, tape node A and tape node B. The tape node A is organized into a group labeled "East Wing." For example, East Wing may be a label that indicates a location of the tape node A. The tape node B is similarly organized into a group labeled "West Wing." A user may interact with the user interface in the tape node selection portion to select which tape node has its sensor data and

properties displayed in the data viewing portion 2120. A user may interact with the data selection portion 2130 to select which types of information are displayed in the data viewing portion 2120. In the example of FIG. 21, the tape node A is selected in the tape node selection portion 2110, and the “node settings” are selected in the data selection portion 2120, resulting in various settings of the tape node A being displayed in the data viewing portion 2120. In some embodiments, the user is able to interact with the settings shown in the data viewing portion 2120 to change various settings and parameters for the selected tape node (e.g., tape node A). Some of the settings and parameters that can be changed using the interface 2100 may affect relevant parameters, logic, and instructions that are included in the distributed intelligent software, according to some embodiments.

FIG. 22 is an example user interface 2200 for setting parameters relevant to the distributed intelligent software. The example user interface 2200 includes the tape node selection portion 2210, the data viewing portion 2220, and the data selection portion 2230. The tape node selection portion 2210, the data viewing portion 2220, and the data selection portion 2230 function similarly to the respective tape node selection portion 2110, the data viewing portion 2120, and the data selection portion 2130 shown in FIG. 21.

In the example of FIG. 22, the tape node A is selected in the tape node selection portion and the “alert settings” are selected in the data selection portion 2230, resulting in the alert settings and information for the tape node A being displayed in the data viewing portion 2220. The alert settings and information for the tape node A are settings and properties relevant to the distributed intelligent software. In particular, the alert settings and information for the tape node partly define rules and logic for when alerts are transmitted to one or more client devices 530 from an embodiment of the central database and control system 520 associated with tape node A, in response to the vibration sensor data that tape node A (an embodiment of the adhesive tape platform 510) collects.

In the example shown in FIG. 22, the central database and control system 520 transmits an alert to one or more client devices 530 associated with the users and email addresses shown in the alert recipient portion 2240. In some embodiments, the alert is transmitted via email to the email addresses shown in the alert recipient portion 2250. The user of the interface 2200 may interact with the alert recipient portion to change which users and/or client devices 530 receive alerts from the central database and control system 520. For example, a user may interact with the add recipient button 2270 to include additional recipients and/or client devices that receive alerts. A row of properties and settings related to an alert 2260 are shown in data viewing portion 2220.

In the example of FIG. 22, the alert 2260 is displayed in the data viewing portion 2220, including the parameters of “Min Frequency (Hz)”, “Max Frequency (Hz)”, and “Velocity Threshold (in/S)” relevant to how the distributed intelligent software determines when to send the alert 2260. In the example shown in FIG. 22, an alert is transmitted when any of selected velocity indicator measurements has a value above the “Velocity Threshold (in/S)” (i.e., 0.2 in/S). The selected velocity indicator measurements are velocity measurement components that occur in the frequency band between the “Min Frequency (Hz)” and the Max Frequency (Hz) (i.e., between 2.31 Hz and 2.4 Hz). The selected velocity indicator measurements may be calculated by performing an FFT on sensor data from a vibration sensor (e.g., an accelerometer) in tape node A, according to some

embodiments. By interacting with the data viewing portion 2220, the user of the interface 2200 may edit and/or change the parameters (e.g., “Min Frequency (Hz)”, “Max Frequency (Hz)”, and “Velocity Threshold (in/S)”) relevant to how the distributed intelligent software determines when to send the alert 2260. By changing the “Min Frequency (Hz)” and the “Max Frequency (Hz)” values, the user may change the frequency band or range of frequencies that are used for the indicator measurement. By changing the “Velocity Threshold (in/S),” the user may change the high threshold value that triggers the alert when the indicator measurement is above the high threshold. The user may also interact with the add alert button 2270 to configure the distributed intelligent software to include additional alerts.

In some embodiments, the client device 530 receives a full range of sensor data from a selected tape node as part of the downloaded sensor data. FIG. 23 is a chart showing example sensor data received by the central database and control system from a tape node and downloaded to the client device 530 associated with the interface 2200, according to some embodiments. The sensor data shown in the chart includes a full range of time-domain acceleration data 2310 (each data pair including an acceleration value and a relative time when the acceleration value was measured) shown on the left two columns of the chart. The sensor data also includes a full range of frequency-domain data 2320 (each data entry including an acceleration value, a velocity value, and a frequency corresponding to the acceleration and the velocity values) in the right 3 columns of the chart. In some embodiments, the frequency-domain data is calculated from a FFT performed on the time-domain acceleration data. **Acoustic Monitoring**

FIG. 24A shows an example of a tape node 2410 with an acoustic sensor collecting audio data relevant to an asset that the tape node is monitoring, according to some embodiments. The tape node 2410 is an embodiment of the tape nodes 102, 103, 105 shown in FIGS. 4 and 5A-5C, according to some embodiments. In some embodiments, the tape node 2410 has a different form factor than the flexible adhesive tape form factor. For example, the tape node 2410 may have a rigid form factor, in some embodiments. The acoustic sensor may be a microphone, a piezoelectric sensor, a piezoresistive sensor, another type of acoustic sensor, or some combination thereof. The tape node 2410 includes a two-dimensional bar code 2411 on the non-adhesive side of the tape node 2410. In this example, the tape node 2410 is adhered to the asset 2420 with the adhesive side contacting a surface of the asset 2420, but in other examples, the tape node 2410 may not be adhered to the asset 2420. A chain 2430 associated with the asset 2420 is in the vicinity of the tape node 2420. In some embodiments, a different component or object is associated with the asset 2420 and in vicinity of the tape node 2420. In some embodiments, the chain 2430 is attached to a portion of the asset 2420. For example, the chain 2430 may secure another object to the asset 2420. A user of the sensing system may be interested in detecting events where the chain 2430 has been broken, to be aware of tampering or damage to the asset 2420 or the chain 2430.

In one example, the tape node 2410 collects audio sensor data using the acoustic sensor of the tape node 2410. In the event of the chain 2430 breaking, the distributed intelligent software determines that the chain has broken based on audio data that is captured by the acoustic sensor of the tape node 2410. In some embodiments, the tape node 2410 itself performs computations for determining that this event has occurred, without communicating with other nodes of an

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associated sensing system **400**. The tape node **2410**, in this case, may also generate software instructions to alter its own state, according to the distributed intelligent software. In other embodiments, the tape node **2410** communicates data (e.g., the audio data or data relevant to the audio data) to other nodes of the sensing system **400**, and the other nodes perform computations for determining that the event has occurred. In this case, the other nodes transmit communications to the tape node **2410** to instruct the tape node **2410** to alter its own state, according to the distributed intelligent software.

In some embodiments, the data from the tape node **2410** is provided to a classifier which determines if an event has occurred. The classifier is a trained machine learning model, as described below with respect to FIGS. **25** and **26A-26C**. Aspects of the classifier may be stored on one or more tape nodes (e.g., tape node **2410**), one or more client devices, one or more gateway devices, one or more other nodes of the sensing system **400**, the server(s)/cloud, or some combination thereof. For example, some combination of analysis functions, parameters, and weighting factors may be stored on the tape node **2410**. The tape node **2410** may access the stored analysis functions, parameters, and weighting functions and execute the functions of the classifier locally, without communicating with other nodes of the sensing system **400**, according to some embodiments. In other embodiments, the aspects of the classifier are distributed across the nodes of the sensing system, and the classifier functionality is executed by communicating between the nodes of the sensing system.

FIG. **24B** is a flow chart showing example steps for a method **2402** of detecting events based on audio data collected by a tape node, according to some embodiments. The tape node **2410** senses **2450** an audio signal collected by an acoustic sensor of the tape node **2410**. In some embodiments, the tape node determines if the process **2402** should be initiated if the audio signal includes a predetermined characteristic for triggering the process **2402**. The characteristic may be a peak or average volume level above a threshold level, an audio level at a specified audio frequency being above a threshold value, or some other characteristic. In other embodiments, the process **2402** is triggered by other events, as discussed above with respect to FIG. **12**, for example. The tape node **2410** performs data compression **2451** on the audio signal, according to a current audio quality state of the tape node **2410**. The current audio quality state of the tape node **2410** defines parameters for compressing the audio signal. The parameters may include a compression technique for compressing the audio signal, a bitrate for encoding the compressed audio signal, a codec for compressing the audio signal, a latency, other parameters for compression, or some combination thereof. The current audio quality state of the tape node **2410** may include a level of compression for compressing the audio signal, with a higher audio quality state corresponding to a lower level of compression. The tape node **2410** may compress the audio signal in order to reduce the bandwidth necessary to transmit the audio signal, reduce the size of the audio data for the audio signal, and/or for other reasons.

The tape node **2410** provides the compressed audio signal to a classifier **2452**. In some embodiments, the tape node **2410** transmits the compressed audio signal to a server of the IOT system **400** which hosts the classifier. The classifier may be an embodiment of the classifier discussed above with respect to FIG. **24A**. The classifier is configured to output a label for the provided input and also a confidence score associated with generated label. If the confidence score

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is below a threshold level, the distributed intelligent software instructs the tape node to enter a higher quality audio state **2454**. The tape node then redoes the compression of the audio signal **2452**, according to the new audio quality state and continues on with the method **2402**, according to some embodiments. In some embodiments, the tape node **2410** collects additional audio data and provides the additional sensing data to the classifier when redoing the classification of the audio signal, in response to the classifier generating a confidence score that is below the threshold. In addition to or alternative to reducing the level of compression when changing to a higher quality audio state **2454**, the tape node **2410** may change a sampling rate or frequency, change a bitrate of compression, send the audio signal to the classifier without any compression, or perform another action to increase the fidelity or quality of the audio signal provided to the classifier.

If the classifier generates a prediction with a confidence level above the threshold, the classifier provides the generated label to the distributed intelligent software **2456**. In response, the distributed intelligent software determines if the label corresponds to an event that requires a state change. If so, the distributed intelligent software instructs the tape node **2410** to change states **2460**. If not, then the distributed intelligent software does not instruct the tape node **2410** to change states. The tape node **2410** then continues monitoring audio sensor data, according to its current state, and starts the process **2402** again when an audio signal is detected that requires classification.

FIG. **25** is a diagram showing an example of using a classifier **2510** to detect events based on sensing data collected by a tape node, according to some embodiments. The classifier receives input sensing data **2530** and outputs a label **2540** based on the received sensing data **2530** and accessed model parameters **2520**. The output label **2540** may include a plurality of output labels. The output label **2540** corresponds to an event that is associated with the input sensing data **2530**, according to some embodiments. One of the output labels **2540** that may be output from the classifier **2510** may include a label that corresponds to no event occurring (e.g., a “normal” label). In some embodiments, the classifier also generates a confidence score **2550** for each output label **2540** that is output. The confidence score **2550** may indicate the probability that the output label **2540** is accurate.

According to some embodiments, the classifier **2510** may optionally receive as an input compression parameters that correspond to data compression that was performed on the input sensing data **1410** and/or a sampling frequency corresponding to the input sensing data **1410**. In this case, the classifier **2510** outputs the label **2540** based on the received sensing data **2530**, the received compression parameters, and accessed model parameters **2520**.

The sensing data **2510** may include data collected by one or more sensors of a tape node associated with a sensing system **400**, according to some embodiments. The sensing data **2510** may include, for example, audio data captured by an acoustic sensor of the tape node.

In some embodiments, the classifier **2510** outputs a “more data” label that indicates that more sensing data is needed, based on the input sensing data **2530** and the accessed parameters **2520**. The “more data” label may indicate that more sensing data **2530** or sensing data **2530** of higher quality/fidelity is necessary for the classifier **2510** to output a label **2540** that is accurate. In some embodiments, the higher quality/fidelity corresponds to a level or type of compression of the data. For example, the sensing data **2530**

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may go through lossy compression before being provided to the classifier **2510** for inference. The “more data” label may indicate that a lower degree of compression or lossless compression should be used for the sensing data **2530**. In some embodiments, the “more data” label is output if an output label **2540** with a high confidence score **2550** cannot be output by the classifier **2510**. A high confidence score **2550** may be a confidence score that is above a threshold level, according to some embodiments. For example, the classifier **2510** may receive input sensing data **2530** that includes audio data from a tape node. If the audio data is heavily compressed, the classifier **2510** may not be able to output a label **2540** with a high confidence score. The classifier **2510** then outputs the “more data” label, requesting more data or data of higher quality/fidelity. In response, the distributed intelligent software may instruct the tape node to provide audio data with lower data compression (e.g. at a higher level of bits) to the classifier **2510**, provide audio data that is not compressed to the classifier **2510**, collect audio data using a higher sampling frequency and provide the audio data to the classifier **2510**, provide a larger amount of audio data to the classifier **2510**, take other actions related to the audio data, or some combination thereof.

FIGS. **26A-26C** are diagrams showing examples of a training process for the classifier **2510** shown in FIG. **25**, according to some embodiments

FIG. **26A** is a block diagram illustrating an example of training the classifier **2510** shown in FIG. **13**. During training, the classifier **2510** is trained using training sensing data **2610** as an input. The classifier **2510** may also optionally receive as an input training labels **2620** that correspond to the training sensing data **2610**. For example, this may be the case when the classifier **2510** is trained using supervised training methods. The training labels **2620** may include labels for events that correspond to the training sensing data **2610**. The classifier **2510** is trained by determining model parameters **2520**, so as to best represent the relationship between the training sensing data **2610** and associated labels that will be generated by the trained classifier **2510**. For example, the classifier **2510** may be a neural network trained using a stochastic gradient descent technique or some other type of machine learning model. Once the model parameter **2520** are known, the trained classifier **2510** may be used to generate output labels based on received input sensing data for detecting events. The distributed intelligent software uses the event detection by the trained classifier **2510** to detect events relevant to assets being monitored by one or more tape nodes and, in response, alters the state of the one or more tape nodes based on the events detected by the trained classifier **2510**.

Referring to FIG. **26B**, the classifier **2510** may be trained by inputting training compressed sensing data **2612** and training compression parameters **2626** to the classifier **2510**, according to some embodiments. The training compression parameters may include information on data compression that was performed on the trained compressed sensing data **2612** and/or a sampling frequency corresponding to the sensing data **2612**. The classifier **2510** is trained by determining model parameters **2520**, so as to best represent the relationship between the training compressed sensing data **2612**, the training compression parameters **2614**, and associated labels that will be generated by the trained classifier **2510**.

Referring to FIG. **26C**, the classifier **2510** may be trained by inputting training compressed sensing data **2612** and training uncompressed sensing data **2616** to the classifier

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2510, according to some embodiments. The training uncompressed sensing data **2616** may include raw sensing data that was used to generate the compressed sensing data **2612**, via data compression. The classifier **2510** is trained by determining model parameters **2520**, so as to best represent the relationship between the training compressed sensing data **2612**, the training uncompressed sensing data **2616**, and associated labels that will be generated by the trained classifier **2510**.

10 Combining Sensor Modalities

The distributed intelligent software and system thereof may combine multiple sensor modalities to detect events, according to some embodiments. For instance, in the example of FIG. **24A**, the tape node **2410** with the acoustic sensor may also include a vibration sensor that is capable of measuring vibrations of the asset **2420**. The activation of a specific sensor modality may be triggered by detecting events using another sensor modality. For example, the tape node **2410** may only activate its vibration sensing functionality, in response to detecting an audio signal that corresponds to the chain **2430** breaking using the acoustic sensor. The vibration sensor of the tape node **2410** may be in a sleep state or deactivated state, prior to the event of the chain **2430** breaking, but may be activated after the tape node **2410** detects an audio signal that corresponds to a potential chain **2430** breaking event. The tape node **2410** may then enter a state, in response, in which the tape node collects sensor data from both the vibration sensor and the acoustic sensor simultaneously. Since the tape node **2410** contains both sensors on the tape node **2410**, the vibration data and the audio data can be captured synchronously with low latency.

In another embodiment, a first tape node is deployed in the field near the asset **2420** to capture audio data using an audio sensor of the first tape node, e.g., to detect the sound of the chain **2430** breaking, and a second tape node is deployed on the asset **2420** to measure the vibration of the asset **2420** using a vibration sensor of the second tape node. The first tape node and the second tape node may communicate with each other using a first type of wireless communication interface (e.g., Bluetooth or BLE), and the first tape node and the second tape node are located at a distance from each other that is within the communication range of the first type of wireless communication range. The first tape node may detect the audio signal corresponding to the chain **2430** breaking, and instruct the second tape node to begin capturing vibration sensor data using wireless communication, in response. Both the first tape node and the second tape node may synchronously measure audio data and vibration data respectively, with precise timing and low latency. Since the tape nodes use low-latency, short-range wireless communication and are able to perform the decision making to initiate the synchronous capture of sensor data without needing to first contact the server of the sensing system **400**, in some embodiments, the tape nodes are able to synchronize their sensor data capture with high precision. In further embodiments, the tape nodes are able to synchronize the timing of their sensor data capture to less than a millisecond precision. In a conventional system, where a server separately instructs a vibration sensor and an audio sensor that are not onboard the same device, the conventional system may suffer from the latency of communicating with a remote server. The latency may result in the vibration sensor and the audio sensor being unable to synchronize their sensor data capture with high frequency. However, in the disclosed system the tape nodes are able to coordinate their sensor capture without first contacting the server, which allows for high precision timing and recording of the data. The two data

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streams (audio and vibration data from the respective first and second tape nodes) can be uploaded together or separately to the sensing system. Additionally, insights and event detection may be enhanced by using more than one sensor modality to detect events.

While the example discussed above includes the two sensor modalities of an acoustic sensor and a vibration sensor, the embodiments are not limited thereto, and the same disclosed principles, methods, and systems thereof are applicable to any sensor modalities. Additionally, the same may be applied to two instances of the same sensor modality, such as two tape nodes measuring vibration at two different locations on an asset.

Example Computer Apparatus

FIG. 27 shows an example embodiment of computer apparatus 320 that, either alone or in combination with one or more other computing apparatus, is operable to implement one or more of the computer systems described in this specification. A client device in the sensing system 400, 500 may be an embodiment of the computer apparatus 320.

The computer apparatus 320 includes a processing unit 322, a system memory 324, and a system bus 326 that couples the processing unit 322 to the various components of the computer apparatus 320. The processing unit 322 may include one or more data processors, each of which may be in the form of any one of various commercially available computer processors. The system memory 324 includes one or more computer-readable media that typically are associated with a software application addressing space that defines the addresses that are available to software applications. The system memory 324 may include a read only memory (ROM) that stores a basic input/output system (BIOS) that contains start-up routines for the computer apparatus 320, and a random-access memory (RAM). The system bus 326 may be a memory bus, a peripheral bus or a local bus, and may be compatible with any of a variety of bus protocols, including PCI, VESA, Microchannel, ISA, and EISA. The computer apparatus 320 also includes a persistent storage memory 328 (e.g., a hard drive, a floppy drive, a CD ROM drive, magnetic tape drives, flash memory devices, and digital video disks) that is connected to the system bus 326 and contains one or more computer-readable media disks that provide non-volatile or persistent storage for data, data structures and computer-executable instructions.

A user may interact (e.g., input commands or data) with the computer apparatus 320 using one or more input devices 330 (e.g., one or more keyboards, computer mice, microphones, cameras, joysticks, physical motion sensors, and touch pads). Information may be presented through a graphical user interface (GUI) that is presented to the user on a display monitor 332, which is controlled by a display controller 334. The computer apparatus 320 also may include other input/output hardware (e.g., peripheral output devices, such as speakers and a printer). The computer apparatus 320 connects to other network nodes through a network adapter 336 (also referred to as a “network interface card” or NIC).

A number of program modules may be stored in the system memory 324, including application programming interfaces 338 (APIs), an operating system (OS) 340 (e.g., the Windows® operating system available from Microsoft Corporation of Redmond, Washington U.S.A.), software applications 341 including one or more software applications programming the computer apparatus 320 to perform one or more of the steps, tasks, operations, or processes of the locationing and/or tracking systems described herein,

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drivers 342 (e.g., a GUI driver), network transport protocols 344, and data 346 (e.g., input data, output data, program data, a registry, and configuration settings).

Examples of the subject matter described herein, including the disclosed systems, methods, processes, functional operations, and logic flows, can be implemented in data processing apparatus (e.g., computer hardware and digital electronic circuitry) operable to perform functions by operating on input and generating output. Examples of the subject matter described herein also can be tangibly embodied in software or firmware, as one or more sets of computer instructions encoded on one or more tangible non-transitory carrier media (e.g., a machine-readable storage device, substrate, or sequential access memory device) for execution by data processing apparatus.

The details of specific implementations described herein may be specific to particular embodiments of particular inventions and should not be construed as limitations on the scope of any claimed invention. For example, features that are described in connection with separate embodiments may also be incorporated into a single embodiment, and features that are described in connection with a single embodiment may also be implemented in multiple separate embodiments. In addition, the disclosure of steps, tasks, operations, or processes being performed in a particular order does not necessarily require that those steps, tasks, operations, or processes be performed in the particular order; instead, in some cases, one or more of the disclosed steps, tasks, operations, and processes may be performed in a different order or in accordance with a multi-tasking schedule or in parallel.

Other embodiments are within the scope of the claims.

Additional Configuration Information

The foregoing description of the embodiments of the disclosure have been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

Some portions of this description describe the embodiments of the disclosure in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In one embodiment, a software module is implemented with a computer program product comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

Embodiments of the disclosure may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program

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stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

Embodiments of the disclosure may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.

Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the disclosure be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments is intended to be illustrative, but not limiting, of the scope of the disclosure, which is set forth in the following claims.

What is claimed is:

1. A method comprising:

instructing a first sensing device to enter a first state, the first state comprising a first set of functions and behavior for the sensing device, the sensing device including at least one sensor that measures a property of the surroundings of the sensing device;

instructing the first sensing device to aggregate sensing data, the sensing data comprising measurements of the property of the surroundings of the first sensing device measured by the at least one sensor;

storing relevant data from the aggregated sensing data on one of a memory of the first sensing device, a database of a central database and control system associated with the first sensing device, and a memory of a client device;

determining that a first event has occurred based on the aggregated sensing data;

responsive to the determining that the first event has occurred, instructing the first sensing device to enter a second state, the second state comprising a second set of a functions and behavior for the first sensing device, wherein the first sensing device comprises a first type of wireless communication system, the first sensing device configured to communicate with one or more other wireless nodes of a sensing system; and responsive to the determining that the first event has occurred, instructing a second sensing device to enter a third state, the third state comprising a third set of a functions and behavior for the second sensing device.

2. The method of claim 1, wherein the at least one sensor comprises an acoustic sensor, and the sensing data comprises audio data.

3. The method of claim 1, wherein the classifier outputs a request for more sensing data, the method further comprising:

instructing the sensing device to enter a third state comprising changing a compression setting or a sampling frequency relevant to the sensing data;

instructing the sensing device to aggregate additional sensing data and provide the additional data to the classifier according to the third state, wherein

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determining that the first event has occurred is further based on an output of the classifier, the output based on the additional sensing data and the accessed parameters.

4. The method of claim 3, wherein the compression setting comprises one of a bitrate, a type of compression, and a codec.

5. The method of claim 1, wherein one or more parameters, one or more functions, or one or more weighting factors of the classifier are stored on a memory of the tape node.

6. The method of claim 1, wherein the first type of wireless communication system comprises at least one of a Bluetooth communication system, a Bluetooth Low Energy (BLE) communication system, a Zigbee communication system, and an ultra-wideband (UWB) communication system.

7. The method of claim 1, wherein the second sensing device comprises the first type of wireless communication system, the second sensing device configured to communicate with the first sensing device using the first type of wireless communication system.

8. The method of claim 7, wherein the second state comprises capturing sensor data measured by the at least one sensor synchronously with a sensor data capture of the second sensing device.

9. The method of claim 8, wherein the third state comprises the second sensing device capturing sensor data using at least one sensor of the second sensing device synchronously with the sensor data capture of the first sensing device in the second state.

10. The method of claim 9, wherein the first sensing device in the second state and the second sensing device in the third state synchronize a timing of their respective sensor data capture based at least in part on communications over the first type of wireless communication system.

11. The method of claim 9, wherein the at least one sensor of the first sensing device is a different type of sensor than the at least one sensor of the second sensing device.

12. The method of claim 8, wherein the second sensing device is located at a distance from the first sensing device that is less than a communication range associated with the first type of wireless communication system.

13. The method of claim 1, wherein the determining that the first event has occurred comprises:

inputting the aggregated sensing data to a classifier;

receiving an output from the classifier that is based on the input aggregated sensing data and based on parameters accessed by the classifier; and

determining that the first event has occurred based on the output from the classifier.

14. The method of claim 13, wherein the, wherein the classifier is a trained machine learning mode.

15. A method comprising:

instructing a first sensing device to enter a first state, the first state comprising a first set of functions and behavior for the sensing device, the sensing device including at least one sensor that measures a property of the surroundings of the sensing device;

instructing the first sensing device to aggregate sensing data, the sensing data comprising measurements of the property of the surroundings of the first sensing device measured by the at least one sensor;

storing relevant data from the aggregated sensing data on one of a memory of the first sensing device, a database

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of a central database and control system associated with the first sensing device, and a memory of a client device;
determining that a first event has occurred based on the aggregated sensing data;
responsive to the determining that the first event has occurred, instructing the first sensing device to enter a second state, the second state comprising a second set of a functions and behavior for the first sensing device; and
responsive to the determining that the first event has occurred, instructing a second sensing device to enter a third state, the third state comprising a third set of a functions and behavior for the second sensing device.

16. The method of claim **15**, wherein the second sensing device comprises the first type of wireless communication system, the second sensing device configured to communicate with the first sensing device using the first type of wireless communication system.

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17. The method of claim **16**, wherein the second state comprises capturing sensor data measured by the at least one sensor synchronously with a sensor data capture of the second sensing device.

18. The method of claim **17**, wherein the third state comprises the second sensing device capturing sensor data using at least one sensor of the second sensing device synchronously with the sensor data capture of the first sensing device in the second state.

19. The method of claim **18**, wherein the first sensing device in the second state and the second sensing device in the third state synchronize a timing of their respective sensor data capture based at least in part on communications over the first type of wireless communication system.

20. The method of claim **18**, wherein the at least one sensor of the first sensing device is a different type of sensor than the at least one sensor of the second sensing device.

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